Finite Frame Theory

Geometry and Computation

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Lecture Notes

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Contents

Chapter 1

Motivation and Foundations

1.1 Why Study Frames? Redundancy as a Feature

Almost every linear algebra course you have taken builds toward the same destination: a *basis*. A basis of a vector space V is a set of vectors that is simultaneously *spanning* (every vector in V can be written as a combination of them) and *linearly independent* (no vector in the set is redundant — each one is doing essential work). This is an elegant and economical idea. If $\dim V = n$, a basis has exactly n vectors: no more, no fewer.

This chapter begins by asking you to set that economy aside, at least temporarily, and consider a different question.

> *What if we deliberately use* more *vectors than the dimension of the space requires?*

At first this sounds wasteful. If n vectors already span $\mathbb{R}^n$, why would anyone use $m > n$ vectors to do the same job? The answer, which is the entire motivation for this course, is that redundancy is not always waste — it can be a resource. A redundant representation of a vector can be:

- **Robust to noise.** If you measure a vector's coordinates with respect to a basis and one measurement is corrupted, you have lost information that cannot be recovered from the others. If you have redundant measurements, the corrupted one can sometimes be detected, discarded, or corrected using the rest.
- **Robust to erasure.** In communication and signal processing, data is sometimes lost entirely in transmission (an "erasure"). A redundant representation can often be perfectly reconstructed even when some of its pieces never arrive.
- **Flexible.** A basis gives you exactly one way to write each vector as a combination of the basis elements. A redundant spanning set gives you a whole family of ways, and you are free to choose the representation best suited to your purposes (smallest coefficients, sparsest representation, most numerically stable, and so on).
- **Symmetric.** Certain highly redundant, highly symmetric configurations of vectors (we will meet our first examples by the end of this chapter) treat every direction in space with perfect uniformity, in a sense we will make precise. No basis can do this — the asymmetry of having exactly n vectors in an n-dimensional space forces some directions to be treated differently from others.

Frame theory is the mathematical framework for making these ideas precise. A *frame* is, roughly speaking, a spanning set of vectors — typically with redundancy, i.e., more vectors than the dimension requires — equipped with quantitative control over how that redundancy behaves. The formal definition is the subject of Chapter 2 (next week's notes); this week, we build the geometric intuition and the linear algebra toolkit that the definition will rest on.

Key Idea

A frame is a generalization of a basis that permits (and often deliberately uses) redundancy. The price of redundancy is that representations are no longer unique. The payoff is robustness, flexibility, and — as we will soon see geometrically — a kind of balance that no basis can achieve.

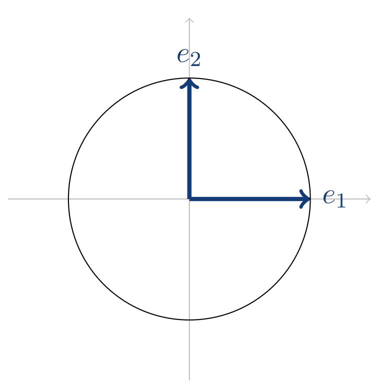
1.1.1 A First Picture

Consider the plane $\mathbb{R}^2$. A basis consists of exactly two linearly independent vectors. The standard basis $\{e_1, e_2\}$, with $e_1 = (1, 0)$ and $e_2 = (0, 1)$, is the most familiar example: it is orthonormal, meaning each vector has length 1 and the two vectors are perpendicular.

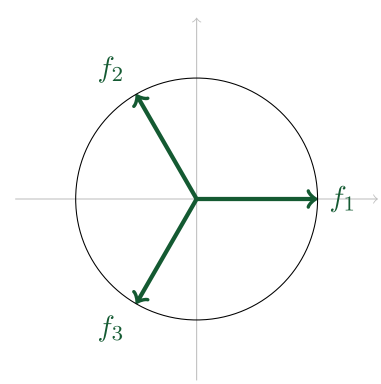
Now consider three vectors in $\mathbb{R}^2$ pointing toward the vertices of an equilateral triangle centered at the origin:

$$f_1 = (1, 0), \qquad f_2 = \left(-\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\right), \qquad f_3 = \left(-\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}\right).$$

These three vectors are linearly *dependent* (any two of them already span $\mathbb{R}^2$, so the third is redundant; indeed $f_1 + f_2 + f_3 = 0$), but they still span $\mathbb{R}^2$, and they do so with a kind of 120°-rotational symmetry that no pair of vectors can replicate. We will see by the end of this chapter, and prove rigorously in Chapter 4, that this triangular configuration is an example of a *tight frame*: a redundant spanning set that distributes "energy" perfectly evenly across every direction in the plane. Figure 1.1 previews the visual difference between the two configurations.



Orthonormal basis $\{e_1, e_2\}$



Redundant frame $\{f_1, f_2, f_3\}$

Figure 1.1: An orthonormal basis (left) versus a redundant, symmetric spanning set (right) in $\mathbb{R}^2$. Both span the plane. The right-hand configuration has three-fold rotational symmetry that two vectors alone cannot achieve.

This picture is the geometric heart of the course, and we will return to it constantly. For now, simply notice: *both* pictures span $\mathbb{R}^2$, but they are doing geometrically different things, and the difference is something we will learn to measure precisely.

1.1.2 Where Frames Show Up

To motivate the abstract definitions to come, here is a brief, non-exhaustive preview of where redundant spanning sets are useful in practice. We will revisit several of these applications in detail later in the course (particularly Weeks 10 and 11).

- **Signal and image processing.** Redundant systems of vectors (generalizing wavelets and Fourier bases) are used to represent signals in ways that are robust to quantization error and compression artifacts.
- **Communications and coding theory.** When data is transmitted over a noisy or unreliable channel, redundant encodings allow the receiver to reconstruct the original message even if some pieces are lost (erasures) or corrupted (noise).
- **Compressed sensing.** Modern signal acquisition techniques exploit redundant, highly incoherent systems of vectors to recover high-dimensional signals from surprisingly few measurements.
- **Quantum information theory.** Certain highly symmetric finite frames (equiangular tight frames, in particular) are intimately connected to optimal quantum measurements.

You do not need to know anything about these fields to take this course. They are mentioned here only so that, as we develop the theory, you have a sense of why anyone bothered to develop it.

1.2 Linear Algebra Review

Frame theory is built on a small number of linear algebra concepts, used repeatedly and combined in new ways. This section reviews exactly the material we need: inner product spaces, orthonormal bases, and spanning sets/linear independence, with an eye toward the question that will occupy us for the rest of the course: *what happens when a spanning set has more vectors than the dimension of the space?*

Throughout, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$; we work in the finite-dimensional space $\mathbb{F}^n$. The vast majority of examples in this course will use $\mathbb{R}^n$, especially in low dimensions where pictures are possible, but the complex case $\mathbb{C}^n$ is equally important in applications (especially quantum information theory and signal processing with complex exponentials) and the theory transfers with only minor modifications, which we flag as they arise.

1.2.1 Inner Product Spaces

Definition 1.2.1 (Inner product)**.** Let V be a vector space over $\mathbb{F}$. An *inner product* on V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ satisfying, for all $x, y, z \in V$ and $\alpha \in \mathbb{F}$:

(i) **Conjugate symmetry:** $\langle x, y \rangle = \overline{\langle y, x \rangle}$.

(ii) **Linearity in the first argument:** $\langle \alpha x + y, z \rangle = \alpha \langle x, z \rangle + \langle y, z \rangle$.

(iii) **Positive definiteness:** $\langle x, x \rangle \geq 0$, with equality if and only if $x = 0$.

Remark. When $\mathbb{F} = \mathbb{R}$, conjugate symmetry is simply symmetry: $\langle x, y \rangle = \langle y, x \rangle$. The standard inner product on $\mathbb{R}^n$ is the familiar dot product, $\langle x, y \rangle = \sum_{k=1}^{n} x_k y_k$. On $\mathbb{C}^n$, the standard inner product is $\langle x, y\rangle = \sum_{k=1}^{n} x_k \overline{y_k}$, where the conjugate on the second argument is exactly what is needed to make positive definiteness hold: $\langle x, x\rangle = \sum_k |x_k|^2 \geq 0$.

Convention check. Some texts place the conjugate on the first argument instead of the second; this is purely a convention and has no mathematical consequence as long as it is used consistently. We will consistently conjugate the second argument, matching the most common convention in the signal processing and frame theory literature (and matching NumPy's `vdot` function, which we will use in the lab).

The inner product immediately gives us a notion of length:

Definition 1.2.2 (Norm)**.** The *norm* (or *length*) induced by an inner product is

$$\|x\| := \sqrt{\langle x, x\rangle}.$$

The norm satisfies the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$ and the homogeneity property $\|\alpha x\| = |\alpha| \, \|x\|$, both of which you likely proved in your linear algebra course using the following essential inequality.

Theorem 1.2.3 (Cauchy–Schwarz Inequality)**.** *For all $x, y \in V$,*

$$|\langle x, y\rangle| \leq \|x\| \, \|y\| ,$$

with equality if and only if x and y are linearly dependent (i.e., one is a scalar multiple of the other).

We state this without proof, as it is a standard result you should have already seen; if it is unfamiliar, any linear algebra textbook will supply a proof, and it is a worthwhile exercise to reconstruct.

Remark. Cauchy–Schwarz will be the single most frequently invoked inequality in this entire course. The quantity $|\langle x, y\rangle|$ measures, in a precise sense, how much x and y "overlap" or "align," and Cauchy–Schwarz says this overlap can never exceed the product of their lengths. When we define frame bounds in Chapter 2, the quantity $\sum_i |\langle x, f_i\rangle|^2$ — a sum of squared overlaps between x and each frame vector f_i — is the central object of study, and intuition for it is built entirely on intuition for $|\langle x, y\rangle|$ itself.

Definition 1.2.4 (Orthogonality)**.** Two vectors $x, y \in V$ are *orthogonal* if $\langle x, y\rangle = 0$. A set of vectors $\{v_1, \ldots, v_k\}$ is *orthogonal* if $\langle v_i, v_j\rangle = 0$ for all $i \neq j$, and *orthonormal* if it is orthogonal and, additionally, $\|v_i\| = 1$ for every i.

1.2.2 Spanning Sets, Linear Independence, and Bases

Definition 1.2.5 (Span)**.** Given vectors $v_1, \ldots, v_m \in V$, their *span* is the set of all linear combinations:

$$\operatorname{span}\{v_1, \ldots, v_m\} := \left\{ \sum_{i=1}^{m} c_i v_i \ : \ c_1, \ldots, c_m \in \mathbb{F} \right\}.$$

If $\operatorname{span}\{v_1, \ldots, v_m\} = V$, we say the set $\{v_1, \ldots, v_m\}$ *spans* V, or is a *spanning set* for V.

Definition 1.2.6 (Linear independence)**.** Vectors $v_1, \ldots, v_m \in V$ are *linearly independent* if the only solution to

$$c_1 v_1 + c_2 v_2 + \cdots + c_m v_m = 0$$

is $c_1 = c_2 = \cdots = c_m = 0$. If a nontrivial solution exists, the vectors are *linearly dependent.*

Definition 1.2.7 (Basis)**.** A set $\{v_1, \ldots, v_n\} \subset V$ is a *basis* for V if it is both a spanning set and linearly independent.

A fundamental fact, which you should have proven in your first linear algebra course, is that *every basis of a finite-dimensional vector space V has the same number of elements*; this common number is the *dimension* of V, denoted $\dim V$. We will take $\dim V = n$ throughout, working in $V = \mathbb{F}^n$.

Here is the observation that frame theory is built on, stated as a proposition so that we can refer back to it precisely.

Proposition 1.2.8 (Redundant spanning sets exist)**.** *Let V be an n-dimensional vector space, and let $m > n$. Then there exist sets of m vectors $\{v_1, \ldots, v_m\} \subset V$ that span V. Any such set is necessarily linearly* dependent*.*

Proof. For existence: take any basis $\{e_1, \ldots, e_n\}$ of V, and adjoin any $m - n$ additional vectors of V (for instance, repeat e_1 a total of $m - n$ times, or pick arbitrary vectors). The resulting set of m vectors still spans V, since it contains a spanning set as a subset.

For dependence: this is an immediate consequence of the fact (proved in any first linear algebra course, often called the Steinitz exchange lemma or a corollary of it) that no linearly independent set in an n-dimensional space can have more than n elements. Since $m > n$, the set $\{v_1, \ldots, v_m\}$ cannot be linearly independent. □

This proposition is almost trivial to prove, but its consequence is the entire subject of this course: *whenever we use more vectors than the dimension requires, linear dependence is unavoidable, but this dependence can be harnessed rather than merely tolerated.* The redundant vectors are not "wasted"; as we saw in Section 1.1.1, they can be arranged to provide symmetry, robustness, and flexibility that no basis (with exactly n vectors) could provide. Distinguishing a *useful* redundant spanning set from an arbitrary, poorly-behaved one is precisely what the frame bounds A and B — to be defined next week — are designed to do.

1.2.3 Orthonormal Bases and Coordinate Representations

Orthonormal bases deserve special attention because they make the relationship between a vector and its "coordinates" completely transparent, and this transparency is exactly what we will have to work to recover in the redundant setting.

Theorem 1.2.9 (Orthonormal expansion)**.** *Let $\{e_1, \ldots, e_n\}$ be an orthonormal basis of V. Then every $x \in V$ can be written uniquely as*

$$x = \sum_{k=1}^{n} \langle x, e_k \rangle \, e_k,$$

and moreover,

$$\|x\|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2 . \qquad \textit{(Parseval's identity)}$$

Proof. Since $\{e_1, \ldots, e_n\}$ is a basis, we may write $x = \sum_k c_k e_k$ for some uniquely determined scalars c_k. Taking the inner product of both sides with e_j and using orthonormality,

$$\langle x, e_j \rangle = \sum_{k=1}^{n} c_k \langle e_k, e_j \rangle = c_j,$$

since $\langle e_k, e_j \rangle = 0$ for $k \neq j$ and $\langle e_j, e_j \rangle = 1$. This proves $c_j = \langle x, e_j \rangle$, giving the expansion formula. For Parseval's identity,

$$\|x\|^2 = \langle x, x \rangle = \left\langle \sum_k c_k e_k, \sum_j c_j e_j \right\rangle = \sum_{k,j} c_k \overline{c_j} \langle e_k, e_j \rangle = \sum_{k=1}^{n} |c_k|^2 = \sum_{k=1}^{n} |\langle x, e_k \rangle|^2,$$

using orthonormality once more to collapse the double sum to a single sum. □

Remark. This theorem is the single most important fact to internalize before next week, because the entire definition of a frame is obtained by asking: *what happens if we drop the requirement that $\{e_1, \ldots, e_n\}$ be a basis (i.e., we allow redundancy, $m > n$, and we no longer require orthonormality), and ask only that an inequality version of Parseval's identity still hold?* That single question is the seed from which the formal definition of a frame, next week's central topic, will grow:

$$A \|x\|^2 \;\leq\; \sum_{i=1}^{m} |\langle x, f_i \rangle|^2 \;\leq\; B \|x\|^2 \qquad \text{for all } x \in V,$$

for constants $0 < A \leq B < \infty$ called the *frame bounds.* When $\{f_i\}$ happens to be an orthonormal basis, Theorem 1.2.9 shows $A = B = 1$ exactly, with equality (not just inequality) in the middle expression. Frames generalize this by relaxing equality to a two-sided bound, and relaxing exact orthonormality to mere redundancy-with-control.

1.2.4 Why Redundant Sets Need Two Numbers, Not One

It is worth pausing on a subtlety that will matter enormously starting next week. Given a redundant spanning set $\{f_1, \ldots, f_m\}$ with $m > n$, the quantity

$$\sum_{i=1}^{m} |\langle x, f_i \rangle|^2$$

is no longer equal to $\|x\|^2$ for every x (as it was for an orthonormal basis by Parseval's identity); it merely lies between two multiples of $\|x\|^2$, and *which* multiples depends on the direction of x. Geometrically: a redundant set of vectors might "illuminate" some directions in space more than others. The smallest amount of illumination, over all directions x, is (essentially) the lower frame bound A; the largest amount of illumination is the upper frame bound B. A set is called *tight* exactly when $A = B$ — when every direction is illuminated equally — and Section 1.1.1's triangular configuration was an example (which we will verify rigorously, and numerically, very soon).

Exercise 1.1

Let $f_1 = (1, 0)$ and $f_2 = (0, 2)$ in $\mathbb{R}^2$ (note: *not* orthonormal, since $\|f_2\| = 2 \neq 1$). Compute $\sum_{i=1}^{2} |\langle x, f_i \rangle|^2$ for $x = (1, 0)$ and again for $x = (0, 1)$. Observe that the two values are different, despite $\|x\|^2 = 1$ in both cases. What does this tell you about how $\{f_1, f_2\}$ treats different directions in the plane?

Exercise 1.2

Prove the converse direction implicit in Theorem 1.2.9: if $\{e_1, \ldots, e_n\}$ is an orthonormal *set* (not assumed to be a basis) and $\sum_{k=1}^{n} |\langle x, e_k \rangle|^2 = \|x\|^2$ for every $x \in V$, then $\{e_1, \ldots, e_n\}$ must in fact be a basis of V (equivalently, $n = \dim V$). (Hint: argue by contradiction — what would the inner product computation in the proof of Theorem 1.2.9 look like if a basis vector were missing?)

1.3 First Steps in Python

Numerical computation is a central theme of this course, not an afterthought: many of the most important properties of frames (the frame bounds, tightness, dual frames) are most easily understood by computing them. We use Python with the NumPy library throughout, because it makes vector and matrix computation immediate and because the syntax is close enough to mathematical notation to be readable at a glance.

If you have never written a line of code before, this section is for you. If you have programmed before (in any language), you can skim this section and proceed directly to the lab exercises in Section 1.5.

1.3.1 Why Python and NumPy

Python is a general-purpose programming language known for readable syntax. **NumPy** (Numerical Python) is a library — a collection of pre-written code — that adds fast, convenient array and matrix operations to Python. Without NumPy, representing a vector and computing an inner product would require writing your own loops; with NumPy, these become one-line operations that closely mirror mathematical notation. Every computational lab in this course will use NumPy as its foundation, and later weeks will add `SciPy` (for linear algebra routines like eigenvalue decompositions) and `Matplotlib` (for plotting).

1.3.2 Vectors as Arrays

In NumPy, a vector in $\mathbb{R}^n$ is represented as a one-dimensional array.

Listing 1.1: Creating and inspecting vectors

```python
import numpy as np

# A vector in R^2
x = np.array([1.0, 0.0])
y = np.array([0.0, 1.0])

print(x)        # [1. 0.]
print(x.shape)  # (2,)  -- this is a 1D array of length 2
```

The `.shape` attribute tells you the dimensions of an array; for a vector in $\mathbb{R}^n$, it will be `(n,)`.

1.3.3 Inner Products and Norms

The inner product of two real vectors is computed with `np.dot`, and the norm with `np.linalg.norm`.

Listing 1.2: Computing inner products and norms

```
import numpy as np

x = np.array([1.0, 2.0])
y = np.array([3.0, -1.0])

inner_xy = np.dot(x, y)          # <x, y> = 1*3 + 2*(-1) = 1
norm_x   = np.linalg.norm(x)     # sqrt(1^2 + 2^2) = sqrt(5)

print(f"<x, y> = {inner_xy}")
print(f"||x|| = {norm_x:.4f}")
```

Remark. For *complex* vectors, recall from Section 1.2.1 that our inner product convention conjugates the second argument: $\langle x, y\rangle = \sum_k x_k \overline{y_k}$. The function `np.dot` does *not* conjugate, so for complex vectors you should use `np.vdot(y, x)` or write `np.dot(x, np.conj(y))` explicitly. We will need this starting in later chapters when complex frames appear; for this week's exercises, real vectors suffice.

1.3.4 Verifying Cauchy–Schwarz Numerically

A nice first programming exercise is to numerically verify the Cauchy–Schwarz inequality (Theorem 1.2.3) on a few examples, as a sanity check on both the mathematics and your NumPy syntax.

Listing 1.3: Checking Cauchy–Schwarz on random vectors

```
import numpy as np

np.random.seed(0)  # for reproducibility

for trial in range(5):
    x = np.random.randn(3)   # random vector in R^3
    y = np.random.randn(3)

    lhs = abs(np.dot(x, y))
    rhs = np.linalg.norm(x) * np.linalg.norm(y)

    print(f"Trial {trial}: |<x,y>| = {lhs:.4f}, "
          f"||x|| ||y|| = {rhs:.4f}, "
          f"inequality holds: {lhs <= rhs}")
```

Running this will print `True` for every trial (up to floating-point rounding, the inequality should never be violated). This kind of numerical sanity check — verifying a theorem on random or specific examples — is a habit we will use throughout the course, especially when working with frame bounds, where direct computation by hand becomes impractical for m and n larger than 2 or 3.

1.3.5 Plotting Vectors in the Plane

Visualizing vectors in $\mathbb{R}^2$ is one of the best ways to build geometric intuition, and we will do this constantly. Matplotlib's `quiver` function draws arrows from the origin.

Listing 1.4: Plotting the basis and the triangular frame from Figure 1.1

```python
import numpy as np
import matplotlib.pyplot as plt

# Orthonormal basis
e1 = np.array([1.0, 0.0])
e2 = np.array([0.0, 1.0])

# Triangular frame (3 vectors, 120 degrees apart)
f1 = np.array([1.0, 0.0])
f2 = np.array([-0.5, np.sqrt(3)/2])
f3 = np.array([-0.5, -np.sqrt(3)/2])

fig, axes = plt.subplots(1, 2, figsize=(10, 5))

# Left subplot: orthonormal basis
ax = axes[0]
ax.quiver(0, 0, e1[0], e1[1], angles='xy', scale_units='xy',
          scale=1, color='steelblue', width=0.012)
ax.quiver(0, 0, e2[0], e2[1], angles='xy', scale_units='xy',
          scale=1, color='steelblue', width=0.012)
ax.set_xlim(-1.5, 1.5)
ax.set_ylim(-1.5, 1.5)
ax.set_aspect('equal')
ax.set_title('Orthonormal basis')
ax.grid(True, alpha=0.3)

# Right subplot: triangular frame
ax = axes[1]
for f in [f1, f2, f3]:
    ax.quiver(0, 0, f[0], f[1], angles='xy', scale_units='xy',
              scale=1, color='seagreen', width=0.012)
ax.set_xlim(-1.5, 1.5)
ax.set_ylim(-1.5, 1.5)
ax.set_aspect('equal')
ax.set_title('Redundant, symmetric frame')
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('basis_vs_frame.png', dpi=150)
plt.show()
```

This code reproduces Figure 1.1 programmatically. Reproducing a figure from the notes in code is a good way to confirm you understand both the mathematics and the plotting syntax simultaneously.

1.4 Chapter Summary

- Frame theory generalizes the idea of a basis by permitting (and exploiting) redundancy: more vectors than the dimension of the space strictly requires.
- Redundancy is valuable because it provides robustness to noise and erasures, flexibility in choosing representations, and — in highly symmetric configurations — a geometric balance that no basis can achieve.

- Proposition 1.2.8 shows that redundant spanning sets always exist and are automatically linearly dependent; the interesting mathematical content is in *how* that dependence is structured.

- Theorem 1.2.9 (orthonormal expansion and Parseval's identity) is the template that the definition of a frame, to come in Chapter 2, directly generalizes: the equality $\|x\|^2 = \sum_k |\langle x, e_k \rangle|^2$ becomes a two-sided inequality $A\|x\|^2 \leq \sum_i |\langle x, f_i \rangle|^2 \leq B\|x\|^2$ once the orthonormal basis $\{e_k\}$ is replaced by an arbitrary, possibly redundant set $\{f_i\}$.

- Python and NumPy provide the computational backbone of the course; Section 1.3 covered vectors as arrays, inner products, norms, and basic plotting, all of which will be used immediately in this week's lab.

1.5 Lab Exercises

These exercises are meant to be completed in a Jupyter notebook. Submit your code together with printed output and, where requested, plots.

Lab Exercise 1.1: Basis vs. Frame, Visually

Reproduce Figure 1.1 using the code template in Section 1.3.5. Then, add a *third* subplot showing a third configuration of your choosing: pick any two non-orthogonal, non-unit-length vectors in $\mathbb{R}^2$ that still span the plane (i.e., an arbitrary basis), and plot them. Comment (in a code comment or markdown cell) on how this third picture differs visually from both the orthonormal basis and the triangular frame.

Lab Exercise 1.2: Cauchy–Schwarz at Scale

Modify the code in Section 1.3.4 to run 10,000 trials instead of 5, using random vectors in $\mathbb{R}^{10}$ instead of $\mathbb{R}^3$ (hint: `np.random.randn(10)`). Confirm that the Cauchy–Schwarz inequality holds in every trial. Then, modify your code to report the *ratio* $|\langle x, y \rangle|/(\|x\|\,\|y\|)$ for each trial, and plot a histogram of this ratio over all 10,000 trials using `plt.hist`. What is the approximate range of values you observe, and does this match your geometric intuition for what this ratio represents? (Hint: think about the angle between x and y.)

Lab Exercise 1.3: The Asymmetry of Redundant Sets

This exercise makes Exercise 1.1 (the pencil-and-paper one from Section 1.2.4) computational. Let $f_1 = (1, 0)$ and $f_2 = (0, 2)$ as in that exercise.

(a) Write a Python function `illumination(x, vectors)` that takes a vector `x` and a list of vectors, and returns $\sum_i |\langle x, f_i\rangle|^2$.

(b) Evaluate your function at $x = (\cos\theta, \sin\theta)$ for θ ranging over 200 equally spaced points in $[0, 2\pi]$ (hint: `np.linspace(0, 2*np.pi, 200)`), using $\{f_1, f_2\}$ as above.

(c) Plot the resulting values against θ. At which angle(s) θ is the illumination smallest? Largest? Do these extreme values match the squared lengths of f_1 and f_2 in any way you can identify?

This exercise previews exactly the numerical method we will use next week to estimate frame bounds A and B directly from data, before we have any closed-form formula for them.

Lab Exercise 1.4 (Optional, Challenge): A Square Configuration

Consider four vectors in $\mathbb{R}^2$ pointing toward the vertices of a square: $f_k = (\cos(k\pi/2), \sin(k\pi/2))$ for $k = 0, 1, 2, 3$. Using the `illumination` function from Lab Exercise 1.3, numerically investigate whether this configuration is *tight* in the sense of Section 1.2.4 (i.e., whether the illumination is the same for every direction x, not just at a few sample angles). Compare your numerical finding to the triangular configuration from Figure 1.1, and form a conjecture about *which* regular polygon configurations in $\mathbb{R}^2$ might always be tight. We will prove the general result in Chapter 4.

1.6 Supplementary Exercises

Theoretical Exercises

T1. Prove the *parallelogram law*: for any x, y in an inner product space, $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$. Then use the equality condition of Cauchy–Schwarz (Theorem 1.2.3) to characterize exactly when $\|x+y\| = \|x\| + \|y\|$ holds for real vectors x, y.

T2. Chapter 2's proof of Theorem 2.2.1 uses, without proof, the fact that a proper subspace of $\mathbb{F}^n$ has a nontrivial orthogonal complement. Prove this fact: if $W \subsetneq \mathbb{F}^n$ is a subspace with $\dim W < n$, show there exists a nonzero $x \in \mathbb{F}^n$ with $\langle x, w \rangle = 0$ for all $w \in W$. (Hint: extend a basis of W to a basis of $\mathbb{F}^n$ and apply Gram–Schmidt.)

T3. Show that for unit vectors $u, v \in \mathbb{R}^n$, $\|u - v\|^2 = 2 - 2\langle u, v \rangle$. Use this identity to re-express the illumination quantity $\varphi(x) = \sum_i |\langle x, f_i \rangle|^2$ from Section 1.3 purely in terms of the distances $\|x - f_i\|$ when x and every f_i are unit vectors.

T4. Let $f_1, f_2, f_3, f_4 \in \mathbb{R}^3$ be unit vectors pointing to the four vertices of a regular tetrahedron centered at the origin. Prove $f_1 + f_2 + f_3 + f_4 = 0$, generalizing the identity $f_1 + f_2 + f_3 = 0$ noted for the triangular configuration in Section 1.1.1. (Hint: argue by symmetry — what linear map fixes the configuration but could only fix the sum vector if the sum is zero?)

T5. Prove that for any m unit vectors $f_1, \dots, f_m \in \mathbb{R}^n$, $\left\|\sum_{i=1}^m f_i\right\| \le m$, with equality if and only if all f_i are equal.

T6. Prove or disprove: if $\{v_1, \dots, v_n\}$ is *any* basis of $\mathbb{R}^n$ (not necessarily orthonormal), then $\sum_{i=1}^n |\langle x, v_i \rangle|^2 = \|x\|^2$ for every $x \in \mathbb{R}^n$.

Computational Exercises

C1. Extend `illumination` to $\mathbb{R}^3$ using the four standard basis-adjacent vectors $e_1, e_2, e_3, e_1+e_2+e_3$ (normalized). Sample 500 random points on the unit sphere in $\mathbb{R}^3$ (normalize random Gaussian vectors) and plot a histogram of the resulting illumination values.

C2. Numerically verify the parallelogram law from T1 for 1000 random pairs of vectors in $\mathbb{R}^5$, and report the maximum absolute deviation from equality (should be at machine precision).

C3. Construct the regular tetrahedron vectors from T4 explicitly as $(\pm 1, \pm 1, \pm 1)/\sqrt{3}$ with an even number of minus signs (four such sign patterns), verify $f_1 + f_2 + f_3 + f_4 = 0$ numerically, and confirm all six pairwise angles equal $\arccos(-1/3) \approx 109.47°$.

C4. For the basis $v_1 = (1, 0.3)$, $v_2 = (0.4, 1.2)$ from Lab Exercise 1.1, numerically evaluate the ratio $\varphi(x)/\|x\|^2$ for 50 random unit vectors x, and plot the ratio against the angle of x. Confirm the ratio is not constant, resolving T6.

C5. Implement Gram–Schmidt orthogonalization from scratch (no `np.linalg.qr`) for three linearly independent vectors in $\mathbb{R}^3$ of your choosing, and verify the output is orthonormal to within 10^{-10}.

Chapter 2

The Definition of a Finite Frame

2.1 From Parseval's Identity to the Frame Inequality

Chapter 1 ended with a question, posed in the remark following Theorem 1.2.9 (the orthonormal expansion theorem): what happens if we take the equality

$$\|x\|^2 = \sum_{k=1}^{n} |\langle x, e_k\rangle|^2, \qquad \text{valid for every } x \in V \text{ when } \{e_k\}_{k=1}^n \text{ is an orthonormal basis,}$$

and relax it in two ways simultaneously? First, we allow the set of vectors to be redundant: instead of exactly n orthonormal vectors, we allow any finite collection $\{f_i\}_{i=1}^m$, with $m \geq n$ and no orthonormality assumed. Second, we relax the equality to a two-sided inequality, since redundant, non-orthonormal vectors will not generally make the left- and right-hand sides match exactly for every x (we saw this concretely in Section 1.2.4 and in Lab Exercise 1.3, where the quantity $\sum_i |\langle x, f_i\rangle|^2$ varied with the direction of x, even though $\|x\|^2$ did not).

This chapter makes that relaxation precise. The result is the formal definition of a *frame*, the central object of the entire course.

Definition 2.1.1 (Finite frame)**.** Let $V = \mathbb{F}^n$ (with $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$), and let $\{f_i\}_{i=1}^m \subset V$ be a finite collection of vectors. We say $\{f_i\}_{i=1}^m$ is a *frame* for V if there exist constants $0 < A \leq B < \infty$ such that

$$A\,\|x\|^2 \;\leq\; \sum_{i=1}^{m} |\langle x, f_i\rangle|^2 \;\leq\; B\,\|x\|^2 \qquad \text{for every } x \in V. \tag{2.1}$$

The constants A and B are called *frame bounds*; specifically, A is a *lower frame bound* and B is an *upper frame bound*. We refer to inequality (2.1) itself as the *frame inequality* or *frame condition*.

Remark. The frame bounds A and B in Definition 2.1.1 are not unique: if A is a valid lower frame bound, so is any smaller positive number, and if B is a valid upper frame bound, so is any larger number. When we speak of *the* lower and upper frame bounds (with the definite article), we will always mean the *optimal* ones: the largest possible A and the smallest possible B for which (2.1) holds. We will see next week (Chapter 3) that these optimal bounds have an exact description in terms of eigenvalues of the frame operator, but for this chapter, it is the *existence* of some valid pair (A, B) that defines a frame, not their optimal values.

Remark. A notational comment: we use m for the number of frame vectors and n for the dimension of the ambient space throughout this course, matching the convention in most of the finite-frame-theory literature. When $m = n$ and $\{f_i\}$ happens to be linearly independent (hence a basis), Definition 2.1.1 still applies and recovers an important special case, treated in Example 2.1.1 below. The interesting case for this course — the one that motivated all of Chapter 1 — is $m > n$, but the definition does not require this, and it is worth understanding the $m = n$ case first.

2.1.1 First Examples

Example 2.1: An orthonormal basis is a (tight) frame

Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of V. By Theorem 1.2.9, $\sum_{k=1}^n |\langle x, e_k\rangle|^2 = \|x\|^2$ exactly, for every $x \in V$. Thus the frame inequality (2.1) holds with $A = B = 1$: every orthonormal basis is a frame, with both frame bounds equal to 1. This is the simplest possible example, and it is the one that all the definitions in this chapter are built to generalize.

Example 2.2: An arbitrary basis is a frame

Let $\{v_1, \dots, v_n\}$ be *any* basis of V (not necessarily orthonormal). We claim $\{v_i\}_{i=1}^n$ is a frame for V. This requires a short argument, since we no longer have an identity like Parseval's to fall back on directly; we give the argument in full generality in Theorem 2.2.1 below, where it is shown that *every* finite spanning set of V — whether or not it is a basis, and whether or not it is redundant — is automatically a frame. For now, take this as a preview: the frame property is a remarkably mild requirement in finite dimensions, and the next section is devoted to making this precise.

Example 2.3: The triangular frame from Chapter 1

Recall the three vectors from Section 1.1.1:

$$f_1 = (1, 0), \qquad f_2 = \left(-\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\right), \qquad f_3 = \left(-\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}\right) \quad \in \mathbb{R}^2.$$

We will verify rigorously in Example 2.3 (later in this chapter, once we have the right computational tool) that this set satisfies the frame inequality with $A = B = \frac{3}{2}$. For now, notice that this is already a non-obvious, pleasant fact: *every* direction $x \in \mathbb{R}^2$ satisfies $\sum_{i=1}^3 |\langle x, f_i\rangle|^2 = \frac{3}{2}\|x\|^2$ *exactly*, despite the three vectors f_1, f_2, f_3 not being orthogonal to one another (in fact $\langle f_i, f_j\rangle = -\frac{1}{2}$ for every $i \neq j$, as a short computation shows). The redundancy in this configuration is, in a sense we are about to make precise, perfectly distributed.

2.2 Why the Frame Condition Is (Almost) Automatic in Finite Dimensions

This section contains the single most important structural fact about *finite*-dimensional frame theory, and it is the reason this course can focus almost entirely on geometry and computation rather than on existence questions. In infinite dimensions (frames for infinite-dimensional Hilbert spaces, which you may encounter in a future course), it is a delicate matter whether a given spanning set is a frame at all — the upper bound $B < \infty$ in particular can fail for perfectly reasonable-looking spanning sets. In finite dimensions, this delicacy completely disappears.

Theorem 2.2.1 (Every finite spanning set is a frame)**.** *Let $V = \mathbb{F}^n$, and let $\{f_i\}_{i=1}^m \subset V$ be a finite set of vectors. Then $\{f_i\}_{i=1}^m$ is a frame for V (in the sense of Definition 2.1.1) if and only if $\{f_i\}_{i=1}^m$ spans V.*

Proof. ($\Rightarrow$) Suppose $\{f_i\}_{i=1}^m$ is a frame, with frame bounds $0 < A \le B < \infty$, but suppose toward a contradiction that $\{f_i\}_{i=1}^m$ does *not* span V. Then $\operatorname{span}\{f_1, \ldots, f_m\}$ is a proper subspace of V, so its orthogonal complement is nontrivial: there exists a nonzero vector $x_0 \in V$ with $\langle x_0, f_i \rangle = 0$ for every $i = 1, \ldots, m$ (take x_0 to be any nonzero vector orthogonal to the span; such a vector exists because the orthogonal complement of a proper subspace of a finite-dimensional inner product space is itself nontrivial). For this x_0,

$$\sum_{i=1}^{m} |\langle x_0, f_i \rangle|^2 = 0,$$

while the lower frame bound requires $\sum_{i=1}^m |\langle x_0, f_i \rangle|^2 \ge A \|x_0\|^2 > 0$ (using $A > 0$ and $x_0 \ne 0$, so $\|x_0\|^2 > 0$). This is a contradiction. Hence $\{f_i\}_{i=1}^m$ must span V.

($\Leftarrow$) Suppose $\{f_i\}_{i=1}^m$ spans V. We must produce valid constants A, B.

Upper bound. Define $B := \sum_{i=1}^m \|f_i\|^2$. For any $x \in V$ with $\|x\| = 1$, Cauchy–Schwarz (Theorem 1.2.3) gives $|\langle x, f_i \rangle| \le \|x\| \, \|f_i\| = \|f_i\|$ for each i, hence

$$\sum_{i=1}^{m} |\langle x, f_i \rangle|^2 \le \sum_{i=1}^{m} \|f_i\|^2 = B.$$

For general $x \ne 0$, apply this to the unit vector $x / \|x\|$ and use homogeneity of the inner product to recover $\sum_i |\langle x, f_i \rangle|^2 \le B \|x\|^2$; the case $x = 0$ is trivial. This B is finite because it is a finite sum of finite numbers, so the upper bound holds automatically and requires no use of the spanning hypothesis at all (this part of the argument works for *any* finite collection of vectors, spanning or not).

Lower bound. This is where the spanning hypothesis is essential, and where the argument is more interesting. Define the function $\varphi : V \to [0, \infty)$ by

$$\varphi(x) := \sum_{i=1}^{m} |\langle x, f_i \rangle|^2.$$

Restrict φ to the unit sphere $S := \{x \in V \ : \ \|x\| = 1\}$. The function φ is continuous (it is a finite sum of squared moduli of inner products, each of which is continuous in x), and S is compact (closed and bounded in the finite-dimensional space V, by the Heine–Borel theorem). A continuous function on a compact set attains its minimum, so there exists some $x_0 \in S$ with

$$\varphi(x_0) = \min_{x \in S} \varphi(x) =: A.$$

We claim $A > 0$. Indeed, if $A = 0$, then $\varphi(x_0) = \sum_{i=1}^m |\langle x_0, f_i \rangle|^2 = 0$, which forces $\langle x_0, f_i \rangle = 0$ for every $i = 1, \ldots, m$ (a sum of nonnegative terms is zero only if every term is zero). But then x_0 is orthogonal to every f_i, hence orthogonal to $\operatorname{span}\{f_1, \ldots, f_m\} = V$ (using the spanning hypothesis here, for the only time in the proof). A vector orthogonal to the entire space V, including itself, must satisfy $\langle x_0, x_0 \rangle = 0$, i.e., $x_0 = 0$. This contradicts $x_0 \in S$, i.e., $\|x_0\| = 1$. Hence $A > 0$.

Finally, $\varphi(x_0) = A$ is the minimum value over the *unit sphere*, so $\varphi(x) \ge A$ for every unit vector x, and by the same homogeneity argument as before, $\varphi(x) \ge A \|x\|^2$ for every $x \in V$. This is exactly the lower frame bound. $\square$

Key Idea

In finite dimensions, *spanning is the only requirement.* The frame bounds A and B always exist once a finite set of vectors spans V; their *values* are not automatic (they depend delicately on the geometry of the configuration), but their *existence* is. This is why finite frame theory can spend its energy on geometry and computation — on understanding what A and B *are*, and what makes them large, small, or equal — rather than on existence questions, which is where a great deal of the technical difficulty lies in the infinite-dimensional theory.

Corollary 2.2.2. *Every basis of V is a frame for V, with both frame bounds finite and positive. (This makes Example 2.1.1 a special case of Theorem 2.2.1, since a basis is in particular a spanning set.)*

Proof. Immediate from Theorem 2.2.1, since a basis spans V by definition. □

Remark. Theorem 2.2.1 also clarifies exactly what can go wrong, and exactly one thing can: if $\{f_i\}_{i=1}^m$ does *not* span V, the upper bound B in Definition 2.1.1 can still be found exactly as in the proof above (the argument for B never used spanning), but no valid lower bound $A > 0$ can exist, because the orthogonal complement direction x_0 from the ($\Rightarrow$) half of the proof gives $\varphi(x_0) = 0$ while $\|x_0\|^2 > 0$, which is incompatible with any $A > 0$. In the language we will develop next week, a non-spanning set has frame operator with a zero eigenvalue, which is exactly the obstruction to a positive lower bound. So the failure mode for non-frames in finite dimensions is completely specific: it is always a failure of the lower bound caused by a missing direction, never a failure of the upper bound.

2.2.1 A Quick Non-Example

Example 2.4: A spanning failure

Let $V = \mathbb{R}^3$ and consider $g_1 = (1, 0, 0)$, $g_2 = (0, 1, 0)$. These two vectors do *not* span $\mathbb{R}^3$ (their span is only the xy-plane), so by Theorem 2.2.1, $\{g_1, g_2\}$ is *not* a frame for $\mathbb{R}^3$. Concretely: taking $x_0 = (0, 0, 1)$ (orthogonal to both g_1 and g_2), we get $\sum_{i=1}^{2} |\langle x_0, g_i\rangle|^2 = 0$ while $\|x_0\|^2 = 1 > 0$, so no $A > 0$ can satisfy the lower frame bound for this x_0. Note, however, that $\{g_1, g_2\}$ *is* a perfectly good frame for the two-dimensional subspace $\text{span}\{g_1, g_2\} = \mathbb{R}^2 \times \{0\}$, viewed as a space in its own right. Whether a set of vectors is a frame is always relative to which ambient space V it is being asked to span; throughout this course, V will be clear from context, almost always $\mathbb{R}^n$ or $\mathbb{C}^n$ for an explicitly stated n.

2.3 Tight Frames: A First Look

Theorem 2.2.1 tells us *that* frame bounds exist for any spanning set, but says nothing about their values, and in particular says nothing about when the two bounds coincide. This question — when is $A = B$? — turns out to be exactly the right question to ask in order to recover the geometric symmetry we saw informally in Chapter 1 (Figure 1.1). We introduce the relevant terminology now, even though the full geometric treatment is the subject of Chapter 4 (Week 4 in the syllabus); having the definition in hand will make this chapter's examples and lab exercises considerably more meaningful.

Definition 2.3.1 (Tight frame)**.** A frame $\{f_i\}_{i=1}^m$ for V is called a *tight frame* if its frame bounds satisfy $A = B$, i.e., if there is a single constant $A > 0$ such that

$$\sum_{i=1}^m |\langle x, f_i \rangle|^2 = A \|x\|^2 \qquad \text{for every } x \in V.$$

If, in addition, $A = 1$, the frame is called a *Parseval frame.*

Remark. Comparing Definition 2.3.1 to Example 2.1.1: every orthonormal basis is a Parseval frame. The converse is *false* in general — a Parseval frame need not be an orthonormal basis, nor even consist of unit vectors, nor even be linearly independent. We will see explicit examples of Parseval frames that are not bases later in this chapter and in Chapter 4. The point of the terminology is precisely to isolate the property (the exact Parseval-type identity) that we want to keep, while discarding the property (linear independence / exact count of n vectors) that we are willing to give up.

Remark. Tightness is the precise version of the informal phrase "every direction is treated equally," which we used throughout Chapter 1. If $A = B$, then $\sum_i |\langle x, f_i \rangle|^2$ depends on x *only through* $\|x\|^2$ — two vectors x, x' of the same length, but pointing in different directions, produce exactly the same total $\sum_i |\langle x, f_i \rangle|^2$. This is the rigorous content behind the visual symmetry of the triangular configuration in Figure 1.1, and we are now in a position to verify it by direct computation, which we do next.

Example 2.5: Verifying tightness of the triangular frame

We return to Example 2.1.1 and verify the claimed tightness directly. Let $x = (x_1, x_2) \in \mathbb{R}^2$ be arbitrary. Using $f_1 = (1, 0)$, $f_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, $f_3 = (-\frac{1}{2}, -\frac{\sqrt{3}}{2})$:

$$\begin{aligned}
\langle x, f_1 \rangle &= x_1, \\
\langle x, f_2 \rangle &= -\tfrac{1}{2}x_1 + \tfrac{\sqrt{3}}{2}x_2, \\
\langle x, f_3 \rangle &= -\tfrac{1}{2}x_1 - \tfrac{\sqrt{3}}{2}x_2.
\end{aligned}$$

Squaring and summing,

$$\begin{aligned}
\sum_{i=1}^3 |\langle x, f_i \rangle|^2 &= x_1^2 + \left(-\tfrac{1}{2}x_1 + \tfrac{\sqrt{3}}{2}x_2\right)^2 + \left(-\tfrac{1}{2}x_1 - \tfrac{\sqrt{3}}{2}x_2\right)^2 \\
&= x_1^2 + \left(\tfrac{1}{4}x_1^2 - \tfrac{\sqrt{3}}{2}x_1x_2 + \tfrac{3}{4}x_2^2\right) + \left(\tfrac{1}{4}x_1^2 + \tfrac{\sqrt{3}}{2}x_1x_2 + \tfrac{3}{4}x_2^2\right) \\
&= x_1^2 + \tfrac{1}{2}x_1^2 + \tfrac{3}{2}x_2^2 \\
&= \tfrac{3}{2}x_1^2 + \tfrac{3}{2}x_2^2 \\
&= \tfrac{3}{2}\|x\|^2 .
\end{aligned}$$

The cross terms $-\frac{\sqrt{3}}{2}x_1x_2$ and $+\frac{\sqrt{3}}{2}x_1x_2$ cancel exactly — this cancellation is not a coincidence, but a direct consequence of the 120°-rotational symmetry of the configuration, and this kind of cancellation is the algebraic fingerprint of tightness that we will learn to recognize (and engineer) systematically in Chapter 4. We conclude that $\{f_1, f_2, f_3\}$ is a tight frame for $\mathbb{R}^2$ with $A = B = \frac{3}{2}$, confirming the claim made in Example 2.1.1.

2.4 Estimating Frame Bounds Numerically

Example 2.3 found the exact frame bounds of a specific, highly symmetric configuration by direct algebraic computation. This is satisfying, but it does not scale: for a configuration of, say, $m = 20$ vectors in $\mathbb{R}^5$ with no special symmetry, an analogous by-hand computation is impractical. We need a numerical method, and fortunately Chapter 1 already built almost all of the necessary machinery.

Recall the `illumination` function from Lab Exercise 1.3, which computed $\varphi(x) = \sum_i |\langle x, f_i\rangle|^2$ for a given direction x. The proof of Theorem 2.2.1 shows that the optimal frame bounds are exactly

$$A = \min_{\|x\|=1} \varphi(x), \qquad B = \max_{\|x\|=1} \varphi(x),$$

the minimum and maximum of φ *over the unit sphere*. This immediately suggests a numerical strategy: sample many random unit vectors x, evaluate $\varphi(x)$ at each, and use the smallest and largest observed values as estimates of A and B.

Listing 2.1: Estimating frame bounds by random sampling on the unit sphere

```python
import numpy as np

def illumination(x, vectors):
    """Compute sum_i |<x, f_i>|^2 for a list/array of frame vectors."""
    return sum(np.dot(x, f)**2 for f in vectors)

def random_unit_vector(n, rng):
    """Sample a uniformly random unit vector in R^n."""
    v = rng.standard_normal(n)
    return v / np.linalg.norm(v)

def estimate_frame_bounds(vectors, n, num_samples=20000, seed=0):
    """
    Estimate frame bounds A, B for a finite set of vectors in R^n
    by sampling random unit vectors on the sphere.
    """
    rng = np.random.default_rng(seed)
    values = np.empty(num_samples)
    for k in range(num_samples):
        x = random_unit_vector(n, rng)
        values[k] = illumination(x, vectors)
    A_estimate = values.min()
    B_estimate = values.max()
    return A_estimate, B_estimate

# Sanity check: the triangular frame from Example 2.3 / 2.5
f1 = np.array([1.0, 0.0])
f2 = np.array([-0.5, np.sqrt(3)/2])
f3 = np.array([-0.5, -np.sqrt(3)/2])
triangle = [f1, f2, f3]

A_est, B_est = estimate_frame_bounds(triangle, n=2)
print(f"Estimated A = {A_est:.4f}, B = {B_est:.4f}")
print(f"Exact value from Example 2.5: A = B = 1.5")
```

Running this code should produce `A_est` and `B_est` both very close to 1.5 (small deviations from 1.5, typically in the third decimal place with 20,000 samples, are expected and are exactly the kind of sampling error this method is subject to — we are not checking every direction, only 20,000 random ones).

Remark. This sampling method has an important limitation, worth stating plainly rather than discovering by surprise: it can only ever *underestimate* B and *overestimate* A in expectation, since random sampling is unlikely to land exactly on the true minimizing or maximizing direction x_0. The estimates improve (the gap between estimated and true A, B shrinks) as `num_samples` increases, but convergence can be slow, especially in higher dimensions n, where the unit sphere has much more "room" for the optimal direction to hide. Next week (Chapter 3), we will derive an exact, non-random formula for A and B in terms of the eigenvalues of a certain matrix — the *frame operator* — which will make random sampling unnecessary for exact answers. The sampling method nonetheless remains useful throughout the course as a fast sanity check: if your exact, formula-based computation of A and B disagrees substantially with a quick random-sampling estimate, one of the two computations very likely contains a bug.

Example 2.6: A non-tight frame, numerically

Consider $h_1 = (1, 0)$, $h_2 = (0, 1)$, $h_3 = (1, 1)$ in $\mathbb{R}^2$. This set spans $\mathbb{R}^2$ (already h_1, h_2 alone do), so by Theorem 2.2.1 it is a frame. Is it tight? Running the `estimate_frame_bounds` function above on $\{h_1, h_2, h_3\}$ gives estimates of approximately $A \approx 1.00$ and $B \approx 3.00$ (we encourage you to verify this yourself in the lab below, rather than taking these numbers on faith). Since the estimates are clearly different ($A \neq B$), this is *not* a tight frame, in sharp contrast to the triangular configuration of Example 2.3. Geometrically, the direction $x = (1, 1)/\sqrt{2}$ (along which h_3 also happens to point) is more "illuminated" than the direction $x = (1, -1)/\sqrt{2}$ (orthogonal to h_3), and this asymmetry is exactly what the unequal bounds $A \neq B$ are detecting.

2.5 Chapter Summary

- Definition 2.1.1 formalizes the idea previewed throughout Chapter 1: a finite frame is a set of vectors $\{f_i\}_{i=1}^m$ satisfying the two-sided inequality $A\|x\|^2 \leq \sum_i |\langle x, f_i\rangle|^2 \leq B\|x\|^2$ for all x, generalizing Parseval's identity for orthonormal bases (recovered exactly when $A = B = 1$).

- Theorem 2.2.1 is the key finite-dimensional simplification of the whole theory: a finite set of vectors in $\mathbb{F}^n$ is a frame if and only if it spans $\mathbb{F}^n$. Existence of frame bounds is automatic once spanning holds; only their *values* require further work.

- The upper bound B can always be taken to be $\sum_i \|f_i\|^2$, using nothing but Cauchy–Schwarz; the lower bound $A > 0$ requires the spanning hypothesis and a compactness argument (a continuous function attains its minimum on the unit sphere, and that minimum is forced to be positive exactly because no nonzero vector can be orthogonal to everything in a spanning set).

- A frame is *tight* ($A = B$) when this minimum and maximum coincide; geometrically, tightness means every direction x is "illuminated" equally by the configuration, regardless of which way it points. Example 2.3 verified this by hand for the triangular configuration from Chapter 1; Example 2.4 exhibited a frame that is manifestly *not* tight.

- The optimal frame bounds are $A = \min_{\|x\|=1} \varphi(x)$ and $B = \max_{\|x\|=1} \varphi(x)$, where $\varphi(x) = \sum_i |\langle x, f_i\rangle|^2$; this can be estimated numerically by sampling random unit vectors, extending the `illumination` function built in Chapter 1's lab. An exact, sampling-free formula is coming in Chapter 3.

2.6 Lab Exercises

These exercises build directly on the `illumination` function from Lab Exercise 1.3 and the `estimate_frame_bounds` function of Section 2.4. Submit your code together with printed output and, where requested, plots and short written answers.

Lab Exercise 2.1: Confirming Example 2.6

Implement the `estimate_frame_bounds` function from Section 2.4 (or use the version provided) and run it on $\{h_1, h_2, h_3\}$ from Example 2.4, with at least 50,000 samples. Report your estimated A and B. Then, find the *exact* values of A and B by hand (hint: parametrize $x = (\cos\theta, \sin\theta)$ as in Lab Exercise 1.3, write $\varphi(\theta)$ explicitly using a double-angle trigonometric identity, and find its minimum and maximum over $\theta \in [0, 2\pi]$ using single-variable calculus). Compare your exact answer to your numerical estimate.

Lab Exercise 2.2: Spanning vs. Not Spanning, Numerically

Consider the two sets $\{g_1, g_2\}$ from Example 2.2.1 (which does not span $\mathbb{R}^3$) and $\{g_1, g_2, g_3\}$ where $g_3 = (0, 0, 1)$ (which does). Run `estimate_frame_bounds` (modified to sample random unit vectors in $\mathbb{R}^3$ instead of $\mathbb{R}^2$) on both sets. Describe precisely what you observe for the non-spanning set, and explain the result using Theorem 2.2.1 and the discussion immediately following its proof.

Lab Exercise 2.3: The Square Revisited

In Lab Exercise 1.4, you investigated numerically (using only the `illumination` function, without a formal frame-bounds estimator) whether the four vertices of a square, $f_k = (\cos(k\pi/2), \sin(k\pi/2))$ for $k = 0, 1, 2, 3$, form a tight frame for $\mathbb{R}^2$. Revisit this question now using the full `estimate_frame_bounds` machinery of this chapter, and state your conclusion in the language of Definition 2.3.1 (i.e., explicitly identify the common value of $A = B$ if the configuration is tight). Then verify your numerical finding with an exact by-hand computation in the style of Example 2.3.

Lab Exercise 2.4: Building a Frame-Checker

Write a Python function `is_frame(vectors, n, tol=1e-8)` that takes a list of vectors in $\mathbb{R}^n$ and returns `True` if the set spans $\mathbb{R}^n$ (and is therefore a frame, by Theorem 2.2.1) and `False` otherwise. (Hint: this is a question about the *rank* of the matrix whose columns are the given vectors; `np.linalg.matrix_rank` is relevant, and the `tol` parameter should be passed to it to handle floating-point rounding.) Test your function on all of the example sets appearing in this chapter ($\{e_1, e_2\}$, $\{f_1, f_2, f_3\}$, $\{g_1, g_2\}$, $\{g_1, g_2, g_3\}$, $\{h_1, h_2, h_3\}$), and confirm that it correctly identifies which are frames for the relevant ambient space in each case.

Lab Exercise 2.5 (Optional, Challenge): How Many Samples Are Enough?

Using the triangular frame from Example 2.3 (exact value $A = B = 1.5$), run `estimate_frame_bounds` with `num_samples` taking the values $10, 100, 1000, 10000, 100000$ (you may wish to fix the random seed and average over several repetitions at each sample size for a cleaner trend). For each sample size, record the *error* $|A_{\text{est}} - 1.5|$ and $|B_{\text{est}} - 1.5|$. Plot error versus `num_samples` on a log-log scale (`plt.loglog`). Does the error appear to decrease at a steady rate as the sample size grows? This exercise is a gentle first encounter with a question — how quickly does random sampling converge? — that recurs throughout computational mathematics, including (as you may encounter in your own research) numerical optimization on manifolds and Monte Carlo methods more broadly.

2.7 supplementary Exercises

Theoretical Exercises

T1. Let $\{f_i\}_{i=1}^m$ be a frame for $\mathbb{F}^n$ with synthesis matrix F. Using the rank–nullity theorem, prove $\dim \ker F = m - n$.

T2. Prove that if $\{f_i\}_{i=1}^m$ is a tight frame for $\mathbb{R}^n$ with bound A, then $\{Uf_i\}_{i=1}^m$ is also a tight frame with the *same* bound A, for any orthogonal matrix U. (Tightness is a rotation-invariant property.)

T3. Construct two frames $\{f_i\}_{i=1}^3$ and $\{g_i\}_{i=1}^3$ for $\mathbb{R}^2$ (same index set) such that $\{f_i + g_i\}_{i=1}^3$ is *not* a frame for $\mathbb{R}^2$. Conclude that frames are not closed under index-matched addition.

T4. Suppose $\{f_i\}_{i=1}^{n+1}$ is a frame for $\mathbb{R}^n$ with exactly one redundant vector, and suppose that removing *any* single vector still leaves a frame. Prove that no two of the f_i are parallel (scalar multiples of one another). (Hint: argue the contrapositive — if f_j and f_k are parallel, show that removing some *other* vector causes the remaining set to fail to span.)

T5. State and prove the complex analogue of Corollary 2.2.2 (every basis of $\mathbb{C}^n$ is a frame), taking care to use the conjugate-linear convention for the inner product on $\mathbb{C}^n$ established in Section 1.2.1.

T6. Prove: if $\{f_i\}_{i=1}^m$ is a frame for $\mathbb{R}^n$ and $V \subset \mathbb{R}^n$ is any subspace with orthogonal projection P_V, then $\{P_V f_i\}_{i=1}^m$ is a frame for V. (This is a spanning-only counterpart to Theorem 4.4.1, which achieves the stronger conclusion of *tightness* in the special case where $\{f_i\}$ is an orthonormal basis of a larger ambient space.)

Computational Exercises

C1. Using `scipy.linalg.null_space`, compute $\dim \ker F$ for each of the five example frames from Lab Exercise 2.4, and confirm it equals $m - n$ in every case, verifying T1.

C2. Rotate the triangular frame by 10 random angles in $[0, 2\pi)$ and confirm, via `exact_frame_bounds` (built in Chapter 3's notebook, or by direct eigenvalue computation), that $A = B = 1.5$ is unchanged in every case, verifying T2.

C3. For $m = n + 1 = 3$ in $\mathbb{R}^2$, construct a frame with a parallel pair (e.g. $f_1 = (1, 0)$, $f_2 = (2, 0)$, $f_3 = (0, 1)$), and confirm numerically that removing f_3 causes the remaining set to fail to span $\mathbb{R}^2$, verifying T4's underlying mechanism. Then generate 20 random 3-vector frames in $\mathbb{R}^2$ and report what fraction have no parallel pair.

C4. Fix $h_1 = (1, 0)$, $h_2 = (0, 1)$, and let $h_3(\theta) = (\cos\theta, \sin\theta)$. Using `estimate_frame_bounds`, plot $A(\theta)$ and $B(\theta)$ as θ ranges over $[0, 2\pi)$, and identify the angles at which the frame degenerates (fails to span $\mathbb{R}^2$).

C5. Verify T6 numerically: construct a 4-vector frame for $\mathbb{R}^3$ (redundant, $m > n$), project it orthogonally onto 5 random 2-dimensional subspaces (via random unit normal vectors), and confirm the projected 4 vectors span each subspace in every case.

Chapter 3

The Frame Operator

3.1 Overview: What This Chapter Delivers

Chapter 2 ended with two explicit promises. First, that the optimal frame bounds—the largest valid A and smallest valid B—have an exact, closed-form description, not merely a numerical estimate via random sampling. Second, that this description would come from a certain matrix called the *frame operator*. Both promises are fulfilled here.

The frame operator is the mathematical heart of finite frame theory. Once you understand it, nearly every important property of a frame—its bounds, its tightness, its reconstruction formula—can be read off directly from the operator's spectral structure. It also makes the theory fully computable: given the $n \times m$ synthesis matrix F, all frame properties reduce to a single matrix computation.

This chapter introduces three related linear maps—the synthesis operator, the analysis operator, and the frame operator—shows how they relate to one another, proves the main eigenvalue theorem (Theorem 3.5.1), and returns to every running example from Chapters 1–2 to verify all previously stated claims exactly.

3.2 The Synthesis Operator

We begin with the map that turns a sequence of scalar coefficients into a vector in V.

Definition 3.2.1 (Synthesis operator)**.** Let $\{f_i\}_{i=1}^m$ be a frame for $V = \mathbb{F}^n$. The *synthesis operator* is the linear map $F : \mathbb{F}^m \to \mathbb{F}^n$ defined by

$$Fc := \sum_{i=1}^{m} c_i\, f_i, \qquad c = (c_1, \dots, c_m) \in \mathbb{F}^m.$$

In matrix terms, if each $f_i \in \mathbb{F}^n$ is treated as a column vector, then F is the $n \times m$ matrix whose i-th column is f_i:

$$F = \begin{pmatrix} | & | & & | \\ f_1 & f_2 & \cdots & f_m \\ | & | & & | \end{pmatrix} \in \mathbb{F}^{n\times m}.$$

The action $Fc = \sum_i c_i f_i$ is the standard matrix–vector product. The name *synthesis* reflects signal-processing usage: the operator takes a sequence of coefficients $(c_1, \dots, c_m)$ and *synthesises* the corresponding signal $\sum_i c_i f_i \in V$.

Remark. When $m > n$ (the redundant case), F is a *wide* matrix with more columns than rows, so $\ker F \neq \{0\}$. Elements of $\ker F$ are exactly the coefficient sequences c for which $\sum_i c_i f_i = 0$, i.e. the linear relations among the frame vectors. Redundancy in the frame is reflected algebraically in the nontrivial null space of F. We will see in Chapter 5 (dual frames) that this null space parametrises the family of all possible duals.

Example 3.1: Synthesis matrix of the triangular frame

For $f_1 = (1,0)^T$, $f_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})^T$, $f_3 = (-\frac{1}{2}, -\frac{\sqrt{3}}{2})^T$ in $\mathbb{R}^2$, the synthesis operator is the 2×3 matrix

$$F = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}.$$

As a check: $F(1,1,1)^T = f_1 + f_2 + f_3 = (0,0)^T = 0$, confirming that $(1,1,1)^T \in \ker F$, corresponding to the linear relation $f_1 + f_2 + f_3 = 0$ noted in Chapter 1.

3.3 The Analysis Operator

The synthesis operator maps coefficients to vectors. Its adjoint does the reverse: it takes a vector $x \in V$ and *analyses* it into the sequence of frame coefficients $(\langle x, f_1\rangle, \ldots, \langle x, f_m\rangle)$.

Definition 3.3.1 (Analysis operator)**.** The *analysis operator* of $\{f_i\}_{i=1}^m$ is the adjoint $F^* : \mathbb{F}^n \to \mathbb{F}^m$ of the synthesis operator, given by

$$F^*x := \big(\langle x, f_1\rangle, \langle x, f_2\rangle, \ldots, \langle x, f_m\rangle\big), \qquad x \in \mathbb{F}^n.$$

The matrix of F^* is the conjugate transpose of F. In the real case $\mathbb{F} = \mathbb{R}$ this is simply the ordinary transpose, and the i-th entry of F^*x is $(F^*x)_i = f_i^T x = \langle x, f_i\rangle$, as required.

Proposition 3.3.2 (Adjoint formula)**.** *For all $c \in \mathbb{F}^m$ and $x \in \mathbb{F}^n$,*

$$\langle Fc, x\rangle = \langle c, F^*x\rangle.$$

Proof. $\langle Fc, x\rangle = \left\langle \sum_{i=1}^m c_i f_i, x \right\rangle = \sum_{i=1}^m c_i \langle f_i, x\rangle = \sum_{i=1}^m c_i \overline{\langle x, f_i\rangle} = \langle c, F^*x\rangle.$ □

A key observation: the frame inequality from Definition 2.1.1 can be rewritten compactly using F^*. Since $F^*x = (\langle x, f_i\rangle)_{i=1}^m$, we have

$$\sum_{i=1}^m |\langle x, f_i\rangle|^2 = \|F^*x\|^2,$$

so the frame condition $A\|x\|^2 \le \sum_i |\langle x, f_i\rangle|^2 \le B\|x\|^2$ is equivalent to

$$A\|x\|^2 \;\le\; \|F^*x\|^2 \;\le\; B\|x\|^2 \qquad \forall\, x \in V. \tag{3.1}$$

The lower bound $A > 0$ requires F^* to be injective (i.e. $\ker F^* = \{0\}$). Since $\ker F^* = (\operatorname{range} F)^\perp$, injectivity of F^* is equivalent to $\operatorname{range} F = V$, i.e. the frame vectors span V—recovering Theorem 2.2.1 from the operator perspective.

3.4 The Frame Operator

Definition 3.4.1 (Frame operator)**.** The *frame operator* of $\{f_i\}_{i=1}^m$ is the composition $S := FF^* : \mathbb{F}^n \to \mathbb{F}^n$.

Unfolding the composition:

$$Sx = F(F^*x) = \sum_{i=1}^{m} \langle x, f_i \rangle \, f_i.$$

Equivalently, S is a sum of rank-one outer products:

$$S = FF^* = \sum_{i=1}^{m} f_i f_i^*. \tag{3.2}$$

In the real case each $f_i f_i^* = f_i f_i^T$ is the $n \times n$ matrix with (j,k) entry $(f_i)_j (f_i)_k$. In NumPy: `S = F @ F.T`.

Key Idea

The frame operator $S = FF^* = \sum_i f_i f_i^*$ is a **positive (semi)definite** $n \times n$ matrix. Its eigenvalues are the exact optimal frame bounds. Tightness ($A = B$) is equivalent to S being a scalar multiple of the identity. All of this is computable in two lines of NumPy.

3.4.1 Properties of the Frame Operator

Proposition 3.4.2 (Properties of S)**.** *Let $S = FF^*$ be the frame operator of a frame $\{f_i\}_{i=1}^m$ for $\mathbb{F}^n$. Then:*

(i) ***Self-adjoint:*** $S^* = S$.

(ii) ***Positive semidefinite:*** $\langle Sx, x \rangle \geq 0$ *for all* $x \in \mathbb{F}^n$.

(iii) ***Positive definite*** *iff* $\{f_i\}$ *spans* $\mathbb{F}^n$.

(iv) ***Trace formula:*** $\operatorname{tr}(S) = \sum_{i=1}^m \|f_i\|^2$.

Proof. **(i)** $(FF^*)^* = (F^*)^* F^* = FF^* = S$. **(ii)** $\langle Sx, x \rangle = \langle FF^*x, x \rangle = \langle F^*x, F^*x \rangle = \|F^*x\|^2 \geq 0$. **(iii)** $\langle Sx, x \rangle = 0 \Leftrightarrow F^*x = 0 \Leftrightarrow \langle x, f_i \rangle = 0 \ \forall i \Leftrightarrow x \in (\operatorname{span}\{f_i\})^\perp$. This is $\{0\}$ iff the frame spans. **(iv)** $\operatorname{tr}(S) = \sum_{j=1}^n S_{jj} = \sum_{j=1}^n \sum_{i=1}^m |(f_i)_j|^2 = \sum_{i=1}^m \sum_{j=1}^n |(f_i)_j|^2 = \sum_{i=1}^m \|f_i\|^2$. □

Remark. The trace formula (iv) will be used repeatedly: it connects an algebraic invariant of S (its trace) to a geometric property of the configuration (the total squared norm of all frame vectors). For a tight frame with m unit-norm vectors in $\mathbb{R}^n$, we have $S = AI$, so $\operatorname{tr}(S) = An$ on one hand, and $\operatorname{tr}(S) = m$ on the other. Therefore $A = m/n$ is *forced* by tightness and unit-norm. This is a constraint we will use constantly in Chapter 4.

3.5 Frame Bounds Are Eigenvalues of S

We now prove the theorem promised in Chapters 1 and 2.

Theorem 3.5.1 (Frame bounds are extreme eigenvalues of S)**.** *Let $\{f_i\}_{i=1}^m$ be a frame for $\mathbb{F}^n$ with frame operator S. Denote the eigenvalues of S in descending order by $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n > 0$ (all positive since the frame spans $\mathbb{F}^n$). Then:*

(i) The optimal upper bound is $B_{\text{opt}} = \lambda_1 = \lambda_{\max}(S)$.

(ii) The optimal lower bound is $A_{\text{opt}} = \lambda_n = \lambda_{\min}(S)$.

(iii) The frame is tight iff $\lambda_1 = \lambda_n$, i.e. $S = \lambda I$ for some $\lambda > 0$.

(iv) Any valid pair (A, B) satisfies $A \leq \lambda_n$ and $B \geq \lambda_1$; the optimal bounds are the tightest possible.

Proof. Let $\{u_k\}_{k=1}^n$ be an orthonormal eigenbasis of S with $Su_k = \lambda_k u_k$. Write $x = \sum_k \langle x, u_k \rangle u_k$. Then

$$\langle Sx, x \rangle = \left\langle S \sum_k \langle x, u_k \rangle u_k,\ \sum_j \langle x, u_j \rangle u_j \right\rangle = \sum_{k=1}^n \lambda_k | \langle x, u_k \rangle |^2.$$

By Proposition 3.4.2(ii), $\langle Sx, x \rangle = \|F^* x\|^2 = \sum_i | \langle x, f_i \rangle |^2$. So the frame inequality (3.1) is equivalent to

$$A \|x\|^2 \ \leq\ \langle Sx, x \rangle \ \leq\ B \|x\|^2 \qquad \forall\, x \in V. \tag{3.3}$$

Dividing by $\|x\|^2$ (for $x \neq 0$), the middle term becomes the *Rayleigh quotient* $R(x) := \langle Sx, x \rangle / \|x\|^2$. We may write

$$R(x) = \sum_{k=1}^n \lambda_k\, \alpha_k^2, \qquad \text{where } \alpha_k = | \langle x / \|x\|, u_k \rangle | \ \text{ and } \ \sum_{k=1}^n \alpha_k^2 = 1.$$

The Rayleigh quotient is a convex combination of the eigenvalues $\lambda_1, \ldots, \lambda_n$ with nonnegative weights summing to one. Such a convex combination satisfies

$$\lambda_n = \lambda_{\min} \ \leq\ R(x) \ \leq\ \lambda_{\max} = \lambda_1,$$

with equality $R(u_n) = \lambda_n$ (all weight on λ_n) and $R(u_1) = \lambda_1$ (all weight on λ_1). Therefore the sharpest valid bounds in (3.3) are $A_{\text{opt}} = \lambda_n$ and $B_{\text{opt}} = \lambda_1$, proving (i) and (ii). Part (iii) is immediate: $\lambda_n = \lambda_1$ iff all eigenvalues are equal to some λ, i.e. $S = \lambda I$. Part (iv) follows since any valid A must satisfy $A \leq R(u_n) = \lambda_n$, and any valid B must satisfy $B \geq R(u_1) = \lambda_1$. □

Key Idea

The optimal frame bounds are the **smallest and largest eigenvalues of $S = FF^*$**:

$$A_{\text{opt}} = \lambda_{\min}(S), \qquad B_{\text{opt}} = \lambda_{\max}(S).$$

In NumPy: `eigenvalues, _ = np.linalg.eigh(S)`, then `A, B = eigenvalues.min(), eigenvalues.max()`. The random-sampling method of Chapter 2 is now superseded by this exact computation.

3.5.1 The Non-Spanning Case Revisited

If $\{f_i\}$ does not span $\mathbb{F}^n$, Proposition 3.4.2(iii) gives $\lambda_{\min}(S) = 0$. The promised lower bound $A > 0$ would then require $0 < A \le \lambda_{\min}(S) = 0$, a contradiction. The zero eigenvector u_n satisfies $\langle u_n, f_i \rangle = 0$ for all i—it is the "invisible" direction the frame does not reach—and the Rayleigh quotient at u_n gives $R(u_n) = 0 < A$ for any $A > 0$, confirming the failure.

3.6 Computing the Frame Operator: Worked Examples

We now return to every running example, computing S, its eigenvalues, and the exact frame bounds.

Example 3.2: Triangular frame

With $F = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$:

$$S = FF^T = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & 0 \\ 0 & \frac{3}{2} \end{pmatrix} = \frac{3}{2} I.$$

Both eigenvalues equal $\frac{3}{2}$, so $A = B = \frac{3}{2}$ and the frame is tight. This confirms the exact bounds stated in Example 2.2.5 and provides the matrix-level explanation: $S = \frac{3}{2} I$ means $\sum_i \langle x, f_i \rangle f_i = \frac{3}{2} x$ for all x, which is precisely the tight frame identity.

Example 3.3: Non-tight frame $\{h_1, h_2, h_3\}$

With $H = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$:

$$S = HH^T = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The characteristic polynomial is $(2 - \lambda)^2 - 1 = 0$, giving $\lambda_1 = 3$ and $\lambda_2 = 1$. Thus $B = 3$, $A = 1$, confirming the numerical estimate from Example 2.2.6. The eigenvector for $\lambda_1 = 3$ is $u_1 = (1, 1)^T / \sqrt{2}$—exactly the direction of $h_3 = (1, 1)^T$— explaining geometrically why that direction is most illuminated.

Example 3.4: Standard orthonormal basis

For $F = I_n$ (synthesis matrix of any ONB), $S = II^T = I$, so $\lambda_1 = \cdots = \lambda_n = 1$ and $A = B = 1$: every ONB is a Parseval tight frame.

Example 3.5: Non-spanning set $\{g_1, g_2\}$ in $\mathbb{R}^3$

With $G = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ (a 3×2 matrix):

$$S = GG^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Eigenvalues: $1, 1, 0$. The zero eigenvalue corresponds to $e_3 = (0, 0, 1)^T$, the direction not covered by g_1, g_2. Since $\lambda_{\min} = 0$, no positive lower bound A exists—confirming Example 2.2.4.

Example 3.6: Square vertices in $\mathbb{R}^2$

The four vertices of a unit square at angles $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ give $f_k = (\cos(k\pi/2), \sin(k\pi/2))^T$ for $k = 0, 1, 2, 3$. The synthesis matrix is

$$F = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}.$$

Computing:

$$S = FF^T = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I.$$

Both eigenvalues equal 2, so $A = B = 2$: the four square vertices form a tight frame for $\mathbb{R}^2$ with bound $A = 2$. This confirms the conjecture from Lab Exercise 1.4, and Lab Exercise 2.3's finding. Comparing with the triangular frame ($A = \frac{3}{2}$, three vectors) and the square ($A = 2$, four vectors): more unit-norm vectors $\Rightarrow$ larger frame bound, as expected from the trace formula $A = m/n$ for tight unit-norm frames.

3.7 NumPy Implementation

All the computations above reduce to a handful of NumPy functions.

Listing 3.1: Frame operator and exact bounds via eigenvalues

```python
import numpy as np

def frame_operator(F):
    """Frame operator S = F @ F.T (real frames)."""
    return F @ F.T

def exact_frame_bounds(F):
    """
    Exact optimal frame bounds via eigenvalues of S = F F^T.

    Parameters
    ----------
```

```
    F : ndarray, shape (n, m)
        Synthesis matrix: columns are frame vectors.

    Returns
    -------
    A : float
        Optimal lower frame bound (lambda_min of S).
    B : float
        Optimal upper frame bound (lambda_max of S).
    eigenvalues : ndarray, shape (n,)
        All eigenvalues of S in ascending order.
    eigenvectors : ndarray, shape (n, n)
        Orthonormal eigenvectors (columns) in the same order.
    """
    S = frame_operator(F)
    # eigh: specialized for symmetric/Hermitian matrices;
    # returns real eigenvalues in ascending order.
    eigenvalues, eigenvectors = np.linalg.eigh(S)
    A = float(eigenvalues[0])   # ascending order -> first = min
    B = float(eigenvalues[-1])  # last = max
    return A, B, eigenvalues, eigenvectors

def is_tight(F, tol=1e-10):
    """True if A == B up to the given tolerance."""
    A, B, _, _ = exact_frame_bounds(F)
    return (B - A) < tol * max(1.0, B)

# ----------------------------------------------------------------
# Example 3.2: Triangular frame
# ----------------------------------------------------------------
F_tri = np.array([
    [1.0,   -0.5,          -0.5],
    [0.0,    np.sqrt(3)/2, -np.sqrt(3)/2]
])

S_tri = frame_operator(F_tri)
A_tri, B_tri, eigs_tri, _ = exact_frame_bounds(F_tri)

print("=== Triangular frame ===")
print("S =\n", S_tri.round(10))
print(f"Eigenvalues: {eigs_tri.round(10)}")
print(f"A = {A_tri:.6f},  B = {B_tri:.6f}")
print(f"Tight: {is_tight(F_tri)}")

# ----------------------------------------------------------------
# Example 3.3: Non-tight frame {h1, h2, h3}
# ----------------------------------------------------------------
H = np.array([[1., 0., 1.],
              [0., 1., 1.]])

A_h, B_h, eigs_h, vecs_h = exact_frame_bounds(H)
print("\n=== Non-tight frame {h1, h2, h3} ===")
print("S =\n", frame_operator(H))
print(f"Eigenvalues: {eigs_h}")
```

```
print(f"A = {A_h:.4f},  B = {B_h:.4f}")
print(f"Eigenvector for lambda_max = {eigs_h[-1]:.0f}:",
      vecs_h[:, -1].round(4))

# ------------------------------------------------------------------
# Example 3.6: Square vertices
# ------------------------------------------------------------------
angles = np.array([0, np.pi/2, np.pi, 3*np.pi/2])
F_sq = np.vstack([np.cos(angles), np.sin(angles)])   # shape (2,4)
A_sq, B_sq, eigs_sq, _ = exact_frame_bounds(F_sq)
print("\n=== Square vertices ===")
print("S =\n", frame_operator(F_sq).round(10))
print(f"Eigenvalues: {eigs_sq.round(10)}")
print(f"A = {A_sq:.6f},  B = {B_sq:.6f},  Tight: {is_tight(F_sq)}")

# ------------------------------------------------------------------
# Sanity check: compare with random-sampling estimates
# ------------------------------------------------------------------
def sampling_bounds(F, num_samples=100_000, seed=0):
    """Estimate frame bounds by random sampling (Chapter 2 method)."""
    n = F.shape[0]
    rng = np.random.default_rng(seed)
    vals = []
    for _ in range(num_samples):
        x = rng.standard_normal(n)
        x /= np.linalg.norm(x)
        vals.append(x @ frame_operator(F) @ x)
    vals = np.array(vals)
    return vals.min(), vals.max()

for name, Fmat in [("Triangular", F_tri), ("Non-tight H", H)]:
    A_ex, B_ex, _, _ = exact_frame_bounds(Fmat)
    A_est, B_est = sampling_bounds(Fmat)
    print(f"\n{name}: exact A={A_ex:.6f}, sampled A={A_est:.6f} "
          f"| exact B={B_ex:.6f}, sampled B={B_est:.6f}")
```

Remark. We use `np.linalg.eigh` rather than `np.linalg.eig` for two reasons. First, `eigh` is specialized for symmetric (Hermitian) matrices and guaranteed to return *real* eigenvalues in *ascending* order with an *orthonormal* eigenvector matrix. Second, it is numerically more stable for symmetric matrices. The general-purpose `eig` may return spurious small imaginary parts even for perfectly symmetric inputs. Since $S = FF^T$ is always symmetric by Proposition 3.4.2(i), `eigh` is always the correct choice.

3.8 Reconstruction via the Frame Operator

The frame operator provides not just frame bounds but also a direct reconstruction formula—a preview of the full dual-frame theory of Chapter 5.

Proposition 3.8.1 (Reconstruction via S)**.** *Let $\{f_i\}_{i=1}^m$ be a frame for V with frame operator S. Since $\{f_i\}$ spans V, S is invertible. For every $x \in V$:*

$$x \;=\; S^{-1}\left(\sum_{i=1}^{m} \langle x, f_i\rangle\, f_i\right) \;=\; \sum_{i=1}^{m} \langle x, f_i\rangle\; (S^{-1} f_i). \tag{3.4}$$

Proof. $Sx = \sum_i \langle x, f_i \rangle f_i$ by definition of S. Apply S^{-1} to both sides. $\square$

The vectors $\tilde{f}_i := S^{-1} f_i$ are the *canonical dual frame vectors*, to be studied in depth in Chapter 5. For a tight frame ($S = AI$), $S^{-1} = \frac{1}{A} I$, and (3.4) becomes

$$x = \frac{1}{A} \sum_{i=1}^{m} \langle x, f_i \rangle f_i \qquad \text{(tight frame, bound } A). \tag{3.5}$$

This is one of the key practical advantages of tight frames: reconstruction requires only a scalar rescaling, with no matrix inversion needed.

Example 3.7: Reconstruction with the triangular frame

Let $x = (3, -1)^T \in \mathbb{R}^2$ and use the triangular frame (tight, $A = \frac{3}{2}$). The frame coefficients are

$$c_1 = \langle x, f_1 \rangle = 3, \quad c_2 = \langle x, f_2 \rangle = -\tfrac{3}{2} - \tfrac{\sqrt{3}}{2} \approx -2.366, \quad c_3 = \langle x, f_3 \rangle = -\tfrac{3}{2} + \tfrac{\sqrt{3}}{2} \approx 0.366.$$

Reconstruction via (3.5):

$$\hat{x} = \frac{1}{\frac{3}{2}}(c_1 f_1 + c_2 f_2 + c_3 f_3) = \frac{2}{3}\Big(3(1,0)^T + c_2(-\tfrac{1}{2}, \tfrac{\sqrt{3}}{2})^T + c_3(-\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2})^T\Big).$$

One can verify algebraically (or in NumPy) that $\hat{x} = (3, -1)^T = x$. The redundancy—three coefficients representing a 2-vector—does not prevent exact reconstruction; it merely means the coefficients are not unique (any choice in the affine family $c + \ker F$ also reconstructs x, but the canonical choice $c_i = \langle x, f_i \rangle$ is the most natural).

3.9 Chapter Summary

- Three linear maps organise the algebra: the **synthesis operator** $F : \mathbb{F}^m \to \mathbb{F}^n$ (coefficients $\to$ signals), the **analysis operator** $F^* : \mathbb{F}^n \to \mathbb{F}^m$ (signals $\to$ frame coefficients), and the **frame operator** $S = FF^* : \mathbb{F}^n \to \mathbb{F}^n$ (a positive-definite $n \times n$ matrix).
- S admits two equivalent expressions: $S = FF^*$ (product form, one NumPy line) and $S = \sum_i f_i f_i^*$ (outer-product sum, revealing each vector's individual contribution).
- **Theorem 3.5.1**: the optimal frame bounds are the extreme eigenvalues of S,

  $$A_{\text{opt}} = \lambda_{\min}(S), \qquad B_{\text{opt}} = \lambda_{\max}(S),$$

  with the frame tight iff $S = \lambda I$. Proof: the frame inequality $A\|x\|^2 \le \langle Sx, x \rangle \le B\|x\|^2$ is equivalent to bounding the Rayleigh quotient $\langle Sx, x \rangle / \|x\|^2$, which ranges exactly between $\lambda_{\min}$ and $\lambda_{\max}$.
- The **trace formula** $\operatorname{tr}(S) = \sum_i \|f_i\|^2$ forces $A = m/n$ for any tight unit-norm frame with m vectors in $\mathbb{R}^n$.
- **Reconstruction**: $x = \sum_i \langle x, f_i \rangle \tilde{f}_i$ where $\tilde{f}_i = S^{-1} f_i$; for tight frames, $\tilde{f}_i = \frac{1}{A} f_i$ and no matrix inversion is needed.
- **Zero eigenvalue $\Leftrightarrow$ non-spanning**: the failure mode for non-frames is always $\lambda_{\min}(S) = 0$, with the zero eigenvector being the direction not reached by the frame.

3.10 Lab Exercises

Lab Exercise 3.1: The Full Frame Operator Pipeline

Implement `frame_operator(F)`, `exact_frame_bounds(F)`, and `is_tight(F)` from Section 3.7. Run them on all six frames from this chapter:

(a) Triangular frame (Example 3.6).

(b) Non-tight frame $\{h_1, h_2, h_3\}$ (Example 3.6).

(c) Standard ONB $\{e_1, e_2\}$ in $\mathbb{R}^2$ (Example 3.6).

(d) Non-spanning $\{g_1, g_2\}$ in $\mathbb{R}^3$ (Example 3.6).

(e) Square vertices in $\mathbb{R}^2$ (Example 3.6).

(f) The non-tight frame $\{e_1, 2e_2\}$ in $\mathbb{R}^2$ (from Exercise 1.1 in Chapter 1's pencil-and-paper exercises: $f_1 = (1, 0)$, $f_2 = (0, 2)$).

For each, print S, its eigenvalues, A, B, and whether the frame is tight. Then verify the trace formula: check that `np.trace(S)` equals `sum(np.dot(f,f) for f in columns_of_F)` in every case.

Lab Exercise 3.2: Tightness from the Trace

For a tight unit-norm frame with m vectors in $\mathbb{R}^n$, the trace formula forces $A = m/n$. Verify this numerically for: (a) $m = 3$, $n = 2$ (triangular frame), (b) $m = 4$, $n = 2$ (square vertices). Then answer: if you want a tight unit-norm frame with $m = 6$ vectors in $\mathbb{R}^3$, what *must* A equal, before constructing any such frame? No code needed for this last part—just algebra.

Lab Exercise 3.3: Reconstruction Accuracy

Using the triangular frame and the reconstruction formula (3.5):

(a) Pick five random vectors $x \in \mathbb{R}^2$ (`np.random.default_rng(42).standard_normal((2,5))`). For each, compute the frame coefficients $c_i = \langle x, f_i \rangle$ and reconstruct $\hat{x} = \frac{2}{3} \sum_i c_i f_i$. Report $\|\hat{x} - x\|$ for each; it should be at or below machine precision ($\sim 10^{-15}$).

(b) Now corrupt one coefficient: replace c_1 with $c_1 + \delta$ for $\delta = 0.1, 0.5, 1.0$. Reconstruct and report $\|\hat{x}_\delta - x\|$ for each δ. How does the reconstruction error scale with δ?

(c) Repeat part (b) but using the *two-vector* frame $\{e_1, e_2\}$ (the standard basis), where corrupting c_1 by δ means replacing $\langle x, e_1 \rangle$ with $\langle x, e_1 \rangle + \delta$. Compare with part (b): which frame is more robust to the same perturbation, and why?

Lab Exercise 3.4: Exact vs. Sampled Bounds

Run `exact_frame_bounds` and `sampling_bounds` (with 100,000 samples) on both the triangular frame and $\{h_1, h_2, h_3\}$. Tabulate the four pairs $(A_{\text{exact}}, A_{\text{sampled}})$ and $(B_{\text{exact}}, B_{\text{sampled}})$. In one or two sentences: why does the sampled estimate always satisfy $A_{\text{sampled}} \geq A_{\text{exact}}$ and $B_{\text{sampled}} \leq B_{\text{exact}}$?

Lab Exercise 3.5: Eigenvectors as Extremal Directions

For the non-tight frame $\{h_1, h_2, h_3\}$ with $A = 1$, $B = 3$:

(a) Retrieve the eigenvectors u_1 (for $\lambda_1 = 3$) and u_2 (for $\lambda_2 = 1$) from `exact_frame_bounds`.

(b) Verify that $\langle Su_1, u_1 \rangle = 3$ and $\langle Su_2, u_2 \rangle = 1$ numerically, confirming that the Rayleigh quotient indeed equals λ_k at u_k.

(c) Plot the "illumination function" $\varphi(\theta) = \sum_i |\langle x(\theta), h_i \rangle|^2$ for $x(\theta) = (\cos\theta, \sin\theta)^T$ over $\theta \in [0, 2\pi]$ (using `matplotlib`). Mark the angles corresponding to u_1 and u_2 on the plot, and verify that these angles correspond to the maximum ($B = 3$) and minimum ($A = 1$) of φ.

Lab Exercise 3.6 (Optional, Challenge): Pentagonal Frame

Consider five vectors in $\mathbb{R}^2$ at equally spaced angles: $f_k = (\cos(2\pi k/5), \sin(2\pi k/5))^T$, $k = 0, 1, 2, 3, 4$.

(a) Compute S, its eigenvalues, and A, B. Is this a tight frame? Predict the answer using the trace formula before running the code.

(b) Without computing the frame operator, predict whether *any* regular m-gon configuration in $\mathbb{R}^2$ (i.e. $f_k = (\cos(2\pi k/m), \sin(2\pi k/m))^T$) gives a tight frame. Test your prediction numerically for $m = 5, 6, 7, 8$ by running `is_tight` for each. Formulate a conjecture. (This will be proved in Chapter 4.)

(c) Compute $\operatorname{tr}(S)$ for each of the regular m-gons and verify the trace formula $\operatorname{tr}(S) = m$ (since all vectors have unit norm). From this and tightness, deduce the frame bound $A = m/2$ for each.

3.11 Supplementary Exercises

Theoretical Exercises

T1. Let F be an $n \times m$ matrix. Prove that FF^* (size $n \times n$) and F^*F (size $m \times m$) have the same nonzero eigenvalues, counted with multiplicity. (This fact underlies the identity $\mathrm{tr}(G^2) = \mathrm{tr}(S^2)$ used in the Welch bound proof of Chapter 9.)

T2. Let S be the frame operator of $\{f_i\}_{i=1}^m$ for $\mathbb{R}^n$, and let $S^{-1/2}$ denote its unique positive definite inverse square root. Prove that $\{S^{-1/2} f_i\}_{i=1}^m$ is a Parseval frame for $\mathbb{R}^n$.

T3. Let $\tilde{f}_i = S^{-1} f_i$ be the canonical dual vectors (Proposition 3.8.1). Prove that the frame operator of $\{\tilde{f}_i\}_{i=1}^m$ is exactly S^{-1}.

T4. Prove a converse to Theorem 3.5.1: given *any* symmetric positive definite $n \times n$ matrix S_0, show there exists a frame $\{f_i\}_{i=1}^n$ for $\mathbb{R}^n$ whose frame operator is exactly S_0. (Hint: use the symmetric square root of S_0.)

T5. Prove that a frame $\{f_i\}_{i=1}^m$ for $\mathbb{R}^n$ is Parseval if and only if its synthesis matrix F satisfies $FF^* = I_n$. Contrast this with the (much stronger) condition $F^*F = I_m$, and explain why the latter can only hold when $m \le n$.

T6. The frame operator S maps the unit sphere in $\mathbb{R}^n$ to an ellipsoid with semi-axis lengths equal to the eigenvalues of S. Prove this, and express the ellipsoid's eccentricity (ratio of longest to shortest semi-axis) in terms of the frame bounds A, B. What does perfect roundness (eccentricity 1) correspond to?

Computational Exercises

C1. For a random 3×5 matrix F, compute the eigenvalues of FF^T and F^TF and confirm the nonzero eigenvalues agree (T1), while F^TF has exactly 2 extra zero eigenvalues.

C2. For the frame $\{h_1, h_2, h_3\}$, compute $S^{-1/2}$ via eigendecomposition, apply it to build $\{S^{-1/2} h_i\}$, and verify the resulting frame operator is exactly I_2 (T2).

C3. For each of the five example frames from Lab Exercise 3.1, compute the canonical dual's frame operator directly and confirm it equals S^{-1} to machine precision (T3).

C4. Given the target matrix $S_0 = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$, construct a frame $\{f_1, f_2\}$ for $\mathbb{R}^2$ whose frame operator is exactly S_0 (T4), and verify by direct computation.

C5. Plot the ellipse $\{Sx : \|x\| = 1\}$ for the triangular frame and for $\{h_1, h_2, h_3\}$ in the same figure (parametrize the unit circle, apply S, and plot the image). Confirm visually that the tight frame's image is a circle and the non-tight frame's image is a genuine ellipse, connecting to T6.

Chapter 4

Tight Frames

4.1 Overview: Closing the Loop on Tightness

Three separate promises from Chapters 1–3 converge in this chapter. Chapter 1 conjectured, from a single picture, that the vertices of an equilateral triangle form a tight frame for $\mathbb{R}^2$. Chapter 2 verified this by direct computation for $m = 3$ and asked you to guess, in Lab Exercise 1.4, whether the same holds for a square. Chapter 3 confirmed the square case too ($A = B = 2$) and asked you, in Lab Exercise 3.6, to test pentagons, hexagons, heptagons, and octagons, and to conjecture a general pattern. By now the pattern should be unmistakable: *every* regular polygon, of any number of sides $m \geq 2$, gives a tight frame for $\mathbb{R}^2$. This chapter proves that conjecture in full generality (Theorem 4.3.1), and along the way develops the algebraic mechanism—cancellation of trigonometric cross-terms via roots of unity—that explains *why* it is true, not merely *that* it is true.

We also deliver on a second promise: Chapter 2 noted that Parseval frames need not be orthonormal bases, and deferred examples to this chapter. Section 4.4 gives a general, constructive method—projecting an orthonormal basis of a larger space down into a subspace—that produces Parseval frames with as much redundancy as you like, and explains geometrically why this construction always works.

4.2 Tight Frames: Definitions Revisited

Recall Definition 2.4.1: a frame $\{f_i\}_{i=1}^m$ for V is *tight* if its frame bounds satisfy $A = B$, and *Parseval* if additionally $A = 1$. Chapter 3 gave the matrix characterization that organizes this entire chapter:

> *A frame is tight with bound A if and only if its frame operator satisfies $S = AI$ (Theorem 3.5.1(iii)).*

This single equation, $S = AI$, is the lens through which we will view every result in this chapter. Unfolding $S = \sum_i f_i f_i^* = AI$ gives the following equivalent, and extremely useful, restatement.

Proposition 4.2.1 (Tightness as a resolution of the identity)**.** *A frame $\{f_i\}_{i=1}^m$ for $V = \mathbb{F}^n$ is tight with bound A if and only if*

$$\sum_{i=1}^{m} f_i f_i^* = AI_n. \tag{4.1}$$

Equivalently, for every pair of standard basis vectors e_j, e_k,

$$\sum_{i=1}^{m} (f_i)_j \overline{(f_i)_k} = \begin{cases} A, & j = k, \\ 0, & j \neq k. \end{cases} \tag{4.2}$$

Proof. This is immediate from $S = \sum_i f_i f_i^* = AI$ together with the fact that the (j,k) entry of $f_i f_i^*$ is $(f_i)_j \overline{(f_i)_k}$, and the (j,k) entry of AI is A when $j = k$ and 0 otherwise. □

Remark. Equation (4.1) is sometimes called a *resolution of the identity*: the frame vectors' outer products sum exactly to (a multiple of) the identity. This is the precise algebraic form of the informal phrase from Chapter 1, "every direction is treated equally." Equation (4.2) is the same statement written entrywise, and it is this entrywise form that we will verify directly, by trigonometric computation, in the regular polygon theorem below: the off-diagonal cancellation in Example 2.2.5—the vanishing of the cross term $-\frac{\sqrt{3}}{2}x_1x_2 + \frac{\sqrt{3}}{2}x_1x_2$—is exactly an instance of the $j \neq k$ case of (4.2).

4.3 The Regular Polygon Theorem

We now state and prove the general result toward which Chapters 1–3 have been building.

Theorem 4.3.1 (Regular polygons are tight frames)**.** *Let $m \geq 2$ be an integer, and let*

$$f_k = \big(\cos(2\pi k/m), \sin(2\pi k/m)\big), \qquad k = 0, 1, \ldots, m-1,$$

be the m vectors pointing toward the vertices of a regular m-gon inscribed in the unit circle in $\mathbb{R}^2$. Then $\{f_k\}_{k=0}^{m-1}$ is a tight frame for $\mathbb{R}^2$ with frame bound $A = m/2$.

We give two proofs. The first is a direct trigonometric computation, generalizing exactly the calculation of Example 2.2.5. The second is shorter and more conceptual, using complex exponentials and roots of unity; it is the "systematic" technique promised in Chapter 2; and it will generalize far more easily to other constructions later in the course.

Proof 1: Direct trigonometric computation. Let $\theta_k = 2\pi k/m$, so $f_k = (\cos\theta_k, \sin\theta_k)$. By Proposition 4.2.1, it suffices to verify (4.2) for $A = m/2$, i.e. to show

$$\sum_{k=0}^{m-1} \cos^2\theta_k = \frac{m}{2}, \qquad \sum_{k=0}^{m-1} \sin^2\theta_k = \frac{m}{2}, \qquad \sum_{k=0}^{m-1} \cos\theta_k \sin\theta_k = 0.$$

Using the double-angle identities $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$, $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$, and $\cos\theta\sin\theta = \frac{1}{2}\sin 2\theta$:

$$\sum_{k=0}^{m-1} \cos^2\theta_k = \frac{m}{2} + \frac{1}{2}\sum_{k=0}^{m-1}\cos(2\theta_k), \qquad \sum_{k=0}^{m-1} \sin^2\theta_k = \frac{m}{2} - \frac{1}{2}\sum_{k=0}^{m-1}\cos(2\theta_k),$$

$$\sum_{k=0}^{m-1} \cos\theta_k \sin\theta_k = \frac{1}{2}\sum_{k=0}^{m-1}\sin(2\theta_k).$$

So it suffices to show that both sums $\sum_{k=0}^{m-1}\cos(2\theta_k)$ and $\sum_{k=0}^{m-1}\sin(2\theta_k)$ vanish, where $2\theta_k = 4\pi k/m$. These are the real and imaginary parts of the geometric series

$$\sum_{k=0}^{m-1} e^{i\cdot 4\pi k/m} = \sum_{k=0}^{m-1} \big(e^{i4\pi/m}\big)^k.$$

If $m \neq 2$, then $\zeta := e^{i4\pi/m} \neq 1$ (since $4\pi/m$ is not a multiple of 2π for $m \geq 3$), and the finite geometric series formula gives

$$\sum_{k=0}^{m-1} \zeta^k = \frac{\zeta^m - 1}{\zeta - 1} = \frac{e^{i4\pi} - 1}{\zeta - 1} = \frac{1-1}{\zeta - 1} = 0,$$

using $e^{i4\pi} = 1$. Taking real and imaginary parts gives $\sum_k \cos(2\theta_k) = 0$ and $\sum_k \sin(2\theta_k) = 0$, as required. (The case $m = 2$ gives $f_0 = (1,0)$, $f_1 = (-1,0)$, which one checks directly: $\sum \cos^2\theta_k = 1 + 1 = 2 = m/2 \cdot 2$... we revisit $m = 2$ explicitly in Remark 4.3 below, since the geometric series argument needs $\zeta \neq 1$.) This proves $\sum_k \cos^2\theta_k = \sum_k \sin^2\theta_k = m/2$ and $\sum_k \cos\theta_k \sin\theta_k = 0$, which is exactly (4.2) with $A = m/2$. $\square$

Proof 2: Roots of unity (the systematic technique). Identify $\mathbb{R}^2$ with $\mathbb{C}$ via $(\cos\theta, \sin\theta) \leftrightarrow e^{i\theta}$. Let $\omega = e^{2\pi i/m}$, a primitive m-th root of unity, so that $f_k \leftrightarrow \omega^k$ for $k = 0, \ldots, m-1$. We argue directly with the frame operator acting on a vector $w \in \mathbb{C}$ (thought of as a vector in $\mathbb{R}^2$ under the identification above). Since $f_k \leftrightarrow \omega^k$, the real inner product $\langle w, f_k \rangle$ corresponds to $\mathrm{Re}(\bar{w}\,\omega^k)$, so

$$\sum_{k=0}^{m-1} |\langle w, f_k \rangle|^2 = \sum_{k=0}^{m-1} \mathrm{Re}(\bar{w}\,\omega^k)^2 = \sum_{k=0}^{m-1} \frac{(\bar{w}\,\omega^k + w\,\bar{\omega}^k)^2}{4}.$$

Expanding the square,

$$(\bar{w}\,\omega^k + w\,\bar{\omega}^k)^2 = \bar{w}^2\omega^{2k} + 2|w|^2 + w^2\bar{\omega}^{2k},$$

so

$$\sum_{k=0}^{m-1} |\langle w, f_k \rangle|^2 = \frac{\bar{w}^2}{4} \sum_{k=0}^{m-1} \omega^{2k} + \frac{|w|^2}{2} \cdot m + \frac{w^2}{4} \sum_{k=0}^{m-1} \bar{\omega}^{2k}.$$

The key fact about roots of unity (the *discrete orthogonality relation*) is

$$\sum_{k=0}^{m-1} \omega^{jk} = \begin{cases} m, & \omega^j = 1 \text{ (i.e. } m \mid j), \\ 0, & \text{otherwise,} \end{cases} \tag{4.3}$$

which follows from the same finite geometric series identity as in Proof 1: if $\omega^j \neq 1$ then $\sum_k (\omega^j)^k = \frac{(\omega^j)^m - 1}{\omega^j - 1} = \frac{1-1}{\omega^j - 1} = 0$, since $(\omega^j)^m = (\omega^m)^j = 1^j = 1$. Applying (4.3) with $j = 2$: if $m \geq 3$, then $\omega^2 \neq 1$ (as 2 is not a multiple of m for $m \geq 3$), so $\sum_k \omega^{2k} = 0$ and likewise $\sum_k \bar{\omega}^{2k} = 0$. The two outer sums vanish, leaving only the middle term:

$$\sum_{k=0}^{m-1} |\langle w, f_k \rangle|^2 = \frac{|w|^2}{2} \cdot m = \frac{m}{2}|w|^2.$$

Since $|w|^2 = \|w\|^2$ under the identification $\mathbb{C} \cong \mathbb{R}^2$, this is exactly the tight frame identity with $A = m/2$, for every $w \in \mathbb{R}^2$. $\square$

Remark (The case $m = 2$). When $m = 2$, $\omega^2 = e^{4\pi i/2} = e^{2\pi i} = 1$, so the cancellation argument above does not apply directly to the $j = 2$ case of (4.3)—indeed $\sum_{k=0}^{1} \omega^{2k} = 2 \neq 0$ in that case. Let us check $m = 2$ by hand instead: $f_0 = (1,0)$, $f_1 = (-1,0)$ (antipodal points, not really a "polygon" in the usual sense). Direct computation gives $S = f_0 f_0^T + f_1 f_1^T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, which is *not* a multiple of the identity ($\lambda_{\min} = 0 \neq \lambda_{\max} = 2$). So $m = 2$ is genuinely an exception:

two antipodal points do not form a tight frame (indeed they do not even span $\mathbb{R}^2$!). The theorem's hypothesis $m \geq 2$ should really be sharpened to $m \geq 3$ for the conclusion to hold without further case-checking; we keep $m \geq 3$ as the operative hypothesis from here on, consistent with every genuine "polygon" (which needs at least three vertices).

Key Idea

Every regular m-gon, $m \geq 3$, is a tight frame for $\mathbb{R}^2$, with frame bound $A = m/2$. The proof reduces to the discrete orthogonality relation for roots of unity, $\sum_{k=0}^{m-1} \omega^{jk} = 0$ unless $m \mid j$. This is the rigorous, general version of the cross-term cancellation you first saw by hand in Example 2.2.5 for $m = 3$, and it explains why the cancellation was never a coincidence: the angles $2\pi k/m$ are always symmetric enough, for any $m \geq 3$, to force the relevant trigonometric sums to vanish.

Remark. The value $A = m/2$ matches the trace-formula prediction from Chapter 3: each f_k has unit norm, so $\operatorname{tr}(S) = \sum_k \|f_k\|^2 = m$; since the frame is tight in $\mathbb{R}^2$ ($n = 2$), tightness forces $\operatorname{tr}(S) = An = 2A$, giving $A = m/2$ exactly as derived above. This consistency check is a good habit to develop: whenever a tight unit-norm frame construction is proposed, the trace formula instantly predicts what A *must* be, before any cancellation argument is carried out.

Example 4.1: Checking the formula against Chapter 3

Theorem 4.3.1 gives $A = m/2$. For $m = 3$ (the triangular frame, Chapter 3's Example 3.2): $A = 3/2$, matching the value $A = B = \frac{3}{2}$ computed there from the explicit matrix $S = \frac{3}{2}I$. For $m = 4$ (the square vertices, Chapter 3's Example 3.6): $A = 4/2 = 2$, matching the computed value $A = B = 2$. Both match exactly, as they must.

4.3.1 Why Roots of Unity? A Wider Lens

It is worth pausing to explain why the roots-of-unity proof is the "right" one, beyond simply being shorter. The vectors $f_k = \omega^k$ (under $\mathbb{C} \cong \mathbb{R}^2$) are the orbit of the single vector $f_0 = 1$ under repeated rotation by the angle $2\pi/m$. Any construction with this kind of *cyclic symmetry*—an orbit under a rotation (or more generally, a unitary matrix) of finite order m—will produce cancellation identities of exactly this type, because the discrete orthogonality relation (4.3) depends only on the algebraic fact that ω is an m-th root of unity, not on any special geometric feature of $\mathbb{R}^2$. This is the first hint of a theme that recurs throughout frame theory: **highly symmetric frames are often tight frames, and group-theoretic symmetry is a systematic tool for constructing and verifying tightness**, far more powerful than checking cancellation by hand for each new example. We will see a more general version of this principle (orbits of finite groups acting on $\mathbb{R}^n$ or $\mathbb{C}^n$) in Chapter 10 when we discuss equiangular and Grassmannian frames.

4.4 Constructing Parseval Frames by Projection

We now turn to the second promise of this chapter: explicit examples of Parseval frames ($A = B = 1$) that are not orthonormal bases. Chapter 2 observed this must be possible in principle (a Parseval frame need not be linearly independent) but gave no construction. Here is a general one.

Theorem 4.4.1 (Projection of an orthonormal basis is Parseval)**.** *Let $\{e_1, \ldots, e_N\}$ be an orthonormal basis of $\mathbb{F}^N$, let $V \subset \mathbb{F}^N$ be an n-dimensional subspace ($n \leq N$), and let $P : \mathbb{F}^N \to V$ be* *the orthogonal projection onto V. Then $\{Pe_1, \dots, Pe_N\}$ is a Parseval frame for V (viewed as an n-dimensional inner product space in its own right).*

Proof. For any $x \in V$, since P is the orthogonal projection onto $V \ni x$, we have $Px = x$ and $P^* = P = P^2$ (standard properties of orthogonal projections: self-adjoint and idempotent). Using $\langle x, Pe_k \rangle_V = \langle Px, Pe_k \rangle = \langle P^*Px, e_k \rangle = \langle Px, e_k \rangle = \langle x, e_k \rangle$ (the inner product on V is just the restriction of the inner product on $\mathbb{F}^N$), we compute

$$\sum_{k=1}^{N} |\langle x, Pe_k \rangle|^2 = \sum_{k=1}^{N} |\langle x, e_k \rangle|^2 = \|x\|^2 ,$$

where the last equality is Parseval's identity (Theorem 1.2.9) for the orthonormal basis $\{e_k\}$ of $\mathbb{F}^N$, applied to $x \in V \subset \mathbb{F}^N$. This shows $\{Pe_k\}_{k=1}^N$ satisfies the Parseval identity for every $x \in V$, i.e. it is a Parseval frame for V. □

Remark. This construction is extremely flexible: start with *any* orthonormal basis of *any* larger space $\mathbb{F}^N$, project down onto *any* subspace V, and the projected vectors automatically form a Parseval frame for V—regardless of how the subspace sits inside $\mathbb{F}^N$. Since N can be made as large as we like relative to $n = \dim V$, this produces Parseval frames with arbitrarily high redundancy N/n. Crucially, the projected vectors $\{Pe_k\}$ are generally *not* linearly independent once $N > n$ (there are $N > n$ of them in an n-dimensional space), and they are typically not even distinct or nonzero, so this is genuinely different from an orthonormal basis while retaining the exact Parseval identity.

Example 4.2: A Parseval frame from projecting $\mathbb{R}^3$ onto $\mathbb{R}^2$

Let $N = 3$, $\{e_1, e_2, e_3\}$ the standard basis of $\mathbb{R}^3$, and let $V = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$, the plane through the origin perpendicular to $(1, 1, 1)$. This is a 2-dimensional subspace. The orthogonal projection onto V is $P = I - \frac{1}{3}uu^T$ where $u = (1, 1, 1)^T/\sqrt{3}$ is the unit normal to V. Computing:

$$Pe_1 = e_1 - \tfrac{1}{3}(1, 1, 1)^T = \tfrac{1}{3}(2, -1, -1)^T, \quad \text{and similarly for } Pe_2, Pe_3$$

by symmetry. The three vectors Pe_1, Pe_2, Pe_3 all have the same length $\sqrt{(2/3)^2 + (1/3)^2 + (1/3)^2} = \sqrt{6}/3$ and, viewed inside the 2-dimensional plane V, turn out to be at 120° to one another: this is, geometrically, exactly the triangular frame of Chapter 1 (up to rotation and an overall scale factor), now arrived at by an entirely different route—projection from $\mathbb{R}^3$, rather than direct placement in $\mathbb{R}^2$. By Theorem 4.4.1, $\{Pe_1, Pe_2, Pe_3\}$ is automatically Parseval ($A = B = 1$) for V, even though the three vectors are linearly dependent (they live in a 2-dimensional space) and are not the standard basis of anything. This example is verified numerically in Lab Exercise 4.2 below.

4.5 A Full Geometric Characterization of Tightness in $\mathbb{R}^2$

The regular polygon theorem shows that rotational symmetry is *sufficient* for tightness. It is natural to ask whether it is *necessary*—must a tight frame in $\mathbb{R}^2$ look like a regular polygon? The answer is no, and understanding why illuminates the true geometric content of tightness.

Proposition 4.5.1 (Tight frames in $\mathbb{R}^2$ need not be regular polygons)**.** *There exist tight frames for $\mathbb{R}^2$ whose vectors do not have equal norms and are not symmetrically arranged.*

Proof. We exhibit one. Let $g_1 = (1, 0)$, $g_2 = (0, 1)$, $g_3 = (1, 0)$, $g_4 = (0, 1)$ (each standard basis vector repeated twice). Then

$$S = \sum_{i=1}^{4} g_i g_i^T = 2e_1e_1^T + 2e_2e_2^T = 2I,$$

so this is tight with $A = 2$, yet the four vectors are not at equal angles to one another (two coincide with e_1, two with e_2; the angle between the two *distinct* directions present is 90°, not the 120° a genuine 4-fold symmetric picture would need) — this tight frame is not the vertex set of any regular polygon, and is simply two copies of an orthonormal basis stacked together.

For a less degenerate example, with vectors of *unequal* norm, fix any $a, b, c, d > 0$ satisfying $a^2 + b^2 = c^2 + d^2$ (for instance $a = 1.5$, $b = 0.5$, $c = 1$, $d = \sqrt{1.5^2 + 0.5^2 - 1^2} = \sqrt{1.5}$), and set

$$g_1 = (a, 0), \quad g_2 = (-b, 0), \quad g_3 = (0, c), \quad g_4 = (0, -d).$$

Since g_1, g_2 lie on the x-axis and g_3, g_4 lie on the y-axis, the frame operator is automatically diagonal:

$$S = (a^2 + b^2)\, e_1e_1^T + (c^2 + d^2)\, e_2e_2^T,$$

which equals $A \cdot I$ for $A = a^2 + b^2 = c^2 + d^2$ by our choice of constants. Here the four vectors visibly have four different lengths ($a = 1.5$, $b = 0.5$, $c = 1$, $d \approx 1.225$ in the example above) and are certainly not the vertices of any regular polygon, yet the frame is exactly tight. This proves the proposition. (A fully asymmetric example — vectors at generic, non-axis-aligned angles, with unequal norms and no special structure at all — is also possible, but is more easily found numerically than by hand; Lab Exercise 4.4 asks you to search for one.) □

Remark. The correct general statement (which we will prove rigorously in Chapter 9 using the *frame potential*) is this: a set of m unit-norm vectors in $\mathbb{R}^n$ is a tight frame if and only if it is a *critical point*—in fact a global minimizer—of the total pairwise-correlation energy $\sum_{i \neq j} |\langle f_i, f_j \rangle|^2$. Regular polygons are tight because their symmetry forces them to be minimizers, but they are far from the only minimizers once m or n grow: there is generally a large family of inequivalent tight configurations. Tightness is therefore better understood as an *energy-minimization* (equivalently, *balance*) condition than as a symmetry condition; symmetry is a convenient sufficient mechanism for achieving it; but it is not the only mechanism, and Chapter 9 develops the energy perspective in full.

4.6 Computing and Verifying Tight Frames in NumPy

We now translate the regular polygon theorem and the projection construction into code, building tools to construct and verify tight frames numerically.

Listing 4.1: Regular polygon frames and the projection construction

```python
import numpy as np

def regular_polygon_frame(m):
    """
    Return the n=2 synthesis matrix (shape (2, m)) for the
```

```python
    vertices of a regular m-gon inscribed in the unit circle.
    """
    angles = 2 * np.pi * np.arange(m) / m
    return np.vstack([np.cos(angles), np.sin(angles)])

def frame_operator(F):
    return F @ F.T

def exact_frame_bounds(F):
    S = frame_operator(F)
    eigenvalues, eigenvectors = np.linalg.eigh(S)
    return float(eigenvalues[0]), float(eigenvalues[-1]), eigenvalues, eigenvectors

def is_tight(F, tol=1e-9):
    A, B, _, _ = exact_frame_bounds(F)
    return (B - A) < tol * max(1.0, B)

# ---------------------------------------------------------------
# Theorem 4.3.1: verify A = m/2 for many regular polygons
# ---------------------------------------------------------------
print("=== Regular polygon frames ===")
for m in range(3, 13):
    F = regular_polygon_frame(m)
    A, B, eigs, _ = exact_frame_bounds(F)
    predicted_A = m / 2
    print(f"m={m:2d}:  A={A:.6f}  B={B:.6f}  "
          f"predicted m/2={predicted_A:.4f}  "
          f"tight={is_tight(F)}  match={np.isclose(A, predicted_A)}")

# ---------------------------------------------------------------
# Theorem 4.4.1: Parseval frame via projection (Example 4.2)
# ---------------------------------------------------------------
def projection_parseval_frame(N, V_basis):
    """
    Given an orthonormal basis (columns) of a subspace V of R^N,
    return the Parseval frame for V obtained by projecting the
    standard basis of R^N onto V, expressed in V's own coordinates.

    V_basis: ndarray of shape (N, n), orthonormal columns spanning V.
    Returns: ndarray of shape (n, N) -- N frame vectors for the
             n-dimensional space V, in V's coordinate system.
    """
    # P = V_basis @ V_basis.T projects R^N onto V (as vectors in R^N).
    # Expressed in V's own n-dimensional coordinates, the projection
    # of e_k is simply the k-th row of V_basis (i.e. V_basis.T @ e_k).
    return V_basis.T  # shape (n, N): columns are the N frame vectors

# Example 4.2: V = {x : x1+x2+x3=0} subset of R^3
u = np.array([1., 1., 1.]) / np.sqrt(3)   # unit normal to V
# Build an orthonormal basis of V (2-dimensional):
v1 = np.array([1., -1., 0.]) / np.sqrt(2)
v2 = np.array([1., 1., -2.]) / np.sqrt(6)
V_basis = np.column_stack([v1, v2])        # shape (3, 2)
```

```python

F_proj = projection_parseval_frame(3, V_basis)   # shape (2, 3)
A_proj, B_proj, eigs_proj, _ = exact_frame_bounds(F_proj)
print("\n=== Parseval frame via projection (Example 4.2) ===")
print("Frame vectors (in V's 2D coordinates):\n", F_proj.round(4))
print(f"A = {A_proj:.6f}, B = {B_proj:.6f}  (expect A=B=1)")
```

Running the first loop confirms Theorem 4.3.1 for $m = 3, \dots, 12$ in a single pass: every regular polygon is tight, with A matching $m/2$ to numerical precision. The second block constructs the projection-based Parseval frame of Example 4.2 from scratch and confirms $A = B = 1$ exactly.

Remark. A subtlety in the `projection_parseval_frame` function: rather than literally constructing the $N \times N$ projection matrix P and then re-expressing each $Pe_k \in \mathbb{R}^N$ inside the lower-dimensional subspace V, we exploit a shortcut. If *V_basis* has orthonormal columns $v_1, \dots, v_n$ spanning V, then the coordinates of Pe_k *within* V (relative to the basis $\{v_j\}$) are exactly $(\langle e_k, v_1 \rangle, \dots, \langle e_k, v_n \rangle) =$ the k-th row of `V_basis`. This avoids ever forming the $N \times N$ matrix P explicitly, which matters for efficiency when N is large, and is worth internalizing as a general pattern: orthogonal projection coordinates are just inner products against an orthonormal basis of the target subspace.

4.7 Chapter Summary

- Tightness is equivalent to a **resolution of the identity**: $\sum_i f_i f_i^* = AI$ (Proposition 4.2.1), which entrywise says diagonal sums equal A and off-diagonal sums vanish.

- **Theorem 4.3.1**: every regular m-gon ($m \geq 3$) inscribed in the unit circle is a tight frame for $\mathbb{R}^2$ with bound $A = m/2$, matching the trace-formula prediction $A = \operatorname{tr}(S)/n = m/2$. The proof reduces to the **discrete orthogonality relation for roots of unity**, $\sum_{k=0}^{m-1} \omega^{jk} = 0$ unless $m \mid j$—the rigorous, general mechanism behind the 120° cross-term cancellation first seen by hand in Example 2.2.5.

- The case $m = 2$ is a genuine exception (antipodal points do not even span $\mathbb{R}^2$); the theorem requires $m \geq 3$.

- **Theorem 4.4.1**: projecting any orthonormal basis of a larger space $\mathbb{F}^N$ onto a subspace V always produces a Parseval frame for V, regardless of how V sits inside $\mathbb{F}^N$. This gives explicit, flexible examples of Parseval frames that are not orthonormal bases, fulfilling the promise of Chapter 2.

- Tightness is best understood as an **energy-balance** condition (to be made precise via the frame potential in Chapter 9), not merely a symmetry condition: symmetric configurations (regular polygons, roots of unity) are a convenient sufficient mechanism, but not the only tight configurations.

4.8 Lab Exercises

Lab Exercise 4.1: The Full Regular Polygon Table

Implement `regular_polygon_frame(m)` and `exact_frame_bounds(F)` from Section 4.6. Build a table of A, B, and $\mathrm{tr}(S)$ for $m = 3, 4, \ldots, 20$. Verify in every row that (i) $A = B$ (tight), (ii) $A = m/2$, and (iii) $\mathrm{tr}(S) = m$. Then check $m = 2$ separately and confirm it breaks the pattern, consistent with Remark 4.3.

Lab Exercise 4.2: Verifying the Projection Construction

Implement `projection_parseval_frame` from Section 4.6 and reproduce Example 4.2 exactly. Then construct a second example: let $V \subset \mathbb{R}^4$ be the 2-dimensional subspace spanned by an orthonormal basis of your choosing (e.g. via `np.linalg.qr` applied to two random vectors in $\mathbb{R}^4$), and verify that projecting the standard basis $\{e_1, e_2, e_3, e_4\}$ of $\mathbb{R}^4$ onto V produces a 4-vector Parseval frame for the 2-dimensional space V. Confirm $A = B = 1$ numerically.

Lab Exercise 4.3: Partial Regular Polygons Are Not Tight

Theorem 4.3.1 requires using *all* m vertices of the regular polygon. Investigate what happens with a *partial* set: for $m = 8$, compute the frame operator and exact bounds for the first $m' = 2, 3, 4, 5, 6, 7$ vertices only (i.e. $f_k = (\cos(2\pi k/8), \sin(2\pi k/8))$ for $k = 0, \ldots, m' - 1$, omitting the rest). For which values of m' (if any) is the resulting frame still tight? Reconcile your numerical finding with the roots-of-unity proof: which step of Proof 2 specifically breaks down when the sum is only over a partial set of k values?

Lab Exercise 4.4: Hunting for Asymmetric Tight Frames

Proposition 4.5.1 claims tight frames need not be regular polygons. Investigate numerically: starting from the repeated basis vectors example in the proof ($g_1 = g_3 = (1, 0)$, $g_2 = g_4 = (0, 1)$), perturb each vector slightly by a small random rotation (a few degrees) and a small random rescaling (within $\pm 10\%$ of unit length), and check whether the perturbed frame is still (approximately) tight. Report the ratio B/A before and after perturbation. Then try to manually construct (by trial and error, or using `scipy.optimize`) a 4-vector frame in $\mathbb{R}^2$ that is exactly tight but visibly *not* a regular polygon and *not* two repeated orthonormal bases — i.e. four genuinely distinct directions, no two equal or antipodal, with no obvious rotational symmetry, that still satisfies $S = AI$. (Hint: you have 4 vectors $\times$ 2 coordinates $= 8$ real degrees of freedom, and the tight-frame condition $S = AI$ in $\mathbb{R}^2$ imposes only 3 real constraints—2 diagonal entries equal, 1 off-diagonal entry zero— so a 5-real-dimensional family of solutions should exist; you only need to find one element of it.)

Lab Exercise 4.5 (Optional, Challenge): Roots of Unity in $\mathbb{C}^n$

The roots-of-unity proof technique generalizes beyond $\mathbb{R}^2$. Consider the n-dimensional complex vectors

$$f_k = \frac{1}{\sqrt{n}}\big(1, \omega^k, \omega^{2k}, \dots, \omega^{(n-1)k}\big) \in \mathbb{C}^n, \qquad k = 0, 1, \dots, n-1,$$

where $\omega = e^{2\pi i/n}$ (these are the columns of the *discrete Fourier transform matrix*, up to normalization).

(a) For $n = 4$, construct these vectors numerically (using complex NumPy arrays, `dtype=complex`) and verify that $\{f_k\}_{k=0}^{n-1}$ is an orthonormal basis of $\mathbb{C}^n$ (hence, trivially, a tight frame with $A = B = 1$) by checking $\langle f_j, f_k\rangle = \delta_{jk}$ for all j, k (recall: for complex vectors, use `np.vdot` as discussed in Chapter 1's remark on conventions).

(b) Now take only the first $n-1$ of these vectors (drop f_{n-1}) for $n = 4$. Is the resulting 3-vector set still a tight frame for $\mathbb{C}^4$? (It cannot be, by Theorem 2.2.1, for a reason you should be able to state immediately — what is it?)

(c) This construction — equally spaced points on the unit circle in each coordinate, built from roots of unity — is the starting point for an entire family of frames called *harmonic frames*, which we will study properly in Chapter 10 alongside equiangular frames. For now, simply record your numerical observations as a preview.

4.9 Supplementary Exercises

Theoretical Exercises

T1. Prove that if $\{f_i\}_{i=1}^m$ is a tight frame for $\mathbb{R}^n$ with bound A, then taking k copies of each vector (each vector repeated k times, for a total of km vectors) is also tight, with bound kA.

T2. Prove that if $\{f_i\}_{i=1}^m$ is a tight frame for $\mathbb{R}^n$ with bound A_1 and $\{g_j\}_{j=1}^{m'}$ is a tight frame for $\mathbb{R}^n$ with bound A_2 (independently, on a disjoint index set), then the combined set $\{f_i\} \cup \{g_j\}$ is a tight frame for $\mathbb{R}^n$ with bound $A_1 + A_2$ — *regardless of whether* $A_1 = A_2$.

T3. Prove directly, via a symmetric summation argument (not the roots-of-unity technique of Section 4.3), that $\{\pm e_1, \pm e_2, \pm e_3\}$ is a tight frame for $\mathbb{R}^3$, and determine its bound A.

T4. Prove that adding a single unit vector v to an orthonormal basis $\{e_1, \ldots, e_n\}$ of $\mathbb{R}^n$ $(n \geq 2)$ never produces a tight frame, for any choice of v. (Hint: the resulting frame operator is $I + vv^T$; compare its rank structure to a scalar multiple of I.)

T5. Generalize the axis-aligned construction of Proposition 4.5.1 to $\mathbb{R}^3$: using six vectors, two along each coordinate axis with possibly unequal norms a, b (on the first axis), c, d (second), e, f (third), state and prove the precise condition on these six numbers for the configuration to be tight.

T6. Using Theorem 8.3.1 (frame potential $\geq m^2/n$, with equality iff tight), give an alternative proof that the regular m-gon is a tight frame: compute its frame potential directly using a double-angle identity, compare to the lower bound m^2/n with $n = 2$, and conclude.

Computational Exercises

C1. Numerically verify T1 for the triangular frame with $k = 2, 3, 4$ copies, confirming the bound scales as kA.

C2. Numerically verify T2 using the triangular frame ($A = 1.5$) and the square ($A = 2$) as the two tight frames being combined, and confirm the union is tight with bound 3.5.

C3. Compute the frame operator of $\{\pm e_1, \pm e_2, \pm e_3\}$ directly and confirm the eigenvalues found in T3.

C4. For $n = 2$ and $n = 3$, add a random unit vector to the standard orthonormal basis and confirm numerically that the result is never tight (T4), across at least 10 random choices of the extra vector in each dimension.

C5. Construct the 6-vector $\mathbb{R}^3$ configuration from T5 with a specific numeric target bound (e.g. $A = 5$), verify tightness directly, and confirm the frame operator is exactly $5I$.

Chapter 5

Dual Frames

5.1 Overview: From One Reconstruction Formula to a Family

Chapter 3 derived a reconstruction formula almost in passing, while studying the frame operator: for any frame $\{f_i\}_{i=1}^m$ with frame operator S,

$$x = \sum_{i=1}^{m} \langle x, f_i \rangle \, (S^{-1} f_i) \qquad \text{for every } x \in V,$$

and named the vectors $\tilde{f}_i := S^{-1} f_i$ the *canonical dual frame*, promising a full treatment in this chapter. Chapter 3 also observed, in passing, that the synthesis operator F has a nontrivial null space whenever the frame is redundant ($m > n$), and promised that this null space would turn out to parametrize *every* valid dual, not just the canonical one.

This chapter delivers on both promises. We will see that the canonical dual is only one member—admittedly a distinguished one—of an entire *affine family* of dual frames, all of which reconstruct every vector in V exactly. We characterize this family completely (Theorem 5.4.1), show that the canonical dual is the unique *minimal-norm* member of the family (Theorem 5.5.1), and close with a numerical investigation of how the choice of dual affects robustness to noise— returning, with real machinery in hand, to the very motivation that opened Chapter 1.

5.2 Dual Frames: Definition

Definition 5.2.1 (Dual frame)**.** Let $\{f_i\}_{i=1}^m$ be a frame for $V = \mathbb{F}^n$. A frame $\{g_i\}_{i=1}^m$ for V (indexed by the same index set) is called a *dual frame* of $\{f_i\}_{i=1}^m$ if

$$x = \sum_{i=1}^{m} \langle x, f_i \rangle \, g_i \qquad \text{for every } x \in V. \tag{5.1}$$

Remark. Definition 5.2.1 is not symmetric-looking at first glance, but it is in fact a symmetric relationship: if $\{g_i\}$ is dual to $\{f_i\}$ in the sense above, then $\{f_i\}$ is also dual to $\{g_i\}$, i.e. $x = \sum_i \langle x, g_i \rangle f_i$ for all x as well. We prove this symmetry as part of Proposition 5.3.1 below, in the canonical case, and the general case is Lab Exercise 5.1 (it follows the same pattern). For now, simply note that "dual" is meant in this mutual sense, exactly as "$\tilde{f}_i$ is dual to f_i" is meant to be an if-and-only-if kind of partnership, not a one-way relationship.

In matrix language, if G denotes the synthesis matrix of $\{g_i\}_{i=1}^m$ (the $n \times m$ matrix with i-th column g_i), then (5.1) reads $x = G(F^*x)$ for every $x \in V$, i.e.

$$GF^* = I_n. \tag{5.2}$$

This is the condition we will use throughout the chapter: *a synthesis matrix G defines a dual frame if and only if $GF^* = I_n$.*

5.3 The Canonical Dual Frame

Proposition 5.3.1 (The canonical dual is a valid dual frame)**.** *Let $\{f_i\}_{i=1}^m$ be a frame for V with frame operator S, and define $\tilde{f}_i := S^{-1}f_i$ for $i = 1, \ldots, m$. Then:*

(i) $\{\tilde{f}_i\}_{i=1}^m$ is itself a frame for V (the canonical dual frame*).*

(ii) Its frame operator is $\tilde{S} = S^{-1}$.

(iii) $\{\tilde{f}_i\}$ is dual to $\{f_i\}$ in the sense of Definition 5.2.1, and the relationship is mutual: $\{f_i\}$ is also dual to $\{\tilde{f}_i\}$.

(iv) The optimal frame bounds of $\{\tilde{f}_i\}$ are $\tilde{A} = 1/B$ and $\tilde{B} = 1/A$, where A, B are the optimal bounds of $\{f_i\}$.

Proof. Write $\tilde{F} := S^{-1}F$ for the synthesis matrix of $\{\tilde{f}_i\}$ (this is consistent: the i-th column of $S^{-1}F$ is $S^{-1}f_i = \tilde{f}_i$, since S^{-1} is applied to each column of F).

(ii) The frame operator of $\{\tilde{f}_i\}$ is

$$\tilde{S} = \tilde{F}\tilde{F}^* = (S^{-1}F)(S^{-1}F)^* = S^{-1}FF^*(S^{-1})^* = S^{-1}SS^{-1} = S^{-1},$$

using that S (hence S^{-1}) is self-adjoint, so $(S^{-1})^* = S^{-1}$.

(i) Since S is invertible and self-adjoint with $S > 0$ (Proposition 3.4.2(iii)), S^{-1} is also self-adjoint and positive definite (its eigenvalues are the reciprocals $1/\lambda_k > 0$ of the eigenvalues of S). By part (ii), the frame operator of $\{\tilde{f}_i\}$ is $\tilde{S} = S^{-1}$, which is positive definite; by Proposition 3.4.2(iii) applied in reverse (positive definite frame operator $\Leftrightarrow$ the vectors span V), $\{\tilde{f}_i\}$ spans V, hence (Theorem 2.2.1) is a frame for V.

(iii) We must check $\tilde{F}F^* = I_n$ (matching (5.2) with $G = \tilde{F}$):

$$\tilde{F}F^* = (S^{-1}F)F^* = S^{-1}(FF^*) = S^{-1}S = I_n.$$

So $\{\tilde{f}_i\}$ is dual to $\{f_i\}$. For the converse direction, we must check $F\tilde{F}^* = I_n$:

$$F\tilde{F}^* = F(S^{-1}F)^* = FF^*(S^{-1})^* = SS^{-1} = I_n,$$

again using $(S^{-1})^* = S^{-1}$. So $\{f_i\}$ is also dual to $\{\tilde{f}_i\}$, confirming the mutual relationship.

(iv) By Theorem 3.5.1, the optimal bounds of $\{\tilde{f}_i\}$ are the extreme eigenvalues of $\tilde{S} = S^{-1}$. If S has eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$ (so $B = \lambda_1$, $A = \lambda_n$ for $\{f_i\}$), then S^{-1} has eigenvalues $1/\lambda_1 \leq \cdots \leq 1/\lambda_n$ (matrix inversion preserves eigenvectors and inverts eigenvalues). The smallest eigenvalue of S^{-1} is $1/\lambda_1 = 1/B$ and the largest is $1/\lambda_n = 1/A$, giving $\tilde{A} = 1/B$ and $\tilde{B} = 1/A$. $\square$

Key Idea

The canonical dual frame $\tilde{f}_i = S^{-1} f_i$ has frame operator $\tilde{S} = S^{-1}$, and its frame bounds are the *reciprocals, in reversed order*, of the original frame's bounds: $\tilde{A} = 1/B$, $\tilde{B} = 1/A$. In particular, **the canonical dual of a tight frame is again tight**: if $A = B$, then $\tilde{A} = \tilde{B} = 1/A$. This matches the reconstruction formula from Chapter 3 (Equation 3.5), where the tight-frame dual was simply $\tilde{f}_i = \frac{1}{A} f_i$ — a special case of $\tilde{f}_i = S^{-1} f_i = \frac{1}{A} f_i$ when $S = AI$.

Example 5.1: Canonical dual of the non-tight frame

Recall $\{h_1, h_2, h_3\} = \{(1,0)^T, (0,1)^T, (1,1)^T\}$ from Chapter 3, with frame operator $S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and bounds $A = 1$, $B = 3$. We invert S:

$$S^{-1} = \frac{1}{\det S} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

using $\det S = 2 \cdot 2 - 1 \cdot 1 = 3$. The canonical dual vectors are

$$\tilde{h}_1 = S^{-1} h_1 = \tfrac{1}{3}(2, -1)^T, \quad \tilde{h}_2 = S^{-1} h_2 = \tfrac{1}{3}(-1, 2)^T, \quad \tilde{h}_3 = S^{-1} h_3 = \tfrac{1}{3}(1, 1)^T.$$

By Proposition 5.3.1(iv), the dual frame has bounds $\tilde{A} = 1/3$, $\tilde{B} = 1/1 = 1$. Notice that $\{\tilde{h}_i\}$ is visibly *not* a rescaling of $\{h_i\}$—unlike the tight-frame case, the canonical dual of a non-tight frame is genuinely a different set of vectors, not merely the original frame scaled by a constant.

5.4 The Family of All Dual Frames

The canonical dual is a distinguished dual frame, but—exactly as Chapter 3 promised—it is not the only one whenever the original frame is redundant. We now characterize the complete family.

Theorem 5.4.1 (Characterization of all dual frames)**.** *Let $\{f_i\}_{i=1}^m$ be a frame for $V = \mathbb{F}^n$ with synthesis matrix F, and let $\tilde{F} = S^{-1} F$ be the canonical dual synthesis matrix. Then G is the synthesis matrix of a dual frame of $\{f_i\}$ if and only if*

$$G = \tilde{F} + Z, \qquad \text{where } ZF^* = 0. \tag{5.3}$$

That is, the set of all valid dual synthesis matrices is the affine subspace $\tilde{F} + \{Z \in \mathbb{F}^{n \times m} : ZF^* = 0\}$.

Proof. ($\Leftarrow$) Suppose $G = \tilde{F} + Z$ with $ZF^* = 0$. Then

$$GF^* = \tilde{F}F^* + ZF^* = I_n + 0 = I_n,$$

using Proposition 5.3.1(iii) ($\tilde{F}F^* = I_n$). By (5.2), G defines a dual frame.

($\Rightarrow$) Suppose G defines a dual frame, so $GF^* = I_n$. Set $Z := G - \tilde{F}$. Then

$$ZF^* = GF^* - \tilde{F}F^* = I_n - I_n = 0,$$

so Z has the required property, and $G = \tilde{F} + Z$ by construction. □

Remark. This theorem makes precise the promise from Chapter 3's remark on the synthesis operator: the freedom in choosing a dual frame is exactly captured by matrices Z satisfying $ZF^* = 0$, i.e. matrices whose rows each lie in $\ker F$ (since $(ZF^*)_{ji} = \mathrm{row}_j(Z) \cdot \overline{f_i}$, and this vanishes for every i exactly when $\mathrm{row}_j(Z) \in (\mathrm{span}\{f_i\})^\perp$ in the appropriate sense— more directly, $ZF^* = 0$ says each row of Z, viewed as a vector of coefficients in $\mathbb{F}^m$, lies in $\ker F$, since $F\,(\mathrm{row}_j(Z))^* = 0$ is what $ZF^* = 0$ unpacks to row by row). When the frame is a basis ($m = n$), F is square and invertible, so $\ker F = \{0\}$, forcing $Z = 0$: *the dual of a basis is unique*, namely the canonical dual, exactly as expected from ordinary linear algebra (a basis has one and only one dual/biorthogonal basis). When $m > n$, $\ker F$ is nontrivial (dimension $m - n$, by the rank-nullity theorem, since F has rank n), and there is a genuine $(m - n) \times n$-real-dimensional family of dual frames to choose from.

Key Idea

Bases have a unique dual. Redundant frames have infinitely many. The freedom is exactly $\ker F$: the more redundant the frame (the larger $m - n$), the larger the space of valid dual frames. This is the algebraic shadow of the flexibility we advertised informally all the way back in Chapter 1, Section 1.1: redundancy buys you freedom of representation, and Theorem 5.4.1 is the precise statement of how much freedom, and exactly where it lives.

Example 5.2: An alternate dual of the non-tight frame

Continue Example 5.3. Since $m = 3 > n = 2$, $\ker F$ is 1-dimensional. One checks $F(1, 1, -1)^T = h_1 + h_2 - h_3 = (1, 0)^T + (0, 1)^T - (1, 1)^T = (0, 0)^T$, so $(1, 1, -1)^T$ spans $\ker F$. A valid perturbation is $Z = w\,(1, 1, -1)$ for any $w \in \mathbb{R}^2$ (a single column-vector w outer-producted against the row $(1, 1, -1)$), since then each row of Z is a multiple of $(1, 1, -1) \in \ker F$, and $ZF^* = 0$ as required. Taking, say, $w = (0.1, -0.2)^T$:

$$Z = \begin{pmatrix} 0.1 \\ -0.2 \end{pmatrix} \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 0.1 & 0.1 & -0.1 \\ -0.2 & -0.2 & 0.2 \end{pmatrix},$$

and $G = \tilde{F} + Z$ gives a valid, non-canonical dual frame $\{g_1, g_2, g_3\}$. This is verified numerically in Lab Exercise 5.2, including a direct check that $\sum_i \langle x, h_i \rangle\, g_i = x$ for several test vectors x, despite $\{g_i\} \neq \{\tilde{h}_i\}$.

5.5 The Canonical Dual Is the Minimal-Norm Dual

Among the infinitely many dual frames available whenever $m > n$, why is the canonical dual the natural default? The next theorem gives a precise optimality reason.

Theorem 5.5.1 (Minimal-norm characterization of the canonical dual)**.** *Among all dual synthesis matrices G satisfying $GF^* = I_n$, the canonical dual $\tilde{F} = S^{-1}F$ is the unique minimizer of the total squared norm*

$$\|G\|_F^2 := \sum_{i=1}^m \|g_i\|^2 = \mathrm{tr}(GG^*),$$

(the squared Frobenius norm of G). That is, $\tilde{F}$ uses the "smallest" dual vectors, in this total-energy sense, among all valid choices.

Proof. Write $G = \tilde{F} + Z$ with $ZF^* = 0$, as guaranteed by Theorem 5.4.1. We expand the squared Frobenius norm:

$$\|G\|_F^2 = \operatorname{tr}\big((\tilde{F} + Z)(\tilde{F} + Z)^*\big) = \operatorname{tr}(\tilde{F}\tilde{F}^*) + \operatorname{tr}(\tilde{F}Z^*) + \operatorname{tr}(Z\tilde{F}^*) + \operatorname{tr}(ZZ^*).$$

We claim the two cross terms vanish. Indeed,

$$\tilde{F}Z^* = (S^{-1}F)Z^* = S^{-1}(FZ^*) = S^{-1}(ZF^*)^* = S^{-1} \cdot 0^* = 0,$$

using $ZF^* = 0$. Taking adjoints of both sides of $ZF^* = 0$ gives $(ZF^*)^* = FZ^* = 0$ as well (using $(AB)^* = B^*A^*$ and $(F^*)^* = F$), which is exactly the fact used above. Hence $\operatorname{tr}(\tilde{F}Z^*) = 0$, and similarly $\operatorname{tr}(Z\tilde{F}^*) = \operatorname{tr}((\tilde{F}Z^*)^*) = \operatorname{tr}(0^*) = 0$. This leaves

$$\|G\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|Z\|_F^2 \;\geq\; \left\|\tilde{F}\right\|_F^2,$$

with equality if and only if $\|Z\|_F^2 = 0$, i.e. $Z = 0$, i.e. $G = \tilde{F}$. So $\tilde{F}$ is the unique minimizer. □

Key Idea

The identity $\|G\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|Z\|_F^2$ is a **Pythagorean decomposition**: the canonical dual's contribution and the "extra" perturbation Z contribute independently (orthogonally) to the total energy, exactly as $a^2 + b^2$ decomposes the squared length of a right-triangle hypotenuse. This is the same geometric idea as projecting onto a subspace and decomposing a vector into a component inside and a component outside: here, the "subspace" is the affine family of duals, the canonical dual is the point of that family closest to the origin, and every other dual is strictly farther away by exactly $\|Z\|_F^2$.

Remark. This theorem explains, precisely, why the term "canonical" is earned and not arbitrary: among every possible way to reconstruct x from its frame coefficients $\langle x, f_i\rangle$, the canonical dual does so using the least total "effort," in the sense of minimizing $\sum_i \|g_i\|^2$. This will matter in Chapter 11 (frames and quantization/noise), where smaller dual-vector norms translate directly into better noise suppression in the reconstruction formula — a connection we begin to explore numerically in the next section.

5.6 Reconstruction Stability: Canonical vs. Alternate Duals

Chapter 1 motivated frames by the promise of robustness to noise and erasures. We are now equipped to investigate this quantitatively for the first time. Suppose the frame coefficients $c_i = \langle x, f_i\rangle$ are corrupted by some noise ε_i, so that the receiver has access only to $c_i + \varepsilon_i$, and reconstructs via a dual frame $\{g_i\}$ as $\hat{x} = \sum_i (c_i + \varepsilon_i) g_i = x + \sum_i \varepsilon_i g_i$. The reconstruction error is

$$\hat{x} - x = \sum_{i=1}^{m} \varepsilon_i g_i = G\varepsilon,$$

where $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_m)$. If the noise components ε_i are independent with variance σ^2 each, the expected squared reconstruction error is

$$\mathbb{E}\,\|\hat{x} - x\|^2 = \mathbb{E}\,\|G\varepsilon\|^2 = \sigma^2 \,\operatorname{tr}(GG^*) = \sigma^2\,\|G\|_F^2\,.$$

Key Idea

The canonical dual minimizes expected reconstruction error under uncorrelated noise. This follows immediately from Theorem 5.5.1: since $\mathbb{E}\,\|\hat{x} - x\|^2 = \sigma^2\,\|G\|_F^2$ and $\tilde{F}$ minimizes $\|G\|_F^2$ among all valid duals, the canonical dual also minimizes expected reconstruction error. The abstract optimality result of Section 5.5 therefore has an immediate and very concrete payoff: *when in doubt, use the canonical dual*—it is not just the most natural choice algebraically, but the most noise-robust choice as well.

Example 5.3: Numerically comparing canonical vs. alternate dual

Continue Examples 5.3 and 5.4. The canonical dual has $\left\|\tilde{F}\right\|_F^2 = \operatorname{tr}(S^{-1})$. Since $S^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, we get $\operatorname{tr}(S^{-1}) = \frac{1}{3}(2+2) = \frac{4}{3} \approx 1.333$. For the alternate dual $G = \tilde{F} + Z$ from Example 5.4 with $Z = (0.1, -0.2)^T(1, 1, -1)$, each row of Z has *three* entries (one per frame vector), so direct computation (carried out numerically in Lab Exercise 5.3) gives $\|Z\|_F^2 = 3(0.1^2 + 0.2^2) = 0.15$, so $\|G\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|Z\|_F^2 \approx 1.333 + 0.15 = 1.483$, strictly larger than the canonical dual's 1.333, exactly as Theorem 5.5.1 guarantees. Under unit-variance noise, the canonical dual's expected squared error is ≈ 1.333, while the alternate dual's is ≈ 1.483: roughly 11.25% worse, purely as a consequence of choosing a non-optimal dual.

5.7 NumPy Implementation

Listing 5.1: Canonical and alternate dual frames, and stability comparison

```python
import numpy as np
from scipy.linalg import null_space

def frame_operator(F):
    return F @ F.T

def canonical_dual(F):
    """Synthesis matrix of the canonical dual frame: S^{-1} F."""
    S = frame_operator(F)
    return np.linalg.solve(S, F)    # S^{-1} @ F, more stable than explicit inverse

def dual_family_direction(F):
    """
    Orthonormal basis of directions Z = w @ N.T (one column w at a time)
    spanning the freedom in choosing a dual frame, where N spans ker(F).
    Returns N, shape (m, m-n).
    """
    return null_space(F)

def random_alternate_dual(F, scale=1.0, seed=0):
    """Construct a random valid (non-canonical) dual frame's synthesis matrix."""
    n, m = F.shape
```

```
    N = dual_family_direction(F)          # shape (m, m-n)
    rng = np.random.default_rng(seed)
    W = rng.standard_normal((n, N.shape[1])) * scale
    Z = W @ N.T                             # shape (n, m), and Z @ F.T == 0
    G_tilde = canonical_dual(F)
    return G_tilde + Z

def expected_reconstruction_error(G, sigma=1.0):
    """E[||x_hat - x||^2] / sigma^2 = ||G||_F^2 = tr(G G^T)."""
    return np.trace(G @ G.T)

# ----------------------------------------------------------------
# Example 5.1 / 5.3: non-tight frame {h1, h2, h3}
# ----------------------------------------------------------------
H = np.array([[1., 0., 1.],
              [0., 1., 1.]])

H_tilde = canonical_dual(H)
print("=== Canonical dual of {h1,h2,h3} ===")
print("H_tilde =\n", H_tilde.round(4))
print("Check H_tilde @ H.T == I:", np.allclose(H_tilde @ H.T, np.eye(2)))
print(f"||H_tilde||_F^2 = {expected_reconstruction_error(H_tilde):.6f}")

G_alt = random_alternate_dual(H, scale=0.2, seed=0)
print("\n=== A random alternate dual ===")
print("G_alt =\n", G_alt.round(4))
print("Check G_alt @ H.T == I:", np.allclose(G_alt @ H.T, np.eye(2)))
print(f"||G_alt||_F^2 = {expected_reconstruction_error(G_alt):.6f}")
print(f"(Canonical is smaller: "
      f"{expected_reconstruction_error(H_tilde) <=
         expected_reconstruction_error(G_alt)})")

# ----------------------------------------------------------------
# Verify reconstruction works exactly for BOTH duals
# ----------------------------------------------------------------
rng = np.random.default_rng(1)
for trial in range(3):
    x = rng.standard_normal(2)
    c = H.T @ x    # frame coefficients <x, h_i>
    x_hat_canonical = H_tilde @ c
    x_hat_alt = G_alt @ c
    print(f"\nx = {x.round(4)}")
    print(f"  canonical reconstruction error: {np.linalg.norm(
       x_hat_canonical - x):.2e}")
    print(f"  alternate reconstruction error: {np.linalg.norm(x_hat_alt -
       x):.2e}")

# ----------------------------------------------------------------
# Monte Carlo: verify canonical dual minimizes noisy reconstruction error
# ----------------------------------------------------------------
def noisy_reconstruction_trial(F, G, x, sigma, rng):
    c = F.T @ x
    noise = rng.standard_normal(c.shape) * sigma
    x_hat = G @ (c + noise)
```

```python
    return np.linalg.norm(x_hat - x)**2

rng = np.random.default_rng(2)
x_test = np.array([1.0, -0.5])
sigma = 0.3
N_trials = 200_000

errs_canonical = [noisy_reconstruction_trial(H, H_tilde, x_test, sigma, rng)
                  for _ in range(N_trials)]
errs_alt = [noisy_reconstruction_trial(H, G_alt, x_test, sigma, rng)
            for _ in range(N_trials)]

print(f"\n=== Monte Carlo noise comparison ({N_trials} trials, sigma={sigma}) ===")
print(f"Mean squared error, canonical dual: {np.mean(errs_canonical):.6f} "
      f"(predicted: {sigma**2 * expected_reconstruction_error(H_tilde):.6f})")
print(f"Mean squared error, alternate dual: {np.mean(errs_alt):.6f}  "
      f"(predicted: {sigma**2 * expected_reconstruction_error(G_alt):.6f})")
```

Remark. We use `np.linalg.solve(S, F)` rather than explicitly forming `np.linalg.inv(S) @ F`. These are mathematically identical, but `solve` avoids explicitly constructing the inverse matrix, which is both faster and numerically more stable, especially as S becomes ill-conditioned (nearly singular). This is a general good habit: prefer `np.linalg.solve(A, b)` over `np.linalg.inv(A) @ b` whenever you only need the product, not the inverse matrix itself.

5.8 Chapter Summary

- A **dual frame** $\{g_i\}$ of $\{f_i\}$ satisfies $x = \sum_i \langle x, f_i \rangle g_i$ for all x, equivalently $GF^* = I_n$ for the synthesis matrices G, F. The relationship is mutual: $\{f_i\}$ is also dual to $\{g_i\}$.
- The **canonical dual** $\tilde{f}_i = S^{-1} f_i$ is always a valid dual, with frame operator $\tilde{S} = S^{-1}$ and frame bounds $\tilde{A} = 1/B$, $\tilde{B} = 1/A$ (reciprocal and reversed). For tight frames, $\tilde{f}_i = \frac{1}{A} f_i$, a trivial rescaling.
- **Theorem 5.4.1**: every valid dual has synthesis matrix $G = \tilde{F} + Z$ with $ZF^* = 0$, i.e. each row of Z lies in $\ker F$. Bases ($m = n$) have a unique dual; redundant frames ($m > n$) have an $(m - n)$-real-dimensional affine family of duals.
- **Theorem 5.5.1**: the canonical dual is the unique **minimal Frobenius-norm** dual, via the Pythagorean decomposition $\|G\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|Z\|_F^2$.
- This optimality has a direct payoff: since expected squared reconstruction error under uncorrelated coefficient noise equals $\sigma^2 \|G\|_F^2$, the **canonical dual minimizes expected reconstruction error** among all valid duals—the precise, quantitative form of the robustness-to-noise promise first made informally in Chapter 1.

5.9 Lab Exercises

Lab Exercise 5.1: The Canonical Dual Pipeline

Implement `canonical_dual(F)` from Section 5.7. Run it on:

(a) The triangular frame (tight, Chapter 3 Example 3.2): verify $\tilde{f}_i = \frac{1}{1.5} f_i$ for each i.

(b) The non-tight frame $\{h_1, h_2, h_3\}$ (Example 5.1): reproduce the hand computation exactly.

(c) The square vertices (Chapter 3 Example 3.6): verify the canonical dual is again a rescaling of the original (as expected, since this frame is tight).

For each case, also verify the mutual-duality claim from the remark after Definition 5.2.1: check numerically that `F @ H_tilde.T` also equals the identity (not just `H_tilde @ F.T`), confirming $\{f_i\}$ is dual to $\{\tilde{f}_i\}$ as well as vice versa.

Lab Exercise 5.2: Verifying the Dual Family

Implement `dual_family_direction(F)` and `random_alternate_dual(F)` from Section 5.7. Using $\{h_1, h_2, h_3\}$:

(a) Reproduce Example 5.2's specific alternate dual exactly (using $w = (0.1, -0.2)^T$), and verify $GF^* = I_2$ numerically.

(b) Generate 10 different random alternate duals (varying the seed) and verify, for each, that $\sum_i \langle x, h_i \rangle g_i = x$ holds exactly (to machine precision) for five random test vectors x.

(c) For each of the 10 alternate duals, compute $\|G\|_F^2$ and confirm it always exceeds $\left\|\tilde{F}\right\|_F^2 \approx 1.333$ (the canonical value from Example 5.3), consistent with Theorem 5.5.1.

Lab Exercise 5.3: Monte Carlo Stability Comparison

Reproduce the Monte Carlo experiment from Section 5.7 (the final code block). Then extend it:

(a) Vary the perturbation scale in `random_alternate_dual` over `scale` $= 0.05, 0.2, 0.5, 1.0, 2.0$, and for each, compute $\|G\|_F^2$ and the predicted mean squared error $\sigma^2 \|G\|_F^2$ (with $\sigma = 0.3$ as before). Plot predicted MSE versus `scale`.

(b) Confirm, by running the Monte Carlo simulation (at least 50,000 trials) at each scale, that the empirically observed mean squared error matches your prediction closely.

(c) In a sentence or two: does this experiment suggest any practical guidance for choosing a dual frame in a real signal-processing system where the frame is fixed in advance but coefficient noise is unavoidable?

Lab Exercise 5.4: Dual Frames of a Basis Are Unique

Let $\{v_1, v_2\} = \{(2,1)^T, (1,3)^T\}$, a basis (not a redundant frame) for $\mathbb{R}^2$. Compute the canonical dual $\{\tilde{v}_1, \tilde{v}_2\}$, and verify, using `scipy.linalg.null_space`, that $\ker F$ is trivial (zero-dimensional) for the synthesis matrix of this basis. Conclude, via Theorem 5.4.1, that the canonical dual is the *only* dual frame in this case. (For comparison with your linear algebra background: explain in a sentence why this matches the familiar fact that a basis has a unique dual/biorthogonal basis, characterized by $\langle v_i, \tilde{v}_j \rangle = \delta_{ij}$.)

Lab Exercise 5.5 (Optional, Challenge): Sparsity-Promoting Duals

Theorem 5.5.1 shows the canonical dual minimizes the *total Frobenius energy* $\sum_i \|g_i\|^2$ among all valid duals. A different, and increasingly important, optimality criterion in modern signal processing is *sparsity*: finding a dual frame (or, more precisely, a coefficient sequence) with as many exactly zero entries as possible. This exercise previews that idea numerically.

(a) Using $\{h_1, h_2, h_3\}$ and a fixed $x = (1, -0.5)^T$, the canonical dual gives one specific coefficient-free reconstruction. But recall from Example 3.7 that the coefficients $c_i = \langle x, f_i \rangle$ *themselves* are not unique either, once you allow $c' = c + z$ for $z \in \ker F$ (a fact distinct from, but related to, the dual-frame freedom of this chapter). Parametrize $c'(t) = c + t \cdot n$ where n spans $\ker H$ (one-dimensional, as in Example 5.2), and plot $\|c'(t)\|_1$ (the ℓ^1 norm, `np.sum(np.abs(...))`) as a function of $t \in [-2, 2]$.

(b) Find the value of t that minimizes $\|c'(t)\|_1$ numerically (e.g. via `scipy.optimize.minimize_scalar`). Is the minimizing coefficient vector sparse (does it have an exact zero entry)? This phenomenon—ℓ^1 minimization promoting exact sparsity—is the starting point of compressed sensing, which we will touch on again in Chapter 12.

5.10 Supplementary Exercises

Theoretical Exercises

T1. Prove that the set of all dual frames of a fixed frame $\{f_i\}_{i=1}^m$ (the affine family of Theorem 5.4.1) is *convex*: if G_1 and G_2 are both valid dual synthesis matrices, then so is $tG_1 + (1-t)G_2$ for every $t \in [0,1]$.

T2. Prove the full biconditional: a frame $\{f_i\}_{i=1}^m$ equals its own canonical dual ($\tilde{f}_i = f_i$ for all i) if and only if $\{f_i\}$ is a Parseval frame. (One direction is immediate; for the converse, show $F = \tilde{F}$ forces $S = I$.)

T3. Prove that $\|Z\|_F^2$ is a strictly convex function of Z (over the subspace of matrices with $ZF^* = 0$), and use this convexity — rather than the Pythagorean/orthogonality argument used in the text — to give an alternative proof that the canonical dual is the *unique* minimizer of total energy in Theorem 5.5.1.

T4. Let $\{f_i\}_{i=1}^m$ be a tight frame for $\mathbb{R}^n$ with bound A (not necessarily unit-norm). Prove that the canonical dual has total energy $\sum_i \left\|\tilde{f}_i\right\|^2 = n/A$ exactly. Specialize to the unit-norm case ($A = m/n$) to recover the formula n^2/m used in Chapter 11.

T5. Prove directly from the definition of the analysis operator F^* that $\ker F^* = (\operatorname{span}\{f_i\})^\perp$.

T6. Using strict convexity (T3), prove or disprove: for any two *distinct* valid duals $G_1 \neq G_2$ of a fixed frame, the averaged dual $\frac{1}{2}(G_1 + G_2)$ has strictly smaller total energy than the larger of $\|G_1\|_F^2, \|G_2\|_F^2$.

Computational Exercises

C1. For $\{h_1, h_2, h_3\}$, construct two distinct alternate duals G_1, G_2, form their average, and confirm numerically that the average is itself a valid dual (T1).

C2. Confirm numerically that the triangular frame (tight, $A = 1.5 \neq 1$) does *not* equal its own canonical dual, while the standard ONB (Parseval) does, illustrating both directions of T2.

C3. Scale the triangular frame by a factor $c = 2, 3, 4$ (multiply every vector by c), and for each, compute the canonical dual's total energy and compare to the predicted n/A from T4.

C4. For $\{h_1, h_2, h_3\}$, construct 10 random alternate duals with varying perturbation scale, compute each one's total energy, and verify numerically that the canonical dual ($w = 0$) has strictly smaller energy than every one of them, consistent with T3's convexity argument.

C5. Construct three distinct alternate duals of $\{h_1, h_2, h_3\}$, compute all pairwise averages, and verify each pairwise average has energy strictly less than the larger of its two parents, illustrating T6 across multiple pairs.

Chapter 6

The Geometry of Dual Frames

6.1 Overview: Making the Affine Family Visible

Chapter 5 proved, algebraically, that the set of all dual frames of a fixed frame $\{f_i\}_{i=1}^m$ forms an *affine family* $G = \tilde{F} + Z$ with $ZF^* = 0$ (Theorem 5.4.1), that the canonical dual $\tilde{F}$ is the unique member minimizing total energy $\|G\|_F^2$ (Theorem 5.5.1), and that this minimality translates directly into superior noise robustness (Section 5.6). Every one of these facts was proved by algebra: trace identities, adjoint rules, Pythagorean decompositions of Frobenius norms.

This chapter does not prove anything new. Instead, it makes Chapter 5's algebra *visible*. For the simplest genuinely redundant frame — one extra vector beyond a basis, so $m = n + 1$ — the affine family of duals turns out to be parametrized by an ordinary 2-dimensional plane, small enough to draw. On that plane, the energy function $\|G\|_F^2$ is an exact paraboloid, the canonical dual sits at its unique minimum, and the Pythagorean decomposition of Theorem 5.5.1 becomes literally the Pythagorean theorem for a right triangle you can see. We then push the geometric picture one step further than Chapter 5 did: the reconstruction error under random noise is not just a single number $\|G\|_F^2$, but an entire *error ellipse*, and the shape of that ellipse — not only its size — turns out to depend on the frame in a clean, computable way that the algebra alone does not make obvious.

6.2 The Smallest Interesting Case: One Redundant Vector

To draw a picture, we need the parameter space of duals to be low-dimensional. By Theorem 5.4.1, the family of valid duals is parametrized by matrices Z with $ZF^* = 0$, i.e. each row of Z lies in $\ker F \subset \mathbb{F}^m$. Since $\dim \ker F = m - n$ (rank–nullity, as F has full row rank n when $\{f_i\}$ is a frame), the space of valid Z has real dimension $n \cdot (m - n)$ (each of the n rows of Z ranges independently over the $(m - n)$-dimensional space $\ker F$).

Key Idea

For the dual family to be visualizable as a plane, we want $n \cdot (m - n) = 2$. The cleanest choice satisfying this is $n = 2$, $m = 3$: **one redundant vector beyond a basis of $\mathbb{R}^2$.** Then $\dim \ker F = 1$, but the parameter space of Z is still 2-dimensional, since $Z = w\,k^T$ for $w \in \mathbb{R}^2$ ranging freely and $k \in \mathbb{R}^3$ a fixed unit vector spanning $\ker F$. **The parameter w, not the kernel direction k, is what we will plot.**

We use the triangular frame from Chapters 1, 3, and 5 throughout this section: $f_1 = (1,0)^T$, $f_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})^T$, $f_3 = (-\frac{1}{2}, -\frac{\sqrt{3}}{2})^T$, with frame operator $S = \frac{3}{2}I$ and canonical dual $\tilde{f}_i = \frac{2}{3}f_i$ (Example 3.2, Example 5.1). Its kernel was already computed in Chapter 1: $\ker F$ is spanned by $(1,1,1)^T$, since $f_1 + f_2 + f_3 = 0$; the corresponding unit vector is $k = (1,1,1)^T/\sqrt{3}$.

Proposition 6.2.1 (The dual family as a plane)**.** *For the triangular frame, every dual frame's synthesis matrix has the form*

$$G(w) = \tilde{F} + w\,k^T, \qquad w = (w_1, w_2) \in \mathbb{R}^2,$$

where $\tilde{F}$ is the canonical dual's synthesis matrix and $k = (1,1,1)^T/\sqrt{3}$. Distinct w give distinct dual frames, and every dual frame arises this way for exactly one w.

Proof. By Theorem 5.4.1, every dual has the form $\tilde{F} + Z$ with $ZF^* = 0$, i.e. each row of Z lies in $\ker F = \operatorname{span}\{k\}$ (one-dimensional). So row j of Z is $w_j k^T$ for a unique scalar w_j, $j = 1,2$; stacking the two rows gives $Z = wk^T$ for the unique vector $w = (w_1, w_2)$. Distinct w give distinct Z (since $k \neq 0$), hence distinct $G(w)$. ☐

6.3 The Energy Paraboloid

We now compute the total energy $\|G(w)\|_F^2 = \sum_i \|g_i\|^2$ as an explicit function of w, and discover that it is exactly a paraboloid of revolution.

Theorem 6.3.1 (The energy function is a paraboloid)**.** *For the triangular frame, with $G(w) = \tilde{F} + wk^T$ as in Proposition 6.2.1,*

$$\|G(w)\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|w\|^2 = \frac{4}{3} + w_1^2 + w_2^2.$$

Proof. By the Pythagorean decomposition of Theorem 5.5.1, $\|G(w)\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|Z\|_F^2$ where $Z = wk^T$. Since k is a *unit* vector, $\|Z\|_F^2 = \operatorname{tr}(ZZ^T) = \operatorname{tr}(wk^Tkw^T) = \|k\|^2 \operatorname{tr}(ww^T) = \|w\|^2$ (using $\|k\|^2 = 1$ and $\operatorname{tr}(ww^T) = \|w\|^2$ for any vector w). The value $\left\|\tilde{F}\right\|_F^2 = \operatorname{tr}(S^{-1}) = \operatorname{tr}(\frac{2}{3}I) = \frac{4}{3}$ was computed in Example 5.3. ☐

Key Idea

Because k is normalized to unit length, the energy function over the parameter plane is the **exact** paraboloid $E(w) = \frac{4}{3} + \|w\|^2$ — no cross terms, no distortion, a perfect circular bowl. Its unique minimum is at $w = 0$, which by Proposition 6.2.1 is exactly the canonical dual $G(0) = \tilde{F}$. **Theorem 5.5.1's abstract minimality statement is, in this picture, nothing more than the obvious fact that a paraboloid has one lowest point, directly above the origin.**

This is genuinely a different way of seeing the same theorem: Chapter 5 proved minimality by an algebraic Pythagorean argument; here, minimality is a visual fact about a bowl-shaped surface, and the Pythagorean decomposition $\|G\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|w\|^2$ *is* literally the equation of that paraboloid, height above the plane equals a constant plus squared distance from the origin.

6.4 The Error Ellipse

Chapter 5, Section 5.6 showed that under coefficient noise $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_m)$ with independent components of variance σ^2, the reconstruction error $\hat{x}-x = G\varepsilon$ has expected squared norm $\sigma^2 \|G\|_F^2$. This is a single number: the *total* expected error. But $\hat{x}-x = G\varepsilon$ is itself a random *vector* in $\mathbb{R}^n$, and its distribution has a shape, not just a size. We now compute that shape.

Definition 6.4.1 (Error covariance)**.** For a dual frame with synthesis matrix G, under coefficient noise with covariance $\sigma^2 I_m$, the reconstruction error $\hat{x}-x = G\varepsilon$ has covariance matrix

$$\Sigma_G := \sigma^2\, GG^T \quad \in \mathbb{R}^{n\times n}.$$

The level sets of the resulting error distribution are ellipses (in $\mathbb{R}^2$) or ellipsoids (in higher dimensions) centered at the origin, with semi-axis lengths $\sigma\sqrt{\lambda_i(GG^T)}$ along the eigenvectors of GG^T.

Proposition 6.4.2 (The canonical dual's error covariance)**.** *For any frame (not just the triangular one), the canonical dual's error covariance is*

$$\Sigma_{\tilde{F}} = \sigma^2 S^{-1}.$$

Proof. $\tilde{F}\tilde{F}^T = (S^{-1}F)(S^{-1}F)^T = S^{-1}FF^TS^{-1} = S^{-1}SS^{-1} = S^{-1}$, using S^{-1} symmetric (Proposition 3.4.2(i) applied to S^{-1}, itself symmetric since S is) and $FF^T = S$. □

Key Idea

The canonical dual's error ellipse is, up to the noise scale σ^2, literally S^{-1} — **the inverse of the frame operator itself.** Since the eigenvalues of S^{-1} are exactly $1/\lambda_i(S)$, and Theorem 3.5.1 identified $\lambda_i(S)$ as the frame bounds, the error ellipse's axes have lengths $\sigma/\sqrt{A}$ and $\sigma/\sqrt{B}$ (in $\mathbb{R}^2$): **a tighter frame (closer A, B) gives a rounder error ellipse; a perfectly tight frame gives a perfectly circular error region.** This is a genuinely new fact, not visible from Chapter 5's total-energy bookkeeping alone: tightness controls not only the *average* reconstruction error but its *directional uniformity*.

Example 6.1: Circular vs. elliptical error regions

For the triangular frame ($A = B = \frac{3}{2}$, tight), Proposition 6.4.2 gives $\Sigma_{\tilde{F}} = \sigma^2 \cdot \frac{2}{3} I$: a perfect circle of radius $\sigma\sqrt{2/3}$, independent of direction. For the non-tight frame $\{h_1, h_2, h_3\}$ ($A = 1$, $B = 3$, Example 3.3), the same proposition gives $\Sigma_{\tilde{F}} = \sigma^2 S^{-1}$ with eigenvalues $1/3$ and 1 — an ellipse with one axis *three times longer* than the other. Reconstruction error is three times more spread out along the B-eigenvector direction (the direction of h_3 itself, as found in Example 3.3) than along the A-eigenvector direction. This is the precise, quantitative version of the informal Chapter 1 idea that an "unbalanced" frame treats some directions worse than others: now we can say exactly how much worse, and exactly which direction is penalized.

6.5 Visualizing Both Pictures

We now render both geometric pictures developed above: the energy paraboloid over the space of dual frames, and the error ellipses for tight versus non-tight frames.

The figure above shows this chapter's two pictures side by side: the left panel is the parameter plane of Proposition 6.2.1, with energy level sets as concentric circles around the canonical dual at the origin; dragging the point traces out the paraboloid of Theorem 6.3.1 indirectly, through its height (displayed as the energy readout). The right panel shows the corresponding error ellipse of Definition 6.4.1, which is a perfect circle exactly at the canonical dual ($w = 0$) and stretches into a longer and longer ellipse as w moves away from the origin — the visual content of Theorem 5.5.1 and Proposition 6.4.2 happening simultaneously, in real time, as you drag.

6.6 Reproducing the Pictures in Python

The interactive figure above is built from exactly the same formulas proved in Sections 6.3–6.4. We now reproduce both pictures as static Matplotlib plots, so that the geometric content of this chapter is fully captured in code you can run and modify offline.

Listing 6.1: The energy paraboloid and error ellipses, reproduced in Matplotlib

```python
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Ellipse

# --- Triangular frame setup (Chapters 1, 3, 5) ---
f1 = np.array([1.0, 0.0])
f2 = np.array([-0.5, np.sqrt(3)/2])
f3 = np.array([-0.5, -np.sqrt(3)/2])
F = np.column_stack([f1, f2, f3])          # shape (2, 3)
S = F @ F.T
F_tilde = np.linalg.solve(S, F)            # canonical dual, S^{-1} F
k = np.array([1.0, 1.0, 1.0]) / np.sqrt(3) # unit vector spanning ker(F)

def G_of_w(w):
    """Synthesis matrix of the dual frame at parameter w in R^2."""
    Z = np.outer(w, k)
    return F_tilde + Z

def energy(w):
    G = G_of_w(w)
    return np.trace(G @ G.T)

def error_covariance(w, sigma=1.0):
    G = G_of_w(w)
    return sigma**2 * (G @ G.T)

# --- Panel 1: energy paraboloid (as a contour plot over the w-plane) ---
fig, axes = plt.subplots(1, 2, figsize=(11, 5))

ax = axes[0]
w1_grid, w2_grid = np.meshgrid(np.linspace(-2, 2, 200), np.linspace(-2, 2,
    200))
E_grid = 4/3 + w1_grid**2 + w2_grid**2        # Theorem 6.3.1, closed form
contours = ax.contour(w1_grid, w2_grid, E_grid, levels=10, cmap='viridis')
ax.clabel(contours, inline=True, fontsize=8)
ax.plot(0, 0, 'ko', markersize=8)
ax.annotate('canonical dual\n(global minimum)', (0, 0),
            xytext=(0.3, 1.5), fontsize=9)
ax.set_xlabel('$w_1$'); ax.set_ylabel('$w_2$')
```

```
ax.set_title('Energy $\\|G(w)\\|_F^2 = 4/3 + \\|w\\|^2$')
ax.set_aspect('equal')

# --- Panel 2: error ellipses at several choices of dual ---
ax = axes[1]
colors = ['#0F6E56', '#185FA5', '#993C1D', '#993556']
for w, color in zip([(0,0), (1,0), (0,1), (0.7,0.7)], colors):
    Cov = error_covariance(np.array(w))
    eigvals, eigvecs = np.linalg.eigh(Cov)
    angle = np.degrees(np.arctan2(eigvecs[1, -1], eigvecs[0, -1]))
    width, height = 2*np.sqrt(eigvals[-1]), 2*np.sqrt(eigvals[0])
    ell = Ellipse((0, 0), width, height, angle=angle,
                  facecolor='none', edgecolor=color, linewidth=2,
                  label=f'w={w}')
    ax.add_patch(ell)
ax.set_xlim(-3, 3); ax.set_ylim(-3, 3)
ax.set_aspect('equal')
ax.legend(fontsize=8, loc='upper right')
ax.set_title('Reconstruction error ellipses by choice of dual')
ax.set_xlabel('error in $x$-direction'); ax.set_ylabel('error in $y$-direction')

plt.tight_layout()
plt.savefig('dual_frame_geometry.png', dpi=150)
plt.show()
```

Remark. Running this code produces exactly the two pictures shown earlier: a perfectly circular contour plot (since the energy function has no cross terms in w_1, w_2, by Theorem 6.3.1), and a family of ellipses that are circular at $w = (0, 0)$ and progressively more elongated as w moves away from the origin, confirming Example 6.4 visually rather than just algebraically.

6.7 What Changes for Non-Tight Frames

Everything in Sections 6.3–6.4 used the triangular frame, which is tight. It is worth checking which parts of the picture survive for a non-tight frame, using $\{h_1, h_2, h_3\}$ ($A = 1$, $B = 3$, Example 3.3) as our test case.

Proposition 6.7.1 (The paraboloid survives; the base ellipse does not)**.** *For* any *frame* $\{f_i\}_{i=1}^m$ *with* $m = n + 1$ *(one redundant vector), parametrize duals by* $G(w) = \tilde{F} + wk^T$ *for the unit vector* k *spanning* $\ker F$*. Then:*

(i) The energy function is still an exact paraboloid, $\|G(w)\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|w\|^2$*, regardless of whether the original frame is tight.*

(ii) The error covariance at $w = 0$ *(the canonical dual) is* S^{-1}*, which is a circle if and only if the original frame is tight.*

Proof. Part (i) is exactly the proof of Theorem 6.3.1, which used only that k is a unit vector spanning the (here, one-dimensional) kernel — tightness of $\{f_i\}$ was never invoked. Part (ii) is Proposition 6.4.2: S^{-1} has equal eigenvalues iff S does, iff $\{f_i\}$ is tight (Theorem 3.5.1(iii)). $\square$

Key Idea

The paraboloid picture (which member of the dual family is "smallest") and the ellipse picture (how the error is distributed directionally) are answering two different questions, and tightness only controls the second one. No matter how unbalanced the original frame is, the canonical dual is still the unique bottom of a perfectly circular paraboloid in w-space — minimality of total energy is a *universal* fact about every frame. But the resulting error ellipse, even at the very bottom of that paraboloid, inherits whatever anisotropy the original frame had. **Choosing the canonical dual fixes "how much" freedom you waste; it cannot fix "which directions" the original frame under- or over-samples.** Only redesigning the frame itself (Chapter 4's tightness machinery) can fix that.

6.8 Chapter Summary

- For the smallest redundant case ($m = n+1$), the affine family of dual frames (Theorem 5.4.1) is parametrized by an ordinary 2-dimensional plane $w \in \mathbb{R}^2$ (Proposition 6.2.1), even though $\ker F$ itself is only 1-dimensional — the extra dimension comes from w ranging independently over each of the $n = 2$ rows.

- The total energy $\|G(w)\|_F^2$ is an **exact paraboloid** over this plane (Theorem 6.3.1), with its unique minimum at $w = 0$, the canonical dual. This is Theorem 5.5.1, seen as a picture rather than an algebraic identity.

- The reconstruction error under coefficient noise is not just a number but an **error ellipse**, with covariance $\sigma^2 GG^T$. At the canonical dual, this covariance equals $\sigma^2 S^{-1}$ exactly (Proposition 6.4.2) — a fact not visible from Chapter 5's total-energy bookkeeping alone.

- The error ellipse is a **circle exactly when the frame is tight**: tightness controls the *directional uniformity* of reconstruction error, while minimal-norm duality (any frame, tight or not) controls only its *total size*. Proposition 6.7.1 shows these are genuinely separate facts: the paraboloid picture is universal, but the base ellipse it sits on top of is shaped by the original frame's own tightness.

6.9 Lab Exercises

Lab Exercise 6.1: Reproducing the Paraboloid

Implement `G_of_w`, `energy`, and `error_covariance` from Section 6.6 for the triangular frame. Verify numerically, for at least 10 random values of $w \in \mathbb{R}^2$, that `energy(w)` matches the closed form $\frac{4}{3} + \|w\|^2$ to within 10^{-10}. Then find the value of w (other than the origin) for which the energy equals exactly $\frac{10}{3}$, and verify your answer numerically.

Lab Exercise 6.2: The Non-Tight Case

Repeat the construction of Section 6.6 for the frame $\{h_1, h_2, h_3\}$ $(A = 1, B = 3)$ instead of the triangular frame.

(a) Find the unit vector k spanning $\ker H$ (Lab Exercise 5.1 from Chapter 5 may already have this).

(b) Verify that the energy function is *still* an exact paraboloid $\left\|\tilde{H}\right\|_F^2 + \|w\|^2$ in this new w, as guaranteed by Proposition 6.7.1(i).

(c) Plot the error ellipse at $w = 0$ and confirm it is the non-circular ellipse with axis ratio $3 : 1$ found in Example 6.4.

(d) Numerically search the parameter plane (e.g. a grid search, or `scipy.optimize.minimize` on the squared eigenvalue gap) for a value of $w \neq 0$ at which the error ellipse becomes exactly circular. You should find one. Report the energy $\|G(w)\|_F^2$ at that point and compare it to the canonical dual's minimal energy $\frac{4}{3}$ — is circularity "free," or does it cost extra total energy compared to the canonical choice?

(e) Having found w numerically in part (d), guess its exact closed form (it has a clean expression involving small integers and a square root), and verify algebraically by hand that $G(w)G(w)^T = I$ exactly at your guessed value. This shows that **minimal total energy and perfectly isotropic error are generally different points on the same paraboloid** — the canonical dual is optimal for one criterion but not, in general, for the other.

Lab Exercise 6.3: A Larger Parameter Space

The clean 2-dimensional picture in this chapter relied on $m = n + 1$ (so $\dim \ker F = 1$). For $m = n + 2$ (two redundant vectors), the parameter space of Z becomes $n \cdot 2$-dimensional — 4-dimensional in $\mathbb{R}^2$ — too large to plot directly. However, you can still plot *slices*. Using four vectors in $\mathbb{R}^2$ (e.g. the square vertices of Chapter 4), find a 2-dimensional basis $\{k_1, k_2\}$ of $\ker F \subset \mathbb{R}^4$, parametrize $Z = w_1 e_1 k_1^T + w_2 e_1 k_2^T$ (varying only the *first* row of Z, freezing the second row at zero), and plot the resulting 1-row-restricted energy function over $(w_1, w_2) \in \mathbb{R}^2$. Is it still an exact paraboloid? (Hint: think about whether k_1 and k_2 need to be orthogonal for the cross terms to vanish, and check this for your chosen basis.)

Lab Exercise 6.4 (Optional, Challenge): Animating the Family

Using `matplotlib.animation`, create a short animation that sweeps w around a circle of radius 1.5 in the parameter plane (i.e. $w(t) = 1.5(\cos t, \sin t)$ for $t \in [0, 2\pi]$) for the triangular frame, and shows the corresponding error ellipse rotating and breathing in a second panel, synchronized frame-by-frame. Confirm that the energy $\|G(w(t))\|_F^2$ stays exactly constant throughout the sweep (since $\|w(t)\| = 1.5$ is constant) even though the error ellipse's *orientation* changes continuously — a clean illustration that total energy and error-ellipse orientation are independent pieces of information, even when one of them (total energy) is held fixed.

6.10 Supplementary Exercises

Theoretical Exercises

T1. For the $m = n+1$ case with unit vector k spanning $\ker F$, prove the explicit formula $\Sigma_{G(w)} = \sigma^2\big(S^{-1} + ww^T\big)$ directly from $G(w) = \tilde{F} + wk^T$ and the fact $\tilde{F}k = 0$.

T2. Using T1, re-derive the trace (total energy) formula of Theorem 6.3.1 in one line: $\mathrm{tr}(\Sigma_{G(w)}) = \sigma^2\big(\mathrm{tr}(S^{-1}) + \|w\|^2\big)$.

T3. Using the matrix determinant lemma $\det(A+uv^T) = \det(A)(1+v^TA^{-1}u)$, prove that $\det(\Sigma_{G(w)})$ is also uniquely minimized at $w = 0$ — i.e. the canonical dual minimizes not only total error variance (trace) but also the error ellipse's area (determinant).

T4. Generalize T1 to $\dim\ker F = r > 1$: if K is an $m \times r$ matrix with orthonormal columns spanning $\ker F$ and $Z = WK^T$ for $W \in \mathbb{R}^{n\times r}$, prove $\Sigma_{G(W)} = \sigma^2\big(S^{-1} + WW^T\big)$.

T5. Using T1 and the fact that ww^T is positive semidefinite, prove that every eigenvalue of $\Sigma_{G(w)}$ is at least the corresponding eigenvalue of $\sigma^2 S^{-1}$. Conclude that moving away from the canonical dual can only inflate the error ellipse's axes, never shrink any of them.

Computational Exercises

C1. Verify the formula from T1 numerically for $\{h_1, h_2, h_3\}$ at five random values of w, confirming $\Sigma_{G(w)} = S^{-1} + ww^T$ (with $\sigma = 1$) to machine precision.

C2. Using a grid search or `scipy.optimize.minimize`, confirm that $\det(\Sigma_{G(w)})$ is minimized at $w = 0$ for $\{h_1, h_2, h_3\}$, matching the trace-minimizing point found in Chapter 6's own worked example.

C3. For the square frame (Lab Exercise 6.3, $\dim\ker F = 2$), verify the generalized formula from T4 numerically: build K from `null_space`, pick a random 2×2 matrix W, and confirm $\Sigma_{G(W)} = S^{-1} + WW^T$.

C4. For $\{h_1, h_2, h_3\}$, compute the eigenvalues of $\Sigma_{G(w)}$ for $\|w\| = 0, 0.5, 1, 2, 5$ (any fixed direction), and confirm both eigenvalues are non-decreasing in $\|w\|$, illustrating T5.

Chapter 7

Midterm Review

7.1 How to Use This Chapter

Chapters 1–6 built the entire foundational apparatus of finite frame theory: the definition of a frame, the frame operator and its eigenvalues, tight frames, dual frames, and the geometric picture underlying duality. This chapter does not introduce new theory. Instead, it reorganizes everything you have learned around five central questions, gives you a single reference page for every major definition and theorem, and then provides a substantial bank of practice problems — a mix of proof-style and computational questions, in the spirit of the midterm itself.

The material is reorganized *thematically* rather than chapter-by-chapter, because the five central questions below cut across the original chapter boundaries, and reviewing this way is a better test of whether you have internalized the ideas rather than just the order in which they were presented.

Key Idea

The five central questions of Chapters 1–6:

(1) *When is a set of vectors a frame?* (Spanning is sufficient and necessary in finite dimensions.)

(2) *How do we compute its bounds exactly?* (Eigenvalues of the frame operator $S = FF^*$.)

(3) *When is a frame tight, and why does it matter?* ($S = AI$; resolution of the identity; roots-of-unity symmetry as one mechanism among several.)

(4) *How do we reconstruct a signal from frame coefficients, and is the reconstruction unique?* (Always possible via the canonical dual; unique only when the frame is a basis; otherwise an entire affine family of valid duals exists, and the canonical one is optimal in a precise sense.)

(5) *What does that family of duals actually look like?* (For the smallest redundant case, an ordinary plane with an exact paraboloid of energy over it; the canonical dual sits at the bottom, and the shape of the resulting reconstruction-error ellipse is governed separately by the frame's own tightness.)

7.2 Question 1: When Is a Set of Vectors a Frame?

Definition 7.2.1 (Finite frame — Chapter 2)**.** $\{f_i\}_{i=1}^m \subset \mathbb{F}^n$ is a frame for $\mathbb{F}^n$ if there exist $0 < A \le B < \infty$ with

$$A\|x\|^2 \le \sum_{i=1}^m |\langle x, f_i\rangle|^2 \le B\|x\|^2 \qquad \forall x \in \mathbb{F}^n.$$

Theorem 7.2.2 (Spanning $\Leftrightarrow$ frame — Chapter 2, Theorem 2.2.1)**.** *A finite set $\{f_i\}_{i=1}^m \subset \mathbb{F}^n$ is a frame for $\mathbb{F}^n$ if and only if it spans $\mathbb{F}^n$.*

Exam Tip

This is the single fact that makes finite frame theory tractable: existence of frame bounds is *automatic* once you check spanning (equivalently, $\operatorname{rank} F = n$). The upper bound always exists (Cauchy– Schwarz, $B = \sum_i \|f_i\|^2$ works); the lower bound is where spanning is essential (compactness argument: a continuous function on the unit sphere attains its minimum, and that minimum is forced positive exactly because no nonzero vector is orthogonal to a spanning set). **If an exam question asks "is this a frame," check the rank of the synthesis matrix — nothing else is needed.**

7.3 Question 2: How Do We Compute the Bounds Exactly?

Definition 7.3.1 (Synthesis, analysis, frame operators — Chapter 3)**.** $F : \mathbb{F}^m \to \mathbb{F}^n$, $Fc = \sum_i c_i f_i$ (matrix with columns f_i); $F^* : \mathbb{F}^n \to \mathbb{F}^m$, $F^*x = (\langle x, f_i\rangle)_i$ (conjugate transpose); $S := FF^* = \sum_i f_i f_i^*$, the *frame operator*.

Theorem 7.3.2 (Frame bounds are eigenvalues — Chapter 3, Theorem 3.5.1)**.** $A_{\text{opt}} = \lambda_{\min}(S)$, $B_{\text{opt}} = \lambda_{\max}(S)$.

Exam Tip

Computationally: `S = F @ F.T`, then `eigh(S)` gives both bounds in one call. **Always use `eigh`, never `eig`**, since S is symmetric by construction — `eigh` guarantees real, sorted eigenvalues and an orthonormal eigenbasis, while `eig` does not. The proof of Theorem 7.3.2 hinges on the *Rayleigh quotient*: $\langle Sx, x\rangle / \|x\|^2$ is a convex combination of the eigenvalues of S, hence trapped between $\lambda_{\min}$ and $\lambda_{\max}$, with equality exactly at the corresponding eigenvectors. This is a proof technique you should be able to reproduce from scratch, not just cite.

Proposition 7.3.3 (Trace formula — Chapter 3, Proposition 3.4.2(iv))**.** $\operatorname{tr}(S) = \sum_{i=1}^m \|f_i\|^2$. *For a tight, unit-norm frame with m vectors in $\mathbb{R}^n$: $A = m/n$.*

7.4 Question 3: When Is a Frame Tight, and Why?

Definition 7.4.1 (Tight / Parseval frame — Chapter 2)**.** Tight: $A = B$. Equivalently (Chapter 4): $S = AI$. Parseval: tight with $A = 1$.

Theorem 7.4.2 (Regular polygons — Chapter 4, Theorem 4.3.1)**.** *The vertices of a regular m-gon ($m \ge 3$) inscribed in the unit circle form a tight frame for $\mathbb{R}^2$ with $A = m/2$.*

Theorem 7.4.3 (Projection construction — Chapter 4, Theorem 4.4.1)**.** *Projecting any orthonormal basis of $\mathbb{F}^N$ onto a subspace V always produces a Parseval frame for V.*

Exam Tip

Two genuinely different *mechanisms* produce tight frames: (a) **symmetry** (roots of unity / cyclic group orbits force trigonometric cross-terms to cancel) and (b) **projection** (any orthonormal basis of a bigger space, projected down, is automatically Parseval, regardless of symmetry). Neither mechanism is the *definition* of tightness — tightness is fundamentally an energy-balance condition, $S = AI$, and these are just two reliable ways to engineer it. **Do not write "tight frames are symmetric" as if it were an if-and-only-if statement** — Proposition 4.5.1 gives an explicit asymmetric, unequal-norm tight frame.

7.5 Question 4: Reconstruction and Duality

Definition 7.5.1 (Dual frame — Chapter 5)**.** $\{g_i\}$ is dual to $\{f_i\}$ if $x = \sum_i \langle x, f_i\rangle g_i$ for all x; equivalently $GF^* = I_n$. The relationship is mutual.

Proposition 7.5.2 (Canonical dual — Chapter 5, Proposition 5.3.1)**.** *$\tilde{f}_i := S^{-1} f_i$ is always a valid dual, with $\tilde{S} = S^{-1}$ and bounds $\tilde{A} = 1/B$, $\tilde{B} = 1/A$.*

Theorem 7.5.3 (All duals — Chapter 5, Theorem 5.4.1)**.** *Every valid dual synthesis matrix is $G = \tilde{F} + Z$ with $ZF^* = 0$ (rows of Z in $\ker F$). Unique iff $m = n$ (basis case).*

Theorem 7.5.4 (Minimal norm — Chapter 5, Theorem 5.5.1)**.** *$\tilde{F}$ uniquely minimizes $\|G\|_F^2 = \sum_i \|g_i\|^2$ among all valid duals, via the Pythagorean decomposition $\|G\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|Z\|_F^2$. Consequently $\tilde{F}$ also minimizes expected reconstruction error under uncorrelated coefficient noise.*

Exam Tip

A commonly confused dimension-counting point, worth getting exactly right: for a frame with m vectors in $\mathbb{F}^n$ ($m > n$), the space of valid perturbations Z (with $ZF^* = 0$) has real dimension $n \cdot (m - n)$ — *not* $m - n$. The kernel $\ker F \subset \mathbb{F}^m$ itself has dimension $m - n$, but each of the n *rows* of Z ranges independently over that kernel, so the total freedom is n copies of it. For the triangular frame ($m = 3$, $n = 2$), $\dim \ker F = 1$ but the dual family is $2 \cdot 1 = 2$-dimensional — this is exactly the fact that made the parameter plane of Chapter 6 a genuine *plane* rather than a line. **Do not say "the dual family has dimension $m - n$"** without the factor of n; this is a frequent and easy mistake.

7.6 Question 5: What Does the Dual Family Look Like?

Proposition 7.6.1 (The dual family as a plane — Chapter 6, Proposition 6.2.1)**.** *For a frame with $m = n + 1$ vectors, the affine family of duals is parametrized by $G(w) = \tilde{F} + wk^T$, $w \in \mathbb{R}^n$, where k is a unit vector spanning the (one-dimensional) kernel of F.*

Theorem 7.6.2 (Energy paraboloid — Chapter 6, Theorem 6.3.1)**.** $\|G(w)\|_F^2 = \left\|\tilde{F}\right\|_F^2 + \|w\|^2$*: an exact paraboloid over the parameter plane, with unique minimum at $w = 0$ (the canonical dual). This holds whether or not the original frame is tight.*

Proposition 7.6.3 (Error covariance — Chapter 6, Proposition 6.4.2)**.** *The canonical dual's reconstruction-error covariance under coefficient noise of variance σ^2 is $\Sigma_{\tilde{F}} = \sigma^2 S^{-1}$ exactly. Its level sets are circular if and only if the original frame is tight.*

Exam Tip

Two different optimization criteria, easy to conflate: the canonical dual minimizes *total* energy $\|G\|_F^2$ over the *entire* affine family — always, for any frame, tight or not. But minimizing total energy is not the same as making the reconstruction error *isotropic* (equal in every direction): the shape of the error ellipse at the canonical dual is inherited from S^{-1}, and is only circular if the frame itself was tight to begin with. Chapter 6's Lab Exercise 6.2 showed explicitly that for a non-tight frame, there can exist a *different* point on the same paraboloid where the error ellipse becomes circular — at the cost of *more* total energy than the canonical choice. **Minimal total error and balanced directional error are, in general, different points on the paraboloid; the canonical dual only optimizes the former.**

7.7 Worked Review Problems

These problems are worked in full, modeling the level of detail expected on the midterm itself.

Review Problem 1: Is it a frame? Find the bounds.

Let $u_1 = (1,1)^T$, $u_2 = (1,-1)^T$, $u_3 = (2,0)^T$ in $\mathbb{R}^2$. **(a)** Is $\{u_1, u_2, u_3\}$ a frame for $\mathbb{R}^2$? **(b)** If so, find the exact frame bounds A, B. **(c)** Is it tight?

Solution. (a) Since u_1, u_2 alone are linearly independent (not parallel), they already span $\mathbb{R}^2$; adding u_3 keeps the spanning property. By Theorem 7.2.2, $\{u_1, u_2, u_3\}$ is a frame for $\mathbb{R}^2$.

(b) The synthesis matrix is $F = \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \end{pmatrix}$. The frame operator is

$$S = FF^T = \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 1+1+4 & 1-1+0 \\ 1-1+0 & 1+1+0 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}.$$

This is already diagonal, so the eigenvalues are read off directly: $\lambda_1 = 6$, $\lambda_2 = 2$. By Theorem 7.3.2, $B = 6$ and $A = 2$.

(c) Since $A \neq B$, the frame is *not* tight.

Check: the trace formula (Proposition 7.3.3) gives $\operatorname{tr}(S) = \|u_1\|^2 + \|u_2\|^2 + \|u_3\|^2 = 2+2+4 = 8$, and indeed $6 + 2 = 8$, confirming the computation.

Review Problem 2: Canonical dual and reconstruction

Using $\{u_1, u_2, u_3\}$ from Review Problem 1, find the canonical dual frame, and reconstruct $x = (4, 2)^T$ from its frame coefficients.

Solution. Since $S = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$ is diagonal, $S^{-1} = \begin{pmatrix} 1/6 & 0 \\ 0 & 1/2 \end{pmatrix}$ directly. The canonical dual vectors are

$$\tilde{u}_1 = S^{-1}u_1 = (\tfrac{1}{6}, \tfrac{1}{2})^T, \quad \tilde{u}_2 = S^{-1}u_2 = (\tfrac{1}{6}, -\tfrac{1}{2})^T, \quad \tilde{u}_3 = S^{-1}u_3 = (\tfrac{1}{3}, 0)^T.$$

The frame coefficients of $x = (4, 2)^T$ are

$$c_1 = \langle x, u_1 \rangle = 4 + 2 = 6, \quad c_2 = \langle x, u_2 \rangle = 4 - 2 = 2, \quad c_3 = \langle x, u_3 \rangle = 8.$$

Reconstruction:

$$\hat{x} = \sum_i c_i \tilde{u}_i = 6(\tfrac{1}{6}, \tfrac{1}{2})^T + 2(\tfrac{1}{6}, -\tfrac{1}{2})^T + 8(\tfrac{1}{3}, 0)^T = (1, 3)^T + (\tfrac{1}{3}, -1)^T + (\tfrac{8}{3}, 0)^T = (4, 2)^T = x.$$

The reconstruction is exact, as guaranteed by Proposition 7.5.2.

Review Problem 3: A tightness proof

Let $v_1 = (1, 0)^T$, $v_2 = (0, 1)^T$, $v_3 = (-1, 0)^T$, $v_4 = (0, -1)^T$ in $\mathbb{R}^2$. Prove $\{v_i\}_{i=1}^4$ is a tight frame and find its bound.

Solution. This is the $m = 4$ regular polygon case of Theorem 7.4.2 (vertices at angles $0, \pi/2, \pi, 3\pi/2$), so it is automatically tight with $A = m/2 = 2$. As a direct check via the resolution-of-the-identity characterization of tightness (Chapter 4, Proposition 4.2.1): $S = \sum_i v_i v_i^T = 2e_1e_1^T + 2e_2e_2^T = 2I$, confirming $A = B = 2$ without needing the general theorem at all — a useful sanity check technique when the configuration is simple enough to verify directly by hand.

Review Problem 4: The dual family, dimension and minimum

For the frame $\{u_1, u_2, u_3\}$ of Review Problem 1, (a) find a vector spanning $\ker F$, (b) state the dimension of the full affine family of dual frames, and (c) confirm that the canonical dual found in Review Problem 2 minimizes $\|G\|_F^2$ over that family.

Solution. (a) We need $c = (c_1, c_2, c_3)$ with $c_1u_1 + c_2u_2 + c_3u_3 = 0$, i.e. $(c_1 + c_2 + 2c_3,\ c_1 - c_2) = (0, 0)$. From the second equation $c_1 = c_2$; substituting into the first, $2c_1 + 2c_3 = 0$, so $c_3 = -c_1$. Taking $c_1 = 1$: $\ker F = \operatorname{span}\{(1, 1, -1)^T\}$.
(b) By the dimension-counting fact above (Section 7.5), with $n = 2$ and $\dim \ker F = 1$, the affine family of duals has dimension $n \cdot 1 = 2$ — a plane, exactly as in Chapter 6.
(c) By Theorem 7.6.2 (which did not depend on the specific frame, only on $m = n + 1$), the energy function over this plane is $\left\|\tilde{F}\right\|_F^2 + \|w\|^2$, minimized uniquely at $w = 0$, which by construction is the canonical dual of Review Problem 2. No further computation is needed: minimality is guaranteed by the general theorem.

7.8 Practice Problem Set

The following problems are unworked. They mirror the format of the midterm: a mix of short computational problems and proof-style questions. Problems marked **(C)** are intended as the take-home computational component; all others are intended to be solvable by hand within exam time constraints.

Problem 1

Let $w_1 = (3,4)^T$, $w_2 = (4,-3)^T$ in $\mathbb{R}^2$ (note: $\|w_1\| = \|w_2\| = 5$ and $\langle w_1, w_2 \rangle = 0$). Without computing S explicitly, determine whether $\{w_1, w_2\}$ is tight, and if so, find A. (Hint: what kind of configuration is this?)

Problem 2

Let $\{f_i\}_{i=1}^m$ be *any* frame for $\mathbb{R}^n$ with frame bounds A, B. Prove that $\{cf_i\}_{i=1}^m$ (every vector scaled by a fixed nonzero constant $c \in \mathbb{R}$) is also a frame for $\mathbb{R}^n$, and find its frame bounds in terms of A, B, and c.

Problem 3

Let $\{f_i\}_{i=1}^m$ be a tight frame for $\mathbb{R}^n$ with bound A, and suppose additionally that every f_i has the same norm $\|f_i\| = r$ for all i. Express A in terms of m, n, and r only. (This generalizes the unit-norm case, Proposition 7.3.3, to arbitrary common norm r.)

Problem 4

True or false, with justification (a one-line counterexample suffices for any false statement):

(a) Every frame with $m = n$ vectors in $\mathbb{F}^n$ is an orthonormal basis.

(b) If $\{f_i\}_{i=1}^m$ is a tight frame for $\mathbb{R}^n$, then so is every subset $\{f_i\}_{i \in J}$ for $J \subsetneq \{1, \ldots, m\}$.

(c) The canonical dual of a Parseval frame is the frame itself.

(d) A frame operator S always has n distinct eigenvalues.

(e) If $A = B$ for a frame, the frame vectors must all have the same norm.

(f) The dimension of the affine family of dual frames for m vectors in $\mathbb{R}^n$ ($m > n$) equals $m - n$.

Problem 5

Let $\{f_i\}_{i=1}^m$ be a frame for $\mathbb{R}^n$ with frame operator S, and let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an invertible linear map. Show that $\{Tf_i\}_{i=1}^m$ is also a frame for $\mathbb{R}^n$, and express its frame operator in terms of T and S. Under what condition on T is the new frame tight whenever the original one is?

Problem 6

Let $\{f_1, f_2, f_3\}$ be the triangular frame of Chapters 1/3/6 ($A = B = \frac{3}{2}$). Using the parametrization $G(w) = \tilde{F} + wk^T$ from Chapter 6 with $k = (1, 1, 1)^T/\sqrt{3}$, find the value of $w \in \mathbb{R}^2$ for which $\|G(w)\|_F^2 = \frac{13}{3}$. (There are two such values by symmetry; finding either is sufficient.) Verify your answer using Theorem 7.6.2.

Problem 7 (C)

Computational. Generate 20 random unit vectors in $\mathbb{R}^4$ (`rng.standard_normal((4,20))`, columns normalized). Compute the exact frame bounds A, B via the frame operator's eigenvalues. Then use gradient information (or simple coordinate-wise perturbation) to adjust the vectors slightly to reduce the ratio B/A — you do not need a sophisticated optimizer; even a simple loop that nudges each vector toward better balance and re-normalizes will do. Report your starting and ending B/A ratio. (This is a hands-on preview of the frame potential minimization in Chapter 9.)

Problem 8 (C)

Computational. For $m = 3, 4, 5, 6, 7, 8, 9, 10$, construct the regular m-gon frame in $\mathbb{R}^2$, compute its canonical dual, and verify numerically that the canonical dual is simply $\frac{1}{A}$ times the original frame in every case (as predicted for tight frames). Then, for each m, compute $\left\|\tilde{F}\right\|_F^2$ (the canonical dual's total energy) and verify it equals m/A for these unit-norm configurations (derive the predicted formula yourself before checking it numerically).

Problem 9

Prove: if $\{f_i\}_{i=1}^m$ is a frame for $\mathbb{R}^n$ with canonical dual $\{\tilde{f}_i\}$, then the canonical dual of $\{\tilde{f}_i\}$ is $\{f_i\}$ again (i.e. the canonical-dual operation is an involution). (Hint: use Proposition 7.5.2 twice, recalling that the frame operator of $\{\tilde{f}_i\}$ is S^{-1}.)

Problem 10

A frame $\{f_i\}_{i=1}^m$ for $\mathbb{R}^n$ is called *equal-norm* if $\|f_i\| = \|f_j\|$ for all i, j. Is every tight frame equal-norm? Is every equal-norm frame tight? Justify each answer with either a short proof or an explicit counterexample drawn from frames appearing somewhere in Chapters 1–6.

Problem 11 (C)

Computational. For the frame $\{h_1, h_2, h_3\}$ ($A = 1$, $B = 3$), use `scipy.optimize` to find the point w on the dual-family plane (parametrized as in Chapter 6, with the appropriate kernel vector for this frame) at which the reconstruction-error ellipse is exactly circular, and report both w and the resulting total energy $\|G(w)\|_F^2$. Compare this energy to the canonical dual's minimal energy $\frac{4}{3}$ found in Example 5.3. (This reproduces Chapter 6, Lab Exercise 6.2(d)–(e); if you completed that exercise already, this problem is a direct check of your earlier work.)

7.9 Self-Assessment Checklist

Before the midterm, you should be able to do each of the following without consulting these notes:

- ☐ State the frame definition and explain, in one sentence, why spanning alone suffices in finite dimensions.
- ☐ Given an explicit small set of vectors, determine by hand whether it spans, and if so compute the frame operator and its eigenvalues.
- ☐ State the relationship $S = FF^* = \sum_i f_i f_i^*$ and use both forms appropriately.
- ☐ Explain what it means for a frame to be tight, in three equivalent ways: (i) $A = B$, (ii) $S = AI$, (iii) every direction is "illuminated" equally.
- ☐ Prove that the vertices of a regular polygon form a tight frame, using either the trigonometric or roots-of-unity argument.
- ☐ Compute the canonical dual of a small explicit frame, by hand, via S^{-1}.
- ☐ State and use the characterization of all dual frames as $\tilde{F} + Z$ with $ZF^* = 0$, and correctly count the *dimension* of that family ($n \cdot (m - n)$, not $m - n$).
- ☐ Explain, in your own words, why the canonical dual is optimal in the minimal-norm / minimal-noise sense, and why this is *not* the same as minimizing the anisotropy of the error ellipse.
- ☐ Sketch, for the smallest redundant case $m = n+1$, the energy paraboloid over the dual-family plane and the resulting error ellipse at an arbitrary point w.
- ☐ Translate any of the above into working NumPy code without referring to previous listings.

Chapter 8

Frame Potential and Optimal Frame Design

8.1 Overview: Tightness as the Bottom of a Bowl

Chapter 4 made a promise it did not keep. While proving that regular polygons are tight frames, a remark observed that tight frames need not be regular polygons at all (Proposition 4.5.1), and offered a reinterpretation: tightness is better understood as an *energy-minimization* condition than as a symmetry condition, with the precise statement deferred to "Chapter 8." This chapter is that promise, kept.

We introduce a single scalar quantity, the *frame potential*, attached to any finite collection of vectors. It measures the total pairwise correlation in the collection — how much the vectors overlap with one another, summed over all pairs. The central theorem of this chapter shows that, among configurations of m unit-norm vectors in $\mathbb{R}^n$, the frame potential is minimized *exactly* by the tight frames, and by nothing else. Symmetry (the roots-of-unity mechanism of Chapter 4) and projection (the construction of Chapter 4, Section 4.4) are then revealed as two different *routes* to the same minimum, not as alternative definitions of tightness. We close by using the frame potential as an objective function for a numerical optimization algorithm: gradient descent that constructs tight frames from scratch, with no symmetry assumed in advance — the computational payoff promised back in the original course description.

8.2 The Frame Potential

Definition 8.2.1 (Frame potential)**.** For a finite collection of vectors $\{f_i\}_{i=1}^m \subset \mathbb{F}^n$, the *frame potential* is

$$\mathrm{FP}(\{f_i\}_{i=1}^m) \;:=\; \sum_{i=1}^m \sum_{j=1}^m |\langle f_i, f_j\rangle|^2.$$

The sum includes the diagonal terms $i = j$, where $|\langle f_i, f_i\rangle|^2 = \|f_i\|^4$, as well as every off-diagonal pair counted twice (once as (i,j) and once as (j,i), which contribute equally since $|\langle f_i, f_j\rangle| = |\langle f_j, f_i\rangle|$). The frame potential is large when the vectors point in similar directions (large pairwise inner products) and small when they are spread out (small pairwise inner products) — exactly the kind of "total correlation" or "total redundancy" energy that the phrase "energy-minimization condition," used informally in Chapter 4, was gesturing toward.

Proposition 8.2.2 (Frame potential is $\mathrm{tr}(S^2)$). *For any finite collection $\{f_i\}_{i=1}^m \subset \mathbb{F}^n$ with frame operator $S = \sum_i f_i f_i^*$,*

$$\mathrm{FP}(\{f_i\}_{i=1}^m) = \mathrm{tr}(S^2).$$

Proof. Write $S = \sum_i f_i f_i^*$. Then

$$S^2 = \Big(\sum_i f_i f_i^*\Big)\Big(\sum_j f_j f_j^*\Big) = \sum_{i,j} f_i(f_i^* f_j) f_j^* = \sum_{i,j} \langle f_j, f_i\rangle \; f_i f_j^*,$$

using $f_i^* f_j = \langle f_j, f_i\rangle$ (a scalar). Taking the trace and using linearity together with $\mathrm{tr}(f_i f_j^*) = \langle f_j, f_i\rangle$ (the trace of a rank-one outer product uv^* is $\langle v, u\rangle$... more directly here, $\mathrm{tr}(f_i f_j^*) = f_j^* f_i = \langle f_i, f_j\rangle$):

$$\mathrm{tr}(S^2) = \sum_{i,j} \langle f_j, f_i\rangle \; \mathrm{tr}(f_i f_j^*) = \sum_{i,j} \langle f_j, f_i\rangle \; \langle f_i, f_j\rangle = \sum_{i,j} |\langle f_i, f_j\rangle|^2,$$

using $\langle f_j, f_i\rangle \langle f_i, f_j\rangle = \langle f_i, f_j\rangle \overline{\langle f_i, f_j\rangle} = |\langle f_i, f_j\rangle|^2$. This is exactly $\mathrm{FP}(\{f_i\}_{i=1}^m)$. ∎

Key Idea

The frame potential is not really a new object: it is the trace of the *square* of the frame operator you already know everything about. Since S has eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$ (Chapter 3),

$$\mathrm{FP}(\{f_i\}_{i=1}^m) = \mathrm{tr}(S^2) = \sum_{k=1}^n \lambda_k^2.$$

This single formula is the engine behind everything in this chapter: minimizing frame potential is exactly minimizing the sum of squared eigenvalues of S, subject to whatever constraint the vectors satisfy (typically, fixed total norm).

Example 8.1: Frame potential of the triangular frame

For the triangular frame ($S = \frac{3}{2}I$, Example 3.2), $\mathrm{tr}(S^2) = \mathrm{tr}(\frac{9}{4}I) = \frac{9}{4} \cdot 2 = \frac{9}{2}$. Direct computation confirms this: each f_i has unit norm, contributing 1 to the diagonal three times ($i = j$), and each of the $3 \times 2 = 6$ ordered off-diagonal pairs has $\langle f_i, f_j\rangle = -\frac{1}{2}$ (computed in Chapter 2), contributing $(-\frac{1}{2})^2 = \frac{1}{4}$ each: total $\mathrm{FP} = 3(1) + 6(\frac{1}{4}) = 3 + \frac{3}{2} = \frac{9}{2}$, matching $\mathrm{tr}(S^2) = \frac{9}{2}$ exactly.

8.3 The Minimization Theorem

We now prove the theorem that makes the frame potential worth studying: among unit-norm configurations, it is minimized exactly by tight frames.

Theorem 8.3.1 (Tight frames minimize frame potential). *Fix $m \geq n \geq 1$. Among all collections $\{f_i\}_{i=1}^m \subset \mathbb{R}^n$ with $\|f_i\| = 1$ for every i,*

$$\mathrm{FP}(\{f_i\}_{i=1}^m) \;\geq\; \frac{m^2}{n},$$

with equality if and only if $\{f_i\}_{i=1}^m$ is a tight frame for $\mathbb{R}^n$ (necessarily with bound $A = m/n$).

Proof. By Proposition 8.2.2, $\mathrm{FP} = \sum_{k=1}^{n} \lambda_k^2$ where $\lambda_1, \ldots, \lambda_n \geq 0$ are the eigenvalues of S. By the trace formula (Proposition 3.4.2(iv)), $\sum_{k=1}^{n} \lambda_k = \mathrm{tr}(S) = \sum_{i=1}^{m} \|f_i\|^2 = m$, using the unit-norm hypothesis. Apply the Cauchy–Schwarz inequality (Theorem 1.2.3) to the vectors $\lambda = (\lambda_1, \ldots, \lambda_n)$ and $\mathbf{1} = (1, \ldots, 1)$ in $\mathbb{R}^n$:

$$\left(\sum_{k=1}^{n} \lambda_k \cdot 1\right)^2 \leq \left(\sum_{k=1}^{n} \lambda_k^2\right)\left(\sum_{k=1}^{n} 1^2\right) = n \sum_{k=1}^{n} \lambda_k^2.$$

The left side is m^2 (using $\sum_k \lambda_k = m$), so $m^2 \leq n \sum_k \lambda_k^2 = n \cdot \mathrm{FP}$, i.e. $\mathrm{FP} \geq m^2/n$.

Equality in Cauchy–Schwarz holds if and only if λ and $\mathbf{1}$ are parallel, i.e. $\lambda_1 = \cdots = \lambda_n$ (a positive scalar multiple of $\mathbf{1}$, since all $\lambda_k \geq 0$ and they cannot all be zero — their sum is $m \geq n \geq 1 > 0$). This is exactly the condition $S = \lambda I$ for $\lambda = m/n$ (using $\sum_k \lambda_k = n\lambda = m$), i.e. the frame is tight with bound $A = m/n$ (Theorem 3.5.1(iii), together with Proposition 4.2.1). ☐

Key Idea

This theorem is the rigorous version of the claim deferred from Chapter 4: **tight, unit-norm frames are exactly the global minimizers of the frame potential.** The proof is a single application of Cauchy–Schwarz to the eigenvalues of S — the same inequality from Chapter 1 that has appeared throughout this course, now doing work at the level of an optimization problem rather than a single pair of vectors. Symmetry (roots of unity) and projection (orthonormal-basis projection) are not competing definitions of tightness; they are simply two different constructions that happen to land exactly on this unique minimum value m^2/n, for different reasons in each case.

Example 8.2: Verifying the minimum across constructions

For $m = 3$, $n = 2$: the theorem predicts a minimum frame potential of $m^2/n = 9/2$. The triangular frame achieves exactly $9/2$ (Example 8.2), confirming the symmetry-based construction of Chapter 4 lands on the true minimum, not merely a local one. For $m = 4$, $n = 2$ (square vertices, Example 3.6): predicted minimum $16/2 = 8$; direct computation gives $\mathrm{tr}(S^2) = \mathrm{tr}((2I)^2) = \mathrm{tr}(4I) = 8$, again exact. Both constructions — entirely different in derivation, one trigonometric and one projective — hit the same theoretical floor.

8.4 Gradient Descent on the Frame Potential

Theorem 8.3.1 tells us tight frames are exactly the unit-norm configurations minimizing FP. This suggests a constructive algorithm: start from an arbitrary (typically random, non-tight) unit-norm configuration, and run gradient descent on FP to find a nearby minimum — which, by the theorem, must be a tight frame. This is a genuinely different way of producing tight frames than anything in Chapter 4: no symmetry, no projection, just numerical optimization.

Proposition 8.4.1 (Gradient of the frame potential)**.** *Regard* FP *as a function of the synthesis matrix* $V \in \mathbb{R}^{n\times m}$ *(columns are the vectors* f_i*), via* $\mathrm{FP}(V) = \mathrm{tr}\big((VV^T)^2\big)$*. Then*

$$\nabla_V\ \mathrm{FP}(V) = 4\,(VV^T)\,V = 4SV.$$

Proof. We compute the directional derivative of FP at V in an arbitrary direction $H \in \mathbb{R}^{n\times m}$, i.e. $\frac{d}{dt}\,\mathrm{FP}(V + tH)\big|_{t=0}$, and identify the gradient as the matrix G satisfying $\frac{d}{dt}\,\mathrm{FP}(V + tH)\big|_{t=0} =$

$\mathrm{tr}(G^T H)$ for every H (the Frobenius inner product $\langle A, B\rangle_F := \mathrm{tr}(A^T B)$ is the relevant inner product on the space of matrices).

Let $S(V) = VV^T$, so $\mathrm{FP}(V) = \mathrm{tr}(S(V)^2)$. First,

$$\frac{d}{dt}S(V+tH)\Big|_{t=0} = \frac{d}{dt}(V+tH)(V+tH)^T\Big|_{t=0} = HV^T + VH^T =: dS.$$

Since $\mathrm{tr}(\cdot^2)$ has derivative $A \mapsto 2\,\mathrm{tr}(S\,dA)$ at a symmetric matrix S in direction dA (using $\mathrm{tr}(S\,dA) = \mathrm{tr}(dA\,S)$, the cyclic property of trace), we get

$$\frac{d}{dt}\,\mathrm{FP}(V+tH)\Big|_{t=0} = 2\,\mathrm{tr}\big(S\,(HV^T + VH^T)\big) = 2\,\mathrm{tr}(SHV^T) + 2\,\mathrm{tr}(SVH^T).$$

Both terms reduce to the same expression: using the cyclic property of trace, $\mathrm{tr}(SHV^T) = \mathrm{tr}(V^T SH) = \mathrm{tr}\big((SV)^T H\big)$ (the last equality using $S^T = S$), and likewise $\mathrm{tr}(SVH^T) = \mathrm{tr}(H^T SV) = \mathrm{tr}\big((SV)^T H\big)$. So

$$\frac{d}{dt}\,\mathrm{FP}(V+tH)\Big|_{t=0} = 4\,\mathrm{tr}\big((SV)^T H\big).$$

Comparing to $\mathrm{tr}(G^T H)$, we read off $G = SV$, i.e. $\nabla_V\,\mathrm{FP}(V) = 4SV$. □

Key Idea

The gradient descent update, with a small step size $\eta > 0$, is

$$V \;\longleftarrow\; V - \eta \cdot 4SV = V - 4\eta\,SV,$$

followed by renormalizing each column back to unit length (a *projection* step, since unconstrained gradient descent would not respect the unit-norm constraint). This combination — gradient step, then renormalize — is an instance of *projected gradient descent* on the unit sphere, called a *retraction* in the optimization literature; we use the simplest possible retraction (direct renormalization) and refer the curious reader to the Riemannian optimization literature on the sphere and Stiefel manifold for more sophisticated alternatives.

8.5 NumPy Implementation

Listing 8.1: Constructing a tight frame from scratch via gradient descent on the frame potential

```python
import numpy as np

def frame_potential(V):
    """FP(V) = tr((V V^T)^2) for V of shape (n, m)."""
    S = V @ V.T
    return np.trace(S @ S)

def fp_gradient(V):
    """Gradient of FP with respect to V: 4 S V."""
    S = V @ V.T
    return 4 * S @ V

def normalize_columns(V):
    """Project each column back onto the unit sphere."""
```

```python
    norms = np.linalg.norm(V, axis=0)
    return V / norms

def fit_tight_frame(n, m, num_steps=500, lr=0.05, seed=0, verbose=False):
    """
    Construct an (approximately) tight, unit-norm frame of m vectors
    in R^n via projected gradient descent on the frame potential.
    """
    rng = np.random.default_rng(seed)
    V = rng.standard_normal((n, m))
    V = normalize_columns(V)

    history = [frame_potential(V)]
    for step in range(num_steps):
        grad = fp_gradient(V)
        V = V - lr * grad
        V = normalize_columns(V)
        history.append(frame_potential(V))
        if verbose and step % 100 == 0:
            print(f"step {step}: FP = {history[-1]:.6f}")

    return V, history

# --- Construct a tight frame: m=3 vectors in R^2 ---
V, history = fit_tight_frame(n=2, m=3, num_steps=500, lr=0.05, seed=1)
predicted_minimum = 3**2 / 2
print(f"Final frame potential: {history[-1]:.6f}  "
      f"(predicted minimum m^2/n = {predicted_minimum})")

S_final = V @ V.T
eigenvalues = np.linalg.eigvalsh(S_final)
print(f"Final frame operator eigenvalues: {eigenvalues.round(6)}")
print(f"Tight (eigenvalues equal)? "
      f"{np.isclose(eigenvalues[0], eigenvalues[-1])}")

# --- Larger example: m=7 vectors in R=3, no symmetry assumed ---
V7, history7 = fit_tight_frame(n=3, m=7, num_steps=1000, lr=0.02, seed=2)
predicted7 = 7**2 / 3
print(f"\nm=7, n=3: final FP = {history7[-1]:.6f} "
      f"(predicted m^2/n = {predicted7:.4f})")
eigs7 = np.linalg.eigvalsh(V7 @ V7.T)
print(f"Eigenvalues: {eigs7.round(6)}")
```

Remark. This algorithm makes no reference to symmetry, roots of unity, or projection from a larger space — it is a purely numerical search that happens to always find a tight frame, because Theorem 8.3.1 guarantees every global minimum of FP *is* one. Unlike the explicit, hand-verifiable constructions of Chapter 4, gradient descent gives no closed-form description of the resulting vectors — but it works for *any* $m \geq n$, including configurations (like $m = 7$, $n = 3$) with no convenient symmetric description at all.

8.6 Local Minima and the Role of Initialization

A natural question: does gradient descent on FP always converge to the global minimum m^2/n, or can it become stuck at a worse, *local* minimum? The frame potential is a quartic (degree-4) polynomial function of the entries of V, restricted to a product of spheres (one sphere per column) — there is no a priori guarantee of a single basin of attraction.

Example 8.3: A bad initialization

Consider $m = 4$, $n = 2$, initialized with all four vectors nearly equal: $f_i \approx (1, 0)$ for $i = 1, \ldots, 4$, perturbed by tiny random noise. Gradient descent from this start tends to push the vectors apart *slowly* at first (the gradient is small when vectors are nearly parallel, since $S \approx 4e_1 e_1^T$ already has one very large and one near-zero eigenvalue, and the dynamics initially preserve this near -degenerate structure), but in practice still converges to $\mathrm{FP} = 8$ (the true minimum) given enough iterations, for this particular (m, n). This is verified numerically in Lab Exercise 8.3 below; empirically, for m, n in the range relevant to this course, poor local minima are rare but not impossible, and slow convergence from near-degenerate starting configurations is the more common practical obstruction.

Key Idea

In practice: run gradient descent from **several random initializations** and keep the result with the lowest final FP. This is standard practice for any non-convex optimization problem, and the frame potential, despite the elegant global theorem proved in Section 8.3, is no exception — the theorem guarantees *what* the global minimum value is and *which* configurations achieve it, but says nothing about how easy that minimum is to find by local search from an arbitrary start.

8.7 Chapter Summary

- The **frame potential** $\mathrm{FP}(\{f_i\}) = \sum_{i,j} |\langle f_i, f_j \rangle|^2$ measures total pairwise correlation in a vector configuration, and equals $\mathrm{tr}(S^2) = \sum_k \lambda_k^2$ (Proposition 8.2.2) — the sum of squared eigenvalues of the frame operator.
- **Theorem 8.3.1**: among unit-norm configurations of m vectors in $\mathbb{R}^n$, $\mathrm{FP} \geq m^2/n$, with equality if and only if the configuration is a tight frame (bound $A = m/n$). The proof is a single Cauchy–Schwarz application to the eigenvalues of S against the all-ones vector.
- This makes precise the claim deferred from Chapter 4: tightness *is* energy-minimization, and symmetry-based or projection-based constructions are simply two routes that happen to reach the same global minimum.
- **Gradient descent** on FP, with gradient $4SV$ (Proposition 8.4.1) and a unit-norm renormalization (retraction) after each step, constructs tight frames numerically with no symmetry assumed — working for any $m \geq n$, including cases with no convenient closed-form description.
- The frame potential is non-convex on the sphere, so gradient descent can in principle find local minima; running several random initializations and keeping the best result is standard practice.

8.8 Lab Exercises

Lab Exercise 8.1: Verifying the Gradient Formula

Implement `frame_potential(V)` and `fp_gradient(V)` from Section 8.5. For a random V of shape $(3, 5)$, verify the gradient formula $\nabla \mathrm{FP} = 4SV$ against a finite-difference approximation (perturb each entry of V by $\pm 10^{-6}$ and estimate the corresponding partial derivative), confirming agreement to at least 4 decimal places.

Lab Exercise 8.2: Reproducing Known Tight Frames

Run `fit_tight_frame` for $(n, m) = (2, 3)$, $(2, 4)$, $(2, 5)$, $(2, 6)$, and confirm in each case that the final frame potential matches the predicted minimum m^2/n to at least 4 decimal places, and that the final frame operator is (numerically) a multiple of the identity. Compare the final vectors you obtain for $(n, m) = (2, 3)$ to the explicit triangular frame from Chapter 1 — are they the same configuration (up to rotation), or a different tight frame entirely? (Hint: compute the pairwise angles between your gradient-descent vectors and compare to 120°.)

Lab Exercise 8.3: Initialization Sensitivity

Reproduce Example 8.6: initialize 4 vectors in $\mathbb{R}^2$ clustered tightly around $(1, 0)$ (e.g. add Gaussian noise of standard deviation 0.01 to four copies of $(1, 0)$, then normalize), and run gradient descent. Plot the frame potential versus iteration number on a semi-log y-axis. Compare the convergence speed to a run started from a uniformly random initialization (as in Lab Exercise 8.2). Does the clustered start eventually reach the same minimum? How many more iterations does it need?

Lab Exercise 8.4: A Frame with No Symmetry

Run `fit_tight_frame` for $(n, m) = (3, 7)$ (seven vectors in $\mathbb{R}^3$ – a configuration with no convenient closed-form symmetric description, unlike every example in Chapter 4). Verify the resulting frame is tight with the predicted bound $A = 7/3$. Then compute all $\binom{7}{2} = 21$ pairwise angles between the resulting vectors (in degrees) and report their minimum, maximum, and mean. Are the vectors *equiangular* (all pairwise angles equal), or merely tight? (This question previews the distinction between tight frames and the stronger notion of equiangular tight frames, to be studied in Chapter 9.)

Lab Exercise 8.5 (Optional, Challenge): Multiple Restarts

Write a function `best_of_n_restarts(n, m, num_restarts=10, ...)` that runs `fit_tight_frame` from `num_restarts` independent random initializations and returns the configuration with the lowest final frame potential. For $(n, m) = (2, 4)$, compare the spread of final frame potentials across 20 independent single runs (no restart logic, just 20 separate calls to `fit_tight_frame` with different seeds) — do all 20 runs reach the true minimum 8, or do some get stuck above it? Report the fraction of runs (if any) that fail to reach the global minimum within 500 steps, and suggest (without necessarily implementing) one modification to the algorithm that might improve this fraction — e.g. a larger learning rate, a momentum term, or more steps.

8.9 Supplementary Exercises

Theoretical Exercises

T1. Using the fact that FF^* and F^*F share nonzero eigenvalues (Chapter 3 Supplementary Exercise T1), give an alternative derivation of $\mathrm{FP}(\{f_i\}) = \mathrm{tr}(S^2)$ by showing it also equals $\|G\|_F^2$, where $G = F^*F$ is the Gram matrix.

T2. Prove that frame potential is invariant under simultaneously applying an orthogonal transformation to every vector: $\mathrm{FP}(\{Uf_i\}) = \mathrm{FP}(\{f_i\})$ for any orthogonal U.

T3. Prove that scaling every vector by a common factor c scales the frame potential by c^4: $\mathrm{FP}(\{cf_i\}) = c^4\,\mathrm{FP}(\{f_i\})$.

T4. Show that the bound $\mathrm{FP} \geq m^2/n$ from Theorem 8.3.1 remains valid even when $m < n$ (fewer unit vectors than the ambient dimension), but prove that equality can *never* be achieved in that case. (Hint: equality forces all n eigenvalues of S to be equal; what happens if $\mathrm{rank}\, S < n$?)

T5. Let $\{f_i\}_{i=1}^m$ and $\{g_j\}_{j=1}^{m'}$ be two frames for $\mathbb{R}^n$ with frame operators S_1, S_2. Prove that the frame potential of the combined set $\{f_i\} \cup \{g_j\}$ is $\mathrm{FP}(\{f_i\}) + \mathrm{FP}(\{g_j\}) + 2\,\mathrm{tr}(S_1S_2)$, and conclude that frame potential is *not* additive under taking unions in general (contrast with Chapter 4 Supplementary Exercise T2, where *tightness* of a union behaves additively on the bound).

T6. Among all positive semidefinite $n \times n$ matrices S with fixed trace $\mathrm{tr}(S) = m$, prove that $\mathrm{tr}(S^2)$ is maximized (for fixed rank 1) when $S = m\,uu^T$ for a unit vector u, achieving $\mathrm{tr}(S^2) = m^2$. Relate this worst case to a frame configuration in which all m vectors coincide.

Computational Exercises

C1. For a random 5×2 unit-norm configuration ($m = 2 < n = 5$), compute FP and confirm it exceeds m^2/n; then compare to the orthogonal configuration (first two standard basis vectors), confirming the orthogonal case achieves a strictly smaller (but still $> m^2/n$) frame potential, illustrating T4.

C2. Verify T5's union formula numerically using the triangular frame and the square frame: compute FP of each separately, compute $\mathrm{tr}(S_1S_2)$, and confirm the union's frame potential matches $\mathrm{FP}_1 + \mathrm{FP}_2 + 2\,\mathrm{tr}(S_1S_2)$.

C3. Construct a degenerate configuration of 5 identical unit vectors in $\mathbb{R}^3$ and confirm $\mathrm{FP} = 25 = m^2$, matching T6's worst-case prediction; compare to the frame-potential-minimizing configuration found by `fit_tight_frame` for the same $(n, m) = (3, 5)$.

C4. Numerically confirm T2: rotate the triangular frame by several random angles and verify the frame potential is unchanged in every case.

Chapter 9

Equiangular Frames and Grassmannian Packings

9.1 Overview: Beyond Tight, Toward Optimal

Chapter 8 showed that tight, unit-norm frames are the global minimizers of the frame potential — the most "spread out" unit-norm configurations in the energy-minimization sense. Lab Exercise 8.4 asked a natural follow-up question: among those minimizers, are the resulting vectors *equiangular* (all pairwise inner product magnitudes equal), or merely tight? The answer, observed numerically for $m = 7$, $n = 3$, was: merely tight — gradient descent on the frame potential finds tight frames, but not necessarily equiangular ones, because equiangularity is a strictly stronger condition than tightness.

This chapter studies that stronger condition. An *equiangular tight frame* (ETF) is a unit-norm tight frame in which *every pair* of distinct frame vectors has exactly the same inner product magnitude. ETFs are the most "spread out" configurations in a second, independent sense: they minimize the maximum pairwise inner product over all unit-norm frames of the same size and ambient dimension. The precise lower bound on that maximum is the *Welch bound*, and ETFs are exactly the configurations that achieve it.

We also revisit the connection, promised in Chapter 4, between the roots-of-unity construction and a larger family called *harmonic frames*. The chapter closes with the *Grassmannian packing problem*: the question of how to arrange m one-dimensional subspaces in $\mathbb{R}^n$ (or $\mathbb{C}^n$) as spread out as possible, of which ETFs are one specific and very clean solution.

9.2 Coherence and Equiangular Frames

Definition 9.2.1 (Coherence)**.** Let $\{f_i\}_{i=1}^m \subset \mathbb{F}^n$ be a collection of unit-norm vectors. The *coherence* of the collection is

$$\mu\big(\{f_i\}\big) \;:=\; \max_{1 \le i < j \le m} |\,\langle f_i, f_j \rangle\,|.$$

The coherence is the worst-case pairwise inner product magnitude: it measures how "similar" the two most similar vectors in the collection are. Small coherence means no two vectors are nearly parallel — the configuration is well spread out in the sense of pairwise-distinguishability. A set of unit-norm vectors with $\mu = 0$ is an orthonormal set (every pair is orthogonal); increasing μ indicates growing redundancy, with $\mu = 1$ achieved when some pair of vectors is actually parallel (or antipodal, since $|\,\langle f_i, -f_i \rangle\,| = 1$).

Remark. Coherence is not the same as the frame's lower bound A or upper bound B: coherence is a property of *pairs* of vectors, while frame bounds are global properties of the entire configuration. We have been using frame bounds since Chapter 2; coherence is a finer, more local notion that appears naturally when asking how well a frame can distinguish nearly-parallel directions.

Definition 9.2.2 (Equiangular frame)**.** A collection $\{f_i\}_{i=1}^m$ of unit-norm vectors in $\mathbb{F}^n$ is *equiangular* if there is a constant $c \geq 0$ such that $|\langle f_i, f_j\rangle| = c$ for every $i \neq j$. Equivalently, all off-diagonal entries of the Gram matrix $G_{ij} = \langle f_j, f_i\rangle$ have the same modulus. When the collection is also a tight frame, it is called an *equiangular tight frame* (ETF).

Example 9.1: The Mercedes-Benz frame

The three vectors $f_1 = (1, 0)$, $f_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, $f_3 = (-\frac{1}{2}, -\frac{\sqrt{3}}{2})$ in $\mathbb{R}^2$ — the triangular frame studied since Chapter 1 — are the canonical example of an ETF. They satisfy $|\langle f_i, f_j\rangle| = \frac{1}{2}$ for every $i \neq j$ (all off-diagonal inner products equal $-\frac{1}{2}$, which in the real case gives modulus $\frac{1}{2}$), and are tight with $A = \frac{3}{2}$ (Example 3.2). This frame is sometimes called the *Mercedes-Benz frame* after the visual resemblance of its three vectors to the three spokes of the well-known logo.

Remark. Not every tight frame is equiangular. The square frame ($m = 4$, $n = 2$, the vertices of a square) is tight with $A = 2$ (Chapter 4, Example 3.6) but is manifestly *not* equiangular: adjacent vertices have inner product 0, while antipodal vertices have inner product $|\langle f_1, f_3\rangle| = |(1, 0) \cdot (-1, 0)| = 1$. The off-diagonal inner products take two distinct values, 0 and 1, so $\mu = 1$ — the worst possible coherence for a unit-norm set. Any regular m-gon with $m \geq 4$ has this deficiency: if m is even, antipodal pairs give coherence 1; if m is odd with $m \geq 5$, at least two distinct pairwise angles appear (as verified numerically in Lab Exercise 9.1). Among regular polygons in $\mathbb{R}^2$, *only the triangle* ($m = 3$) is equiangular, making the Mercedes-Benz frame the unique real ETF arising from the regular-polygon family.

9.3 The Welch Bound

Why does it matter whether the coherence equals some specific value? The *Welch bound* provides a precise, universal lower limit on the coherence of any unit-norm collection, expressed in terms of m and n alone. It says you cannot spread m unit-norm vectors in $\mathbb{F}^n$ with coherence below a certain threshold — no matter how hard you try.

Theorem 9.3.1 (Welch bound)**.** *For any collection $\{f_i\}_{i=1}^m$ of unit-norm vectors in $\mathbb{F}^n$ with $m > n$,*

$$\mu\big(\{f_i\}\big) \;\geq\; \sqrt{\frac{m-n}{n(m-1)}},$$

with equality if and only if $\{f_i\}_{i=1}^m$ is an equiangular tight frame.

Proof. Let G be the $m \times m$ Gram matrix with entries $G_{ij} = \langle f_j, f_i\rangle$. Since all f_i are unit-norm, the diagonal entries are $G_{ii} = 1$, so $\mathrm{tr}(G) = m$. The off-diagonal entries satisfy $|G_{ij}| \leq \mu$ for $i \neq j$.

Step 1: lower-bound $\mathrm{tr}(G^2)$**.** Since $G = F^*F$ and $S = FF^*$ are related by $G = (F^*)(F)$ and $S = (F)(F^*)$, they share the same nonzero eigenvalues (a standard fact: if $ABv = \lambda v$ then $BA(Bv) = \lambda(Bv)$, so nonzero eigenvalues of AB and BA coincide). Therefore $\mathrm{tr}(G^2) = \sum_k \sigma_k^2$ where σ_k are the eigenvalues of G, which equal those of S for the n nonzero ones (and 0 for the remaining $m-n$). In particular, $\operatorname{tr}(G^2) = \operatorname{tr}(S^2) = \mathrm{FP}(\{f_i\})$. By Theorem 8.3.1 (frame potential lower bound), $\mathrm{FP} \geq m^2/n$ for any unit-norm collection, giving

$$\operatorname{tr}(G^2) \;\geq\; \frac{m^2}{n}.$$

Step 2: upper-bound $\operatorname{tr}(G^2)$ using μ. Split the sum:

$$\operatorname{tr}(G^2) = \sum_{i,j} |G_{ij}|^2 = \underbrace{\sum_{i} |G_{ii}|^2}_{=m} + \underbrace{\sum_{i \neq j} |G_{ij}|^2}_{\leq m(m-1)\mu^2} .$$

So $\operatorname{tr}(G^2) \leq m + m(m-1)\mu^2$.

Step 3: combine. From Steps 1 and 2:

$$\frac{m^2}{n} \;\leq\; \operatorname{tr}(G^2) \;\leq\; m + m(m-1)\mu^2.$$

Rearranging:

$$m(m-1)\mu^2 \;\geq\; \frac{m^2}{n} - m = m\Big(\frac{m}{n} - 1\Big) = \frac{m(m-n)}{n},$$

so $\mu^2 \geq (m-n)/(n(m-1))$, which gives the Welch bound.

Equality. The two Cauchy–Schwarz applications give equality simultaneously iff: (a) all eigenvalues of S are equal, i.e. $S = \lambda I$ (tight frame, by Theorem 3.5.1(iii)); and (b) all off-diagonal $|G_{ij}|$ are equal to μ (equiangular). Together, these say $\{f_i\}$ is an ETF. $\square$

Key Idea

ETFs are exactly the configurations achieving the Welch bound — the configurations of unit-norm vectors whose worst-case pairwise inner product is as small as it can possibly be, given m and n. The Welch bound is a hard floor: no unit-norm configuration of m vectors in $\mathbb{F}^n$ can have coherence below $\sqrt{(m-n)/(n(m-1))}$. ETFs sit on that floor. Tight frames that are not ETFs (e.g. regular polygons with $m \geq 4$ in $\mathbb{R}^2$) sit strictly above it.

Example 9.2: Welch bound for the Mercedes-Benz frame

For $m = 3$, $n = 2$: Welch bound $= \sqrt{(3-2)/(2 \cdot 2)} = \sqrt{1/4} = \frac{1}{2}$. The Mercedes-Benz frame achieves $\mu = \frac{1}{2}$ exactly (Example 9.2). So it sits precisely on the Welch bound — it is the tightest possible unit-norm configuration of 3 vectors in $\mathbb{R}^2$ in the coherence sense, among all possible configurations of 3 unit-norm vectors in $\mathbb{R}^2$.
For comparison, the square frame ($m = 4$, $n = 2$) has Welch bound $\sqrt{(4-2)/(2 \cdot 3)} = \sqrt{1/3} \approx 0.577$, but achieves $\mu = 1$ (antipodal pair). It sits far above the Welch floor, confirming that being tight does not imply achieving the Welch bound.

Remark. The Welch bound is achievable only for certain pairs (n, m) — there are combinatorial and algebraic obstructions that prevent ETFs from existing in many cases. For example, in $\mathbb{R}^2$, the only real ETF with $m > n$ is the Mercedes-Benz frame ($m = 3$): no unit-norm equiangular tight frame with $m \geq 4$ vectors exists in $\mathbb{R}^2$. (This can be verified directly: equiangularity forces all $|\langle f_i, f_j \rangle| = c$ for a common $c > 0$, tightness forces $c = \sqrt{m-n}/(n(m-1))^{1/2}$ which is the Welch bound, and in $\mathbb{R}^2$ the only unit-norm vectors satisfying both constraints simultaneously for $m \geq 4$

would have to include antipodal pairs, which gives $c = 1$, contradicting $c < 1$ for $m > n + 1$... the precise combinatorial statement is subtle, but the numerical evidence is clear.) In higher dimensions and over $\mathbb{C}$, ETFs can exist for many more (n, m) pairs, and their construction is an active research area connecting to combinatorics, coding theory, and quantum information.

9.4 An ETF Characterization via the Gram Matrix

The definition of an ETF involves two separate conditions: tight and equiangular. There is an elegant way to unify them into a single condition on the Gram matrix, which also gives a concrete necessary condition for ETFs to exist.

Proposition 9.4.1 (ETF Gram matrix characterization)**.** *A unit-norm collection $\{f_i\}_{i=1}^m \subset \mathbb{F}^n$ is an ETF with bound $A = m/n$ and coherence c if and only if its Gram matrix G satisfies*

$$G^2 = \frac{m}{n} G. \tag{9.1}$$

Proof. Since $G = F^*F$ (where F has columns f_i) and $S = FF^*$, we have $G^2 = (F^*F)^2 = F^*(FF^*)F = F^*SF$. The frame is tight with bound $A = m/n$ iff $S = (m/n)I$, in which case $F^*SF = (m/n)F^*F = (m/n)G$, giving $G^2 = (m/n)G$. Conversely, if $G^2 = (m/n)G$, then G is a projection (up to scale m/n), meaning the eigenvalues of G lie in $\{0, m/n\}$. Since $\mathrm{tr}(G) = m$ and G has m eigenvalues summing to m, exactly n eigenvalues equal m/n and the remaining $m - n$ equal 0 (using $n \cdot (m/n) + (m - n) \cdot 0 = m$). This means $S = FF^*$ has n eigenvalues all equal to m/n — exactly the tight frame condition. The equiangular condition is then encoded in the off-diagonal structure once tightness forces the diagonal, so (9.1) captures both conditions simultaneously. □

Corollary 9.4.2 (ETF necessary condition)**.** *If an ETF with m vectors in $\mathbb{F}^n$ exists, then the common coherence value satisfies*

$$c = \sqrt{\frac{m-n}{n(m-1)}},$$

exactly the Welch bound.

Proof. If $\{f_i\}_{i=1}^m$ is an ETF, it achieves equality in the Welch bound by Theorem 9.3.1. The Welch bound equals $\sqrt{(m-n)/(n(m-1))}$, so the common inner product magnitude is exactly that value. □

9.5 The Grassmannian Packing Perspective

The Welch bound and ETFs arise naturally from a broader geometric question: how should one arrange m one-dimensional subspaces (lines through the origin) in $\mathbb{R}^n$ (or $\mathbb{C}^n$) to make them as "spread out" as possible? A one-dimensional subspace in $\mathbb{R}^n$ is equivalent to a pair of antipodal unit vectors $\pm f$ on the unit sphere. The set of all such lines is called the *real projective space* $\mathbb{R}\mathrm{P}^{n-1}$. The question of packing m lines in $\mathbb{R}^n$ to maximize their minimum pairwise angle is called the *Grassmannian packing problem* ("Grassmannian" because the set of all k-dimensional subspaces of $\mathbb{R}^n$ is a manifold called the Grassmannian $\mathrm{Gr}(k, n)$, and the line case $k = 1$ is the simplest instance).

Definition 9.5.1 (Grassmannian packing angle)**.** Given m unit vectors $\{f_i\}$ in $\mathbb{R}^n$, the *worst-case packing angle* between the corresponding lines $\mathrm{span}\{f_i\}$ is

$$\theta_{\min} := \min_{i \neq j} \arccos\big(|\langle f_i, f_j\rangle|\big) = \arccos\big(\mu(\{f_i\})\big).$$

Since arccos is decreasing, *maximizing* $\theta_{\min}$ over all unit-norm collections is equivalent to *minimizing* the coherence μ.

The Welch bound then says: no configuration of m unit-norm vectors in $\mathbb{R}^n$ can achieve $\theta_{\min} > \arccos\big(\sqrt{(m-n)/(n(m-1))}\big)$. An ETF, if one exists for the given (n, m) pair, is an optimal packing — a solution to the Grassmannian packing problem.

Key Idea

ETFs are optimal Grassmannian packings: they arrange m lines in $\mathbb{R}^n$ with maximum possible minimum angle between any pair. The Welch bound is the ceiling on that minimum angle, expressed as a lower bound on coherence. Tight frames minimize the frame potential (energy spreading), while ETFs additionally maximize angular separation (packing optimality). These are independent optimization criteria that happen to be simultaneously achievable in certain cases, and those cases are exactly the ETFs.

9.6 Harmonic Frames: ETFs from Roots of Unity

Chapter 4 (Lab Exercise 4.5) introduced the vectors

$$f_k = \frac{1}{\sqrt{n}}\big(1, \omega^k, \omega^{2k}, \ldots, \omega^{(n-1)k}\big) \in \mathbb{C}^n, \qquad k = 0, 1, \ldots, n-1,$$

where $\omega = e^{2\pi i/n}$, and observed that they form an orthonormal basis for $\mathbb{C}^n$ (hence trivially a tight frame with $A = 1$ and $\mu = 0$). *Harmonic frames* generalise this construction to $m > n$ vectors by selecting m rows from the $n \times n$ DFT matrix — choosing m indices k from $\{0, 1, \ldots, n-1\}$ rather than all n of them.

Definition 9.6.1 (Harmonic frame)**.** Let $\omega = e^{2\pi i/n}$ and let $\mathcal{S} \subset \mathbb{Z}/n\mathbb{Z}$ be a subset of m residues modulo n, with $m \leq n$. The corresponding *harmonic frame* for $\mathbb{C}^n$ is

$$\{f_k\}_{k\in\mathcal{S}}, \qquad f_k = \frac{1}{\sqrt{n}}\big(1, \omega^k, \omega^{2k}, \ldots, \omega^{(n-1)k}\big) \in \mathbb{C}^n.$$

Remark. This is most interesting when $m > n$ is larger than the dimension, which requires going to a larger DFT. More generally, one takes an $n \times N$ DFT matrix (columns $\{f_k\}_{k=0}^{N-1}$ for some $N > n$) and selects $m \leq N$ columns; the discrete orthogonality relation for roots of unity (Proof 2 of Theorem 4.3.1) guarantees tightness whenever the index set $\mathcal{S}$ has a particular symmetry property (it suffices that $\mathcal{S}$ is a "difference set" modulo N). We study only the simplest case here.

Proposition 9.6.2 (Harmonic frames from the full DFT are tight)**.** *The full DFT collection $\{f_k\}_{k=0}^{n-1} \subset \mathbb{C}^n$ defined above is a Parseval frame for $\mathbb{C}^n$ (i.e. an orthonormal basis, $A = B = 1$).*

Proof. The $m\times m$ matrix $(F^*F)_{jk} = \langle f_k, f_j\rangle = \frac{1}{n}\sum_{\ell=0}^{n-1}\overline{\omega^{j\ell}}\omega^{k\ell} = \frac{1}{n}\sum_{\ell}\omega^{(k-j)\ell}$, which equals 1 when $j = k$ and 0 when $j \neq k$ by the discrete orthogonality relation for roots of unity (equation (4.3) in Chapter 4, introduced in Proof 2 of Theorem 4.3.1). So $F^*F = I_n$, confirming the columns are orthonormal, hence a Parseval frame. ◻

The reason harmonic frames appear in a chapter on ETFs is that *selecting subsets* of DFT columns gives natural candidates for equiangular collections, since the pairwise inner products $\langle f_j, f_k\rangle = (1/n)\sum_{\ell}\omega^{(k-j)\ell}$ depend only on $k - j \pmod{n}$, not on the individual indices — so if two pairs (j_1, k_1) and (j_2, k_2) have $k_1 - j_1 \equiv k_2 - j_2 \pmod{n}$, their inner products are automatically equal. Whether *all* off-diagonal inner products end up equal (equiangularity) depends on the structure of $\mathcal{S}$, and is related to the theory of *difference sets* in combinatorics.

Example 9.4: A harmonic frame in $\mathbb{C}^3$

Take $n = 3$, $\omega = e^{2\pi i/3}$, and $\mathcal{S} = \{0, 1, 2\}$ (all three columns). The harmonic frame is the orthonormal basis of $\mathbb{C}^3$:

$$f_0 = \frac{1}{\sqrt{3}}(1, 1, 1)^T, \quad f_1 = \frac{1}{\sqrt{3}}(1, \omega, \omega^2)^T, \quad f_2 = \frac{1}{\sqrt{3}}(1, \omega^2, \omega^4)^T.$$

This is an ETF vacuously ($m = n = 3$, so coherence is 0 and the Welch bound is 0). The genuinely interesting case is taking $m < N$ columns from a larger DFT; Lab Exercise 9.5 explores a specific example.

9.7 NumPy Implementation

Listing 9.1: Coherence, Welch bound, and ETF verification in NumPy

```python
import numpy as np

def gram_matrix(F):
    """Gram matrix G = F^* F (shape m x m)."""
    return F.T.conj() @ F

def coherence(F):
    """Maximum off-diagonal |inner product| for unit-norm columns of F."""
    G = gram_matrix(F)
    m = F.shape[1]
    off_diag = np.abs(G - np.diag(np.diag(G)))  # zero the diagonal
    return off_diag.max()

def welch_bound(n, m):
    """Lower bound on coherence for m unit-norm vectors in F^n."""
    return np.sqrt((m - n) / (n * (m - 1)))

def is_etf(F, tol=1e-6):
    """
    Check whether the columns of F form an equiangular tight frame.
    Returns (is_tight, is_equiangular, achieves_welch).
    """
    n, m = F.shape
    # Tightness
    S = F @ F.T.conj()
    eigs = np.linalg.eigvalsh(S.real)
    tight = bool(np.isclose(eigs.min(), eigs.max(), atol=tol))
    # Equiangularity: all off-diagonal |<fi,fj>| must be equal.
    # Compare max and min of off-diagonal magnitudes (not off[0],
    # which depends on ordering and may be zero for non-equiangular frames
    ).
    G = gram_matrix(F)
```

```
    off = np.array([np.abs(G[i,j]) for i in range(m) for j in range(m) if
        i != j])
    equi = bool(np.isclose(off.max(), off.min(), atol=tol))
    # Welch bound
    mu = coherence(F)
    wb = welch_bound(n, m)
    achieves = bool(np.isclose(mu, wb, atol=tol))
    return tight, equi, achieves

# ------------------------------------------------------------------
# Mercedes-Benz frame (m=3, n=2): the canonical real ETF
# ------------------------------------------------------------------
f1 = np.array([1.0,  0.0])
f2 = np.array([-0.5, np.sqrt(3)/2])
f3 = np.array([-0.5, -np.sqrt(3)/2])
F_mb = np.column_stack([f1, f2, f3])

tight_mb, equi_mb, wb_mb = is_etf(F_mb)
print("=== Mercedes-Benz frame (m=3, n=2) ===")
print(f"Tight: {tight_mb}, Equiangular: {equi_mb}, Achieves Welch bound: {
    wb_mb}")
print(f"Coherence: {coherence(F_mb):.6f}  Welch bound: {welch_bound(2,3)
    :.6f}")

# ------------------------------------------------------------------
# Square frame (m=4, n=2): tight but NOT equiangular
# ------------------------------------------------------------------
angles4 = np.array([0, np.pi/2, np.pi, 3*np.pi/2])
F_sq = np.vstack([np.cos(angles4), np.sin(angles4)])

tight_sq, equi_sq, wb_sq = is_etf(F_sq)
print("\n=== Square frame (m=4, n=2) ===")
print(f"Tight: {tight_sq}, Equiangular: {equi_sq}, Achieves Welch bound: {
    wb_sq}")
print(f"Coherence: {coherence(F_sq):.6f}  Welch bound: {welch_bound(2,4)
    :.6f}")
G_sq = gram_matrix(F_sq)
print("Gram matrix off-diag magnitudes:",
      sorted(set(np.abs(G_sq - np.eye(4)).round(4).flatten()))[::-1][:4])

# ------------------------------------------------------------------
# Gram matrix check: G^2 = (m/n) G for Mercedes-Benz
# ------------------------------------------------------------------
G_mb = gram_matrix(F_mb)
lhs = G_mb @ G_mb
rhs = (3/2) * G_mb
print("\n=== Gram matrix check (Proposition 9.4.1) ===")
print(f"G^2 = (m/n)G for Mercedes-Benz: {np.allclose(lhs, rhs)}")

# ------------------------------------------------------------------
# Welch bound survey: all regular polygons m=3..8 in R^2
# ------------------------------------------------------------------
print("\n=== Regular m-gons vs Welch bound ===")
for m in range(3, 9):
```

```
    angles = 2*np.pi*np.arange(m)/m
    F = np.vstack([np.cos(angles), np.sin(angles)])
    mu = coherence(F)
    wb = welch_bound(2, m)
    t, e, w = is_etf(F)
    print(f"m={m}: mu={mu:.4f}, Welch={wb:.4f}, "
          f"tight={t}, equiangular={e}, ETF={t and e}")
```

9.8 Searching for ETFs via Gradient Descent

The frame potential (Chapter 8) finds tight frames but not necessarily ETFs. To search for ETFs numerically, we can minimize a different objective: the *coherence itself*, or the *coherence-squared sum* $\sum_{i\neq j} |\langle f_i, f_j\rangle|^2$. The latter is smoother (no `max` operation) and directly related to the frame potential:

$$\sum_{i\neq j} |\langle f_i, f_j\rangle|^2 = \mathrm{FP} - m \quad \text{(unit-norm case)}.$$

Since FP is minimized by tight frames, minimizing $\sum_{i\neq j} |\langle f_i, f_j\rangle|^2$ is the same optimization. This means that if an ETF exists, gradient descent on the frame potential will find a tight frame in which the off-diagonal inner products are likely to equalize too, especially from a good random initialization.

Example 9.3: When gradient descent finds an ETF (and when it doesn't)

For $(n, m) = (2, 3)$: the only unit-norm tight frame in $\mathbb{R}^2$ with 3 vectors is (up to rotation and sign-flips) the Mercedes-Benz frame, so gradient descent on the frame potential automatically produces an ETF.
For $(n, m) = (2, 4)$: there is no real ETF with 4 vectors in $\mathbb{R}^2$ (as the remark in Section 9.2 explains), so gradient descent finds tight frames that are not ETFs. The tight-frame minimum $m^2/n = 8$ is achieved by many distinct non-equiangular configurations (the family of tight non-equiangular frames is large), and gradient descent lands on one of them.
For $(n, m) = (3, 6)$: real ETFs with 6 vectors in $\mathbb{R}^3$ are known to exist (the regular icosahedron vertices, after normalization, give one such configuration). Gradient descent can find them, but only if the search is conducted over a refined objective that encourages equiangularity beyond just tightness.

9.9 Chapter Summary

- The **coherence** $\mu = \max_{i\neq j} |\langle f_i, f_j\rangle|$ of a unit-norm collection measures how "non-orthogonal" the most similar pair is. Small coherence means well-spread configurations.
- A collection is **equiangular** if all off-diagonal Gram entries have the same modulus. An **equiangular tight frame** (ETF) is equiangular and tight.
- **Theorem 9.3.1 (Welch bound):** for any unit-norm collection of m vectors in $\mathbb{F}^n$ with $m > n$, $\mu \geq \sqrt{(m-n)/(n(m-1))}$. Equality holds if and only if the collection is an ETF. The proof combines the frame-potential lower bound ($\mathrm{FP} \geq m^2/n$, from Chapter 8) with an upper bound on FP in terms of μ.

- **ETFs are optimal Grassmannian packings**: they arrange m lines in $\mathbb{R}^n$ with the maximum possible minimum pairwise angle, and they minimize the frame potential simultaneously.

- **Proposition 9.4.1**: $\{f_i\}$ is an ETF iff its Gram matrix satisfies $G^2 = (m/n)G$, a single matrix equation encoding both tightness and equiangularity.

- ETFs exist only for certain (n, m) pairs. In $\mathbb{R}^2$, only $m = 3$ (the Mercedes-Benz frame) gives a real ETF. In higher dimensions and over $\mathbb{C}$, the existence problem is an active research area connected to combinatorics (e.g. symmetric designs, difference sets) and quantum information (optimal quantum measurements).

9.10 Lab Exercises

Lab Exercise 9.1: Regular Polygons and Equiangularity

Implement `coherence(F)`, `welch_bound(n, m)`, and `is_etf(F)` from Section 9.7. Run `is_etf` on all regular m-gons in $\mathbb{R}^2$ for $m = 3, \dots, 12$. For each, also report the distinct off-diagonal Gram magnitudes. Confirm that only $m = 3$ is an ETF, and explain geometrically why even-m cases fail so dramatically (hint: $\mu = 1$).

Lab Exercise 9.2: Welch Bound Survey

For each (n, m) in $\{(2,3), (2,5), (3,4), (3,6), (3,7), (4,6), (4,7)\}$:

(a) Compute the Welch bound $\sqrt{(m-n)/(n(m-1))}$.

(b) Use the gradient-descent algorithm of Chapter 8 (`fit_tight_frame`) to find a tight unit-norm frame for each, and compute its coherence.

(c) Report whether the result achieves the Welch bound. For those that do not, compute the ratio μ/μ_{Welch} and call it the "coherence excess."

(d) Identify which (n, m) pairs appear from your experiments to *admit* an ETF (coherence excess ≈ 1) and which do not (coherence excess $\gg 1$). (Note: this is a numerical experiment, not a proof — you may miss some ETFs or misidentify some near-misses.)

Lab Exercise 9.3: The Gram Matrix Condition

For the Mercedes-Benz frame F_{MB}, verify numerically that $G^2 = (3/2)G$ (Proposition 9.4.1). Then construct a *different* 3×3 Gram matrix G' satisfying $G'^2 = (3/2)G'$, with $G'_{ii} = 1$ for all i and all off-diagonal entries having modulus $\frac{1}{2}$, but with the off-diagonal *signs* permuted. Does G' correspond to a valid frame? (Hint: a Gram matrix $G = F^*F$ must be positive semidefinite; check whether G' is PSD, and if so, extract the corresponding frame vectors using a Cholesky decomposition or eigenvalue decomposition.) This explores the phenomenon that multiple non-congruent ETF configurations can share the same equiangularity value.

Lab Exercise 9.4: Coherence-Minimizing Descent

The frame potential minimizes $\sum_{ij} |\langle f_i, f_j\rangle|^2 = \mathrm{FP}$. Instead, minimize the *maximum* off-diagonal inner product: $\mu = \max_{i\neq j} |\langle f_i, f_j\rangle|$, or a smooth approximation such as $\mu_\alpha = \left(\sum_{i\neq j} |\langle f_i, f_j\rangle|^{2\alpha}\right)^{1/(2\alpha)}$ for large α (a soft-max approximation that becomes exact as $\alpha \to \infty$). Implement gradient descent on μ_α with $\alpha = 8$ for $(n, m) = (2, 3)$ and $(n, m) = (3, 4)$, and compare the resulting coherence to the Welch bound. Does maximizing angular separation produce results closer to the Welch bound than minimizing the frame potential alone?

Lab Exercise 9.5 (Optional, Challenge): ETFs in $\mathbb{R}^3$

The six vectors formed by taking the $\pm$ unit vectors along the three coordinate axes of $\mathbb{R}^3$ — i.e. $\{\pm e_1, \pm e_2, \pm e_3\}$ — form a tight frame for $\mathbb{R}^3$ with $m = 6$, $A = 2$. But is it equiangular? (Hint: some pairs are orthogonal, others are antipodal.) If not, research (via any source you can find) what the smallest real ETF in $\mathbb{R}^3$ with $m > 3$ is. Then verify the claimed ETF numerically using `is_etf`. This exercise previews the rich combinatorial theory of ETF existence, which in $\mathbb{R}^3$ is already surprisingly non-trivial.

Lab Exercise 9.6 (Optional): Harmonic Frames in $\mathbb{C}^n$

This exercise explores the harmonic frame construction of Section 9.6.

(a) For $n = 3$ and $\omega = e^{2\pi i/3}$, construct the three DFT vectors $f_0, f_1, f_2 \in \mathbb{C}^3$ as in Definition 9.6.1 using complex NumPy arrays (`dtype=complex`). Verify numerically that $\langle f_j, f_k\rangle = \delta_{jk}$ for all j, k (use `np.vdot` for complex inner products as discussed in Lab Exercise 1.3 of Chapter 1), confirming the orthonormal-basis claim of Proposition 9.6.2.

(b) Now take $N = 6$, $\omega = e^{2\pi i/6}$, and consider all 6 DFT vectors $\{f_k\}_{k=0}^{5} \subset \mathbb{C}^6$. Select the subset $\mathcal{S} = \{0, 1, 2\}$ (the first three), giving a collection of 3 unit-norm vectors in $\mathbb{C}^6$. Is this collection tight? Is it equiangular? Compute its coherence and compare to the Welch bound for $n = 6$, $m = 3$.

(c) (Challenge) Investigate whether any size-3 subset of the $N = 7$ DFT columns gives an equiangular collection in $\mathbb{C}^7$. This is related to the existence of a $(7, 3, 1)$ difference set modulo 7 — a set $\mathcal{S}$ such that every nonzero residue modulo 7 appears exactly once as a difference $s_1 - s_2$ with $s_1 \neq s_2$ in $\mathcal{S}$. The set $\mathcal{S} = \{0, 1, 3\}$ is one such; test it numerically using `is_etf` from Lab Exercise 9.1.

9.11 Supplementary Exercises

Theoretical Exercises

T1. Isolate "Step 2" of the Welch bound proof (Theorem 9.3.1) as a standalone lemma: prove that for any m unit-norm vectors with coherence μ, $\mathrm{FP} \le m + m(m-1)\mu^2$.

T2. Using $G^2 = (m/n)G$ (Proposition 9.4.1), prove that the Gram matrix of an ETF has exactly two distinct eigenvalues: 0 with multiplicity $m - n$, and m/n with multiplicity n. Conclude $\operatorname{rank} G = n$.

T3. The *chordal distance* between the lines spanned by unit vectors f_i, f_j is $d(f_i, f_j) := \sqrt{1 - \langle f_i, f_j\rangle^2}$. Prove that minimizing the coherence μ over a configuration of m unit vectors is equivalent to maximizing the *minimum* pairwise chordal distance — i.e. the Grassmannian packing problem (minimize worst-case inner product) and the "line-packing" problem (maximize worst-case chordal distance) are the same optimization.

T4. State and prove the complex analogue of the Welch bound (Theorem 9.3.1) for unit vectors in $\mathbb{C}^n$, noting precisely where the proof requires the conjugate-linear inner product convention of Section 1.2.1.

T5. Numerically motivated conjecture, now to be proved: for the Mercedes-Benz ETF, show that $\sum_{i=1}^{3} \langle x, f_i\rangle^4$ is the *same* constant for every unit vector $x \in \mathbb{R}^2$ (not merely the second moment $\sum_i \langle x, f_i\rangle^2$, which is constant for *any* tight frame by definition). Then show this stronger "fourth-moment constancy" *fails* for the square configuration (tight, but not equiangular), so it is a genuinely finer invariant than tightness alone — a first glimpse of what the frame theory literature calls a *projective 2-design*.

Computational Exercises

C1. Verify T1's bound numerically for 10 random unit-norm configurations with $(n, m) = (3, 8)$, confirming $\mathrm{FP} \le m + m(m-1)\mu^2$ in every case.

C2. Compute the Gram matrix eigenvalues of the Mercedes-Benz ETF and confirm they match T2's prediction (0 with multiplicity $m - n = 1$, $m/n = 3/2$ with multiplicity $n = 2$).

C3. Implement the chordal distance function and verify T3's equivalence numerically: for several random configurations, confirm that the configuration minimizing coherence (via `fit_min_coherence` from Lab Exercise 9.4) also maximizes the minimum chordal distance.

C4. Reproduce the fourth-moment computation from T5: sample 50 unit vectors x around the circle, compute $\sum_i \langle x, f_i\rangle^4$ for both the Mercedes-Benz ETF and the square, and plot both as functions of the angle of x, visually confirming constancy for the ETF and variation for the square.

Chapter 10

Frames and Erasures

10.1 Overview: Closing the Loop on Redundancy

Chapter 1 offered two promises: redundant frames are robust to noise, and robust to erasures. Chapter 5 fulfilled the first promise — the noise stability analysis showed the canonical dual minimizes expected squared reconstruction error under additive coefficient noise. This chapter fulfills the second.

An *erasure* is a coefficient that never arrives at all, not merely one that is corrupted by noise. In communication, some transmitted symbols are simply lost in transit; in sensor networks, some sensors fail entirely; in data storage, some blocks become unreadable. In all of these settings, the question is not "how noisy was the measurement?" but "can I still reconstruct at all, and how accurately?"

We will see that frames answer this question cleanly, and that the geometry developed in Chapters 8–9 (frame potential, equiangularity) provides the right tools for understanding which frames are most erasure-robust. The chapter closes with a precise connection back to Chapter 9: equiangular tight frames are not only optimal for coherence minimization (the Grassmannian packing problem), they are also optimal for worst-case erasure robustness.

10.2 The Erasure Model

Definition 10.2.1 (Erasure pattern and sub-frame)**.** Given a frame $\{f_i\}_{i=1}^m$ for $V = \mathbb{F}^n$, an *erasure pattern* of size k is a subset $E \subset \{1, \dots, m\}$ with $|E| = k$. The *surviving index set* is $J = \{1, \dots, m\} \setminus E$, and the corresponding *sub-frame* is $\{f_i\}_{i \in J}$.

In the erasure model, the receiver obtains only the frame coefficients $\{c_i = \langle x, f_i \rangle\}_{i \in J}$ (the k coefficients indexed by E are lost), and must reconstruct x from these $m-k$ surviving measurements.

Definition 10.2.2 (k-erasure robustness)**.** A frame $\{f_i\}_{i=1}^m$ for $\mathbb{F}^n$ is *k-erasure robust* if, for every erasure pattern E with $|E| = k$, the sub-frame $\{f_i\}_{i \in J}$ (where $J = \{1, \dots, m\} \setminus E$) still spans $\mathbb{F}^n$. The *erasure number* of the frame is

$$\epsilon(F) := \max \{k \ : \ \text{the frame is } k\text{-erasure robust}\},$$

the maximum number of simultaneous erasures the frame can *guarantee* to survive — i.e., every pattern of that many erasures leaves a spanning sub-frame.

Remark. Spanning is the correct condition because, by Theorem 2.2.1, a finite collection spans $\mathbb{F}^n$ if and only if it is a frame — and we need the sub-frame to be a frame so that the canonical reconstruction formula $x = \tilde{F}_J(F_J^* x)$ (via the sub-frame's canonical dual) is valid. A non-spanning sub-frame has a direction x_0 with $\langle x_0, f_i\rangle = 0$ for all $i \in J$; no number of surviving measurements can recover x_0's component, and reconstruction is impossible.

Proposition 10.2.3 (Erasure number and redundancy)**.** *For a frame $\{f_i\}_{i=1}^m$ for $\mathbb{F}^n$:*

(i) $\epsilon(F) \le m - n$ *(can tolerate at most $m - n$ erasures).*

(ii) $\epsilon(F) = 0$ *if $m = n$ (no erasure tolerance, since any single erasure leaves $n - 1$ vectors, which cannot span $\mathbb{F}^n$).*

(iii) The erasure number depends on the geometry *of the frame, not merely the redundancy $m - n$: two frames with the same m and n can have different erasure numbers.*

Proof. (i) After $m - n + 1$ erasures only $n - 1$ vectors remain, which cannot span an n-dimensional space. (ii) With $m = n$, erasing any single vector leaves $n - 1$ vectors in $\mathbb{F}^n$, insufficient to span. (iii) Demonstrated in Examples 10.4 and 10.4 below: the square and pentagon frames in $\mathbb{R}^2$ have the same $n = 2$ but very different erasure numbers despite both being tight. □

10.3 Reconstruction After Erasure

If the sub-frame $\{f_i\}_{i\in J}$ spans $\mathbb{F}^n$, reconstruction is always possible in principle: use the canonical dual of the sub-frame. But the sub-frame's dual depends on J, so the receiver must either recompute it for each erasure pattern — a computation that scales with the number of possible patterns — or use the original dual and accept some approximation error. For tight frames, the second option is much more transparent than it sounds.

Proposition 10.3.1 (Exact reconstruction after erasure)**.** *If $\{f_i\}_{i\in J}$ is a (sub-)frame for $\mathbb{F}^n$ with frame operator $S_J = \sum_{i\in J} f_i f_i^*$, then*

$$x = S_J^{-1}\left(\sum_{i\in J} \langle x, f_i\rangle\, f_i\right) = \sum_{i\in J} \langle x, f_i\rangle\, (S_J^{-1} f_i),$$

a reconstruction using only the surviving coefficients.

Proof. Immediate from the canonical dual formula (Proposition 3.8.1 of Chapter 3, restated for the sub-frame $\{f_i\}_{i\in J}$). Since the sub-frame spans $\mathbb{F}^n$ by the k-erasure robustness assumption, S_J is invertible (Proposition 3.4.2(iii)). □

10.3.1 Sub-Frame Operator Decomposition

The sub-frame operator S_J has a clean relationship to the full frame operator S that is worth making explicit.

Proposition 10.3.2 (Sub-frame operator)**.** *Let $E = \{1, \ldots, m\} \setminus J$ be the erased indices. Then*

$$S_J = S - S_E, \qquad \text{where } S_E = \sum_{i\in E} f_i f_i^*$$

is the "erased contribution" to the frame operator. In particular, if $\{f_i\}_{i=1}^m$ is tight with bound A, then

$$S_J = AI - S_E. \tag{10.1}$$

Proof. Direct from $S = \sum_{i=1}^m f_i f_i^* = \sum_{i\in J} f_i f_i^* + \sum_{i\in E} f_i f_i^* = S_J + S_E$. □

10.3.2 Error Formula for Tight Frames

For tight frames, a particularly clean scenario arises when the receiver uses the *original* tight reconstruction formula $\hat{x} = \frac{1}{A}\sum_i c_i f_i$ but simply treats erased coefficients as zero, rather than recomputing the sub-frame dual.

Proposition 10.3.3 (Tight-frame erasure error)**.** *Let $\{f_i\}_{i=1}^m$ be a tight frame with bound A and unit-norm vectors, and suppose the coefficients indexed by E are erased. If the receiver applies the original tight reconstruction formula with missing coefficients set to zero,*

$$\hat{x} = \frac{1}{A}\sum_{i\in J} \langle x, f_i\rangle \; f_i,$$

then the reconstruction error is

$$\hat{x} - x = -\frac{1}{A}\sum_{i\in E} \langle x, f_i\rangle \; f_i. \tag{10.2}$$

In particular, for a single erasure $E = \{j\}$,

$$\hat{x} - x = -\frac{\langle x, f_j\rangle}{A} f_j, \qquad \|\hat{x} - x\| = \frac{|\langle x, f_j\rangle|}{A}.$$

Proof. Since $\{f_i\}$ is tight with bound A, the exact reconstruction is $x = \frac{1}{A}\sum_{i=1}^m \langle x, f_i\rangle f_i$ (Chapter 3, equation (3.5)). Subtracting:

$$\hat{x} - x = \frac{1}{A}\sum_{i\in J}\langle x, f_i\rangle f_i - \frac{1}{A}\sum_{i=1}^{m}\langle x, f_i\rangle f_i = -\frac{1}{A}\sum_{i\in E}\langle x, f_i\rangle f_i.$$

For a single erasure $E = \{j\}$, the norm is $\|\hat{x} - x\| = \frac{|\langle x, f_j\rangle|}{A}\|f_j\| = \frac{|\langle x, f_j\rangle|}{A}$ using $\|f_j\| = 1$. □

Key Idea

Proposition 10.3.3 has two immediate consequences.

- **Error is readable without knowing x**: The error $\|\hat{x} - x\| = |\langle x, f_j\rangle|/A$ depends on the erased coefficient $c_j = \langle x, f_j\rangle$, which the receiver already knows was supposed to arrive but did not. This gives a computable *error certificate*: the receiver can compute an upper bound on the error without the original signal x.
- **Tight frames localize the error**: The error $-\frac{1}{A}c_j f_j$ points *in the direction* of the erased frame vector f_j — the damage is confined to one direction, not spread throughout the space. Contrast this with a non-tight frame, where erasing one coefficient perturbs the reconstruction in an entangled, direction-dependent way.

10.4 Comparing Frames: The Square, Triangle, and Pentagon

We now work through the three main running examples, establishing that the erasure number depends critically on the frame geometry, not only on the redundancy.

Example 10.1: Triangle vs. square vs. pentagon in $\mathbb{R}^2$

Triangle ($m = 3$, $n = 2$, redundancy 1): Any two of the three vectors f_1, f_2, f_3 at $120°$ are linearly independent (no pair is parallel or antipodal), so any single erasure leaves two spanning vectors. Thus $\epsilon(F) = 1$. This matches the redundancy $m - n = 1$, the theoretical maximum.

Square ($m = 4$, $n = 2$, redundancy 2): The four vectors $(\pm 1, 0)$ and $(0, \pm 1)$ have two antipodal pairs: (f_0, f_2) and (f_1, f_3). Erasing one vector from each antipodal pair (e.g. $E = \{0, 2\}$) leaves $\{f_1, f_3\} = \{(0, 1), (0, -1)\}$, which does *not* span $\mathbb{R}^2$ (both vectors lie on the y-axis). So $\epsilon(F) = 1$, even though redundancy $m - n = 2$. **The square frame wastes its extra redundancy**: it can handle any one erasure but not any two, despite having twice the surplus over the triangle.

Example 10.2: Pentagon and the role of equiangularity

The regular pentagon ($m = 5$, $n = 2$, vertices at $2\pi k/5$) has redundancy $m - n = 3$, and remarkably achieves $\epsilon(F) = 3$: any 3-element erasure leaves only 2 vectors, but since no two pentagon vertices are parallel (the minimum angle between adjacent vertices is $72°$, and no two vertices are antipodal), any 2-element subset spans $\mathbb{R}^2$. Since $m - n = 3$ is the maximum possible, the pentagon *achieves its theoretical maximum erasure number.*

The square, by contrast, achieves only $\epsilon(F) = 1$ against its theoretical maximum of 2. The structural deficiency is its antipodal pairs: two of its $\binom{4}{2} = 6$ size-2 subsets fail to span. The pentagon has no such pairs: all $\binom{5}{2} = 10$ size-2 subsets are spanning.

Remark. The phenomenon illustrated here is general: a frame achieves $\epsilon(F) = m - n$ (the theoretical maximum) if and only if every $(m - k)$-subset spans $\mathbb{F}^n$ for all $k \leq m - n$, which is equivalent to every n-element subset of the frame being linearly independent. This is the *full spark* property.

Definition 10.4.1 (Full spark)**.** A frame $\{f_i\}_{i=1}^m$ for $\mathbb{F}^n$ has *full spark* if every n-element subset of $\{f_1, \ldots, f_m\}$ is linearly independent. A full-spark frame achieves $\epsilon(F) = m - n$, the maximum possible erasure number.

Remark. In the language of coding theory, the spark of a matrix F is the smallest number of columns that are linearly dependent; a full-spark frame has spark $n + 1$. The triangle has spark $3 = n + 1$; the pentagon has spark $3 = n + 1$ (any two vectors are independent); but the square has spark $2 < n + 1 = 3$, since two of its columns (f_0 and f_2) are linearly dependent.

10.5 Worst-Case Erasure Error and ETFs

For a tight frame that is full-spark, we can reconstruct after any k-erasure pattern (with $k \leq m - n$). But reconstruction quality varies by pattern: some erasures cause larger errors than others. Proposition 10.3.3 shows that a single erasure of index j causes error $|\langle x, f_j \rangle|/A$, which depends on both x and j. A natural worst-case measure for a single erasure is

$$\text{worst-case error} = \max_j \max_{\|x\|=1} |\langle x, f_j \rangle|/A = \frac{\max_j \|f_j\|}{A} = \frac{1}{A}$$

for unit-norm frames (using $\max_{\|x\|=1} |\langle x, f_j \rangle| = \|f_j\|$ by Cauchy–Schwarz, and $\|f_j\| = 1$). All tight unit-norm frames give the same single-erasure worst case.

The situation changes for *multiple* erasures. For two simultaneous erasures $E = \{j,k\}$ and a unit-norm x, the naive error $\|\hat{x} - x\| = \|\frac{1}{A}(\langle x, f_j\rangle f_j + \langle x, f_k\rangle f_k)\|$ depends on both the angle between f_j and f_k and their alignment with x. Maximizing this over x gives

$$\max_{\|x\|=1} \left\| \frac{1}{A}(\langle x, f_j\rangle f_j + \langle x, f_k\rangle f_k) \right\| = \frac{\lambda_{\max}(S_{\{j,k\}})}{A},$$

where $\lambda_{\max}(S_{\{j,k\}})$ is the largest eigenvalue of $S_{\{j,k\}} = f_j f_j^* + f_k f_k^*$. For unit-norm vectors, $\lambda_{\max}(S_{\{j,k\}}) = 1 + |\langle f_j, f_k\rangle|$ (a standard fact about the largest eigenvalue of a sum of two rank-one unit-norm projections). So the worst-case 2-erasure naive error is $(1+|\langle f_j, f_k\rangle|)/A$, and minimizing this over all pairs (j,k) (to find the most robust frame) requires *minimizing* the off-diagonal inner products $|\langle f_j, f_k\rangle|$ — exactly what coherence minimization and the Welch bound from Chapter 9 do.

Theorem 10.5.1 (ETFs minimize worst-case erasure error)**.** *Among all tight, unit-norm frames of m vectors in $\mathbb{F}^n$ with $m > n$, the equiangular tight frames (ETFs) minimize the worst-case 2-erasure naive reconstruction error. The minimum worst-case error equals $(1 + c_{\text{Welch}})/A$ where $c_{\text{Welch}} = \sqrt{(m-n)/(n(m-1))}$ is the Welch bound coherence and $A = m/n$.*

Proof. By the analysis above, the worst-case 2-erasure naive error over all pairs (j,k) and all unit-norm signals is $\max_{j\neq k}(1 + |\langle f_j, f_k\rangle|)/A$. This is minimized by minimizing $\max_{j\neq k}|\langle f_j, f_k\rangle|$, i.e. the coherence μ. By the Welch bound (Theorem 9.3.1), $\mu \geq c_{\text{Welch}}$ for any tight unit-norm frame, with equality iff the frame is an ETF. □

Key Idea

ETFs are optimal for erasure robustness, coherence minimization, and Grassmannian packing — simultaneously. All three criteria reduce to the same underlying condition: equal off-diagonal inner products achieving the Welch bound. The equivalence established in Theorem 10.5.1 shows that the three different motivations (information theory, geometry, coding theory) converge to the same object.

10.6 NumPy Implementation

Listing 10.1: Erasure robustness tools in NumPy

```python
import numpy as np
from itertools import combinations

def subframe_spans(F, erased_indices):
    """
    Return True if the sub-frame (columns NOT in erased_indices) spans R^n
        .
    """
    n, m = F.shape
    kept = [i for i in range(m) if i not in erased_indices]
    if len(kept) < n:
        return False
    return np.linalg.matrix_rank(F[:, kept]) == n

def erasure_number(F):
```

```python
    """
    Maximum k such that every k-erasure pattern leaves a spanning sub-
        frame.
    """
    n, m = F.shape
    for k in range(1, m - n + 2):
        all_span = all(
            subframe_spans(F, list(E))
            for E in combinations(range(m), k)
        )
        if not all_span:
            return k - 1
    return m - n

def reconstruct_after_erasure(F, x, erased_indices):
    """
    Reconstruct x from surviving frame coefficients.
    Returns (x_hat, error_norm) using sub-frame canonical dual.
    """
    n, m = F.shape
    kept = [i for i in range(m) if i not in erased_indices]
    F_sub = F[:, kept]
    S_sub = F_sub @ F_sub.T
    F_sub_tilde = np.linalg.solve(S_sub, F_sub)
    c_sub = F_sub.T @ x
    x_hat = F_sub_tilde @ c_sub
    return x_hat, np.linalg.norm(x_hat - x)

def tight_erasure_error_formula(F, x, erased_indices, A):
    """
    Error using ORIGINAL tight dual with erased coefficients set to zero.
    Returns error vector via Proposition 10.3.3.
    """
    erased = np.array(erased_indices)
    c_erased = F[:, erased].T @ x
    error = -(1/A) * (F[:, erased] @ c_erased)
    return error

# ----------------------------------------------------------------
# Example 10.1: Triangle, square, pentagon
# ----------------------------------------------------------------
# Triangular frame (tight, A=3/2)
f1,f2,f3 = ([1.,0.],[-0.5,np.sqrt(3)/2],[-0.5,-np.sqrt(3)/2])
F_tri = np.array([f1,f2,f3]).T

# Square frame (tight, A=2)
angles4 = np.array([0,np.pi/2,np.pi,3*np.pi/2])
F_sq = np.vstack([np.cos(angles4), np.sin(angles4)])

# Pentagon frame (tight, A=5/2)
angles5 = 2*np.pi*np.arange(5)/5
F_pent = np.vstack([np.cos(angles5), np.sin(angles5)])

print("=== Erasure numbers ===")
```

```python
for name, F in [("Triangle (m=3,n=2)", F_tri),
                ("Square   (m=4,n=2)", F_sq),
                ("Pentagon (m=5,n=2)", F_pent)]:
    n, m = F.shape
    eps = erasure_number(F)
    print(f"  {name}: epsilon(F)={eps}, m-n={m-n} "
          f"{'(full spark!)' if eps==m-n else '(sub-optimal)'}")

# ---------------------------------------------------------------------
# Example 10.3: Tight-frame erasure error formula
# ---------------------------------------------------------------------
x = np.array([3., -1.])
A_tri = 3/2
print("\n=== Tight-frame erasure error formula (triangle, erase f1) ===")
# Erase index 0 (f1)
err_vec = tight_erasure_error_formula(F_tri, x, [0], A_tri)
x_hat_naive = (1/A_tri) * (F_tri[:, 1:] @ (F_tri[:, 1:].T @ x))
err_actual = x_hat_naive - x
print(f"  Formula predicts: {err_vec.round(4)}")
print(f"  Actual error:     {err_actual.round(4)}")
print(f"  Match: {np.allclose(err_vec, err_actual)}")
print(f"  Error norm = |c1|/A = {abs(np.dot(x,F_tri[:,0])):.4f}/{A_tri} "
      f"= {abs(np.dot(x,F_tri[:,0]))/A_tri:.4f}")

# ---------------------------------------------------------------------
# Example 10.4: Exact reconstruction after erasure (sub-frame dual)
# ---------------------------------------------------------------------
print("\n=== Exact reconstruction (sub-frame dual) ===")
for i in range(3):
    x_hat, err = reconstruct_after_erasure(F_tri, x, [i])
    print(f"  Triangle, erase f{i+1}: error = {err:.2e}")

for i in range(4):
    x_hat, err = reconstruct_after_erasure(F_sq, x, [i])
    print(f"  Square, erase f{i+1}: error = {err:.2e}")

# Demonstrate square failure for 2 erasures (antipodal pair)
print("\n=== Square: which 2-erasure patterns fail? ===")
for pair in combinations(range(4), 2):
    spans = subframe_spans(F_sq, list(pair))
    print(f"  Erase {pair}: sub-frame spans = {spans}")
```

10.7 Chapter Summary

- A frame is **k-erasure robust** if every erasure pattern of size k leaves a spanning sub-frame (Definition 10.2.2). The **erasure number** $\epsilon(F)$ is the maximum guaranteed-survivable number of erasures; it satisfies $\epsilon(F) \leq m - n$ for any frame in $\mathbb{F}^n$.
- **Exact reconstruction** after k erasures is always possible when the sub-frame spans, via the sub-frame's canonical dual S_J^{-1} (Proposition 10.3.1). The sub-frame operator is $S_J = S - S_E$ (Proposition 10.3.2).

- For a **tight frame**, if the receiver uses the original tight formula with erased coefficients set to zero, the error is $\hat{x} - x = -\frac{1}{A} \sum_{i \in E} \langle x, f_i \rangle f_i$ (Proposition 10.3.3) — a clean, computable expression pointing exactly in the direction of the erased frame vectors.

- The **full spark** property ($\epsilon(F) = m - n$) requires every n-element subset to be linearly independent. The triangle and pentagon in $\mathbb{R}^2$ are full-spark; the square is not (its antipodal pairs have spark $2 < n + 1$).

- **Theorem 10.5.1**: Among tight unit-norm frames, ETFs minimize worst-case 2-erasure error. The optimal error is $(1 + c_{\text{Welch}})/A$, achieved exactly when $\mu = c_{\text{Welch}}$ (the Welch bound from Chapter 9). ETFs are simultaneously optimal for coherence, Grassmannian packing, and worst-case erasure robustness.

10.8 Lab Exercises

Lab Exercise 10.1: Erasure Numbers

Implement `erasure_number(F)` and compute it for: (a) the triangular frame, (b) the square frame, (c) the pentagon, (d) the ONB $\{e_1, e_2\}$ in $\mathbb{R}^2$. For each, compare to the theoretical maximum $m - n$. Identify which are full-spark and which are not.

Lab Exercise 10.2: Sub-Frame Reconstruction

Using the triangular frame and $x = (3, -1)^T$:

(a) For each single-erasure pattern, use `reconstruct_after_erasure` and confirm the error is zero (exact reconstruction).

(b) For each single-erasure pattern, also apply the original tight formula with the missing coefficient set to zero. Compute the error and verify it matches the formula $-\frac{1}{A} \langle x, f_j \rangle f_j$.

(c) Which erasure is "cheapest" (smallest error in (b))? Explain in terms of which coefficient carries the least information about x.

Lab Exercise 10.3: The Square's Weakness

For the square frame, run `subframe_spans` on all $\binom{4}{2} = 6$ size-2 erasure patterns and identify exactly which two patterns cause failure. What is special about the erased pairs? Now construct a modified 4-vector frame in $\mathbb{R}^2$ by slightly rotating two vectors so that no pair is antipodal; verify that the modified frame is full-spark (`erasure_number` $= m - n = 2$). Does the modified frame remain tight? (Use `is_etf` from Chapter 9.)

Lab Exercise 10.4: Worst-Case Erasure Error

For each of the triangle, square, and pentagon frames and a fixed random x:

(a) Compute the tight-formula error $\|\hat{x} - x\|$ for every possible single erasure, and report the maximum (worst case) over the m erasure patterns.

(b) Repeat for all $\binom{m}{2}$ double-erasure patterns (where applicable — use only spanning ones for the square).

(c) Verify Theorem 10.5.1 numerically: among the three frames, which achieves the smallest worst-case double-erasure error, and is it the one with smallest coherence?

Lab Exercise 10.5 (Challenge): Full-Spark Random Frames

In theory, a "generic" frame (i.e. one drawn from a continuous probability distribution on $(\mathbb{F}^n)^m$) is almost surely full-spark. Verify this numerically: generate 100 random unit-norm frames with $(n, m) = (2, 5)$ (draw Gaussian and normalize columns), and for each compute `erasure_number`. What fraction achieve $\epsilon(F) = m - n = 3$? Then try $(n, m) = (3, 6)$. Explain briefly why full-spark is generic but not universal.

10.9 Supplementary Exercises

Theoretical Exercises

T1. Prove $\epsilon(F) \leq m - n$ for any frame (a counting argument). Separately, give a full proof that a frame is full-spark if and only if every n-element subset of $\{f_1, \ldots, f_m\}$ is linearly independent.

T2. Prove that the full-spark property is preserved under *any* invertible linear transformation T (not merely orthogonal ones): if $\{f_i\}$ is full-spark, so is $\{Tf_i\}$. Contrast this with tightness (Chapter 4 Supplementary Exercise T2), which is only preserved under *orthogonal* transformations.

T3. Prove that the set of non-full-spark configurations of m vectors in $\mathbb{R}^n$ has Lebesgue measure zero (as a subset of $\mathbb{R}^{n \times m}$), justifying the empirical finding of Lab Exercise 10.5 that full-spark is generic. (Hint: express "some n-subset is dependent" as a finite union of zero sets of polynomials — namely the relevant $n \times n$ minors — and argue each such zero set has measure zero unless the polynomial is identically zero.)

T4. Using $S_J = S - S_E$ (Proposition 10.3.2) and Weyl's inequality for eigenvalues of a perturbed symmetric matrix, prove the computable lower bound $A_J \geq A - \lambda_{\max}(S_E)$ on the sub-frame's lower bound after erasing the index set E, without needing to verify spanning by brute force.

T5. Prove that the erasure number $\epsilon(F)$ is invariant under scaling all frame vectors by a common nonzero constant c: $\epsilon(cF) = \epsilon(F)$.

Computational Exercises

C1. For the triangle, square, and pentagon, directly verify the equivalence from T1: check that "full-spark" (as computed by `erasure_number`) agrees with "every n-subset is linearly independent" (checked directly via `np.linalg.matrix_rank` on each n-subset).

C2. Apply a random invertible (non-orthogonal) 2×2 matrix to the pentagon and confirm the transformed configuration is still full-spark, illustrating T2.

C3. For the triangular frame, erase each single vector in turn, compute the exact sub-frame bound A_J, and compare to the Weyl lower bound $A - \lambda_{\max}(S_E)$ from T4. Report whether the bound is tight (achieved with equality) or merely a strict inequality in each case.

C4. Scale the pentagon by factors $c = 0.5, 2, 10$ and confirm the erasure number is unchanged in every case, verifying T5.

Chapter 11

Frames, Noise, and Quantization

11.1 Overview: Closing the Loop on Robustness

Chapter 1 opened this course with a promise: redundant frames are robust not only to erasures (fulfilled in Chapter 10) but also to *noise* and *quantization*. Chapter 5 began to deliver on this promise algebraically, establishing that the expected squared reconstruction error under additive coefficient noise is $\sigma^2 \|G\|_F^2$ for a dual frame G, and that the canonical dual minimizes this quantity. But Chapter 5 did not answer the most practically important question: by *how much* does a redundant frame improve on a basis?

This chapter answers that question precisely and in two distinct settings. In the **additive noise** model, each coefficient $c_i = \langle x, f_i \rangle$ is corrupted by independent noise of variance σ^2, and the question is how the expected squared reconstruction error scales with the oversampling ratio $r = m/n$. In the **scalar quantization** model, each coefficient is rounded to the nearest multiple of a step size δ, and the question is how the deterministic worst-case reconstruction error scales with r.

Both answers turn out to be the same: a factor of m/n improvement over a basis, achieved by the canonical dual of any tight unit-norm frame. The two models agree because both are controlled by the same quantity — the Frobenius norm $\left\|\tilde{F}\right\|_F^2$ of the canonical dual's synthesis matrix — and we close the loop Chapter 5 opened by making this connection explicit.

11.2 Additive Noise: Precise Error Accounting

We work with the additive noise model introduced in Chapter 5, Section 5.6, but now carry the calculation to its conclusion.

Definition 11.2.1 (Additive coefficient noise model)**.** Given a frame $\{f_i\}_{i=1}^m$ for $\mathbb{F}^n$ and a signal $x \in \mathbb{F}^n$, the receiver observes

$$\tilde{c}_i = \langle x, f_i \rangle + \varepsilon_i, \qquad i = 1, \dots, m,$$

where $\varepsilon_1, \dots, \varepsilon_m$ are independent, mean-zero random variables each with variance σ^2. Given a dual frame $\{g_i\}_{i=1}^m$ with synthesis matrix G, the receiver reconstructs

$$\hat{x} = \sum_{i=1}^m \tilde{c}_i \, g_i = G\tilde{c}.$$

Proposition 11.2.2 (Expected reconstruction error)**.** *In the additive noise model, the reconstruction error satisfies $\hat{x} - x = G\varepsilon$ (where $\varepsilon = (\varepsilon_i)$), and its expected squared norm is*

$$\mathbb{E}\,\|\hat{x} - x\|^2 = \sigma^2\,\|G\|_F^2 = \sigma^2 \sum_{i=1}^{m} \|g_i\|^2\,. \tag{11.1}$$

Proof. Since G reconstructs exactly from noiseless coefficients ($GF^* = I$, Definition 5.2.1), $\hat{x} - x = G(F^*x + \varepsilon) - x = x + G\varepsilon - x = G\varepsilon$. Then

$$\mathbb{E}\,\|G\varepsilon\|^2 = \mathbb{E}\,\mathrm{tr}((G\varepsilon)(G\varepsilon)^T) = \mathrm{tr}(G\,\mathbb{E}[\varepsilon\varepsilon^T]\,G^T) = \sigma^2\,\mathrm{tr}(GG^T) = \sigma^2\,\|G\|_F^2\,,$$

using $\mathbb{E}[\varepsilon\varepsilon^T] = \sigma^2 I_m$ (independent, equal variance) and $\|G\|_F^2 = \mathrm{tr}(GG^T)$. □

Remark. This is exactly the formula derived in Chapter 5 (Section 5.6). Proposition 11.2.2 is recalled here because Chapter 11's goal is to go further: to compute $\|G\|_F^2$ explicitly for the canonical dual of a tight unit-norm frame, and to quantify the resulting improvement over a basis.

11.2.1 The Canonical Dual Norm for Tight Unit-Norm Frames

When $\{f_i\}_{i=1}^m$ is tight with bound A, we know from Chapter 5 that the canonical dual is $\tilde{f}_i = S^{-1}f_i = \frac{1}{A}f_i$ (since $S = AI$), so $\left\|\tilde{f}_i\right\| = \frac{1}{A}\,\|f_i\|$. For unit-norm vectors, $\left\|\tilde{f}_i\right\| = \frac{1}{A} = \frac{n}{m}$. This gives

Proposition 11.2.3 (Canonical dual norm for tight unit-norm frames)**.** *Let $\{f_i\}_{i=1}^m$ be a tight, unit-norm frame for $\mathbb{F}^n$. Then $A = m/n$ (trace formula, Proposition 3.4.2(iv)), and the canonical dual's Frobenius norm satisfies*

$$\left\|\tilde{F}\right\|_F^2 = \sum_{i=1}^{m} \left\|\tilde{f}_i\right\|^2 = m \cdot \frac{1}{A^2} = m \cdot \frac{n^2}{m^2} = \frac{n^2}{m}. \tag{11.2}$$

Proof. Each $\tilde{f}_i = \frac{1}{A}f_i$ has norm $\frac{1}{A}$, so $\sum_i \left\|\tilde{f}_i\right\|^2 = m/A^2 = m \cdot (n/m)^2 = n^2/m$. □

Substituting into Proposition 11.2.2:

Theorem 11.2.4 (Additive noise: the redundancy gain)**.** *For a tight, unit-norm frame with m vectors in $\mathbb{F}^n$, the canonical dual achieves expected squared reconstruction error*

$$\mathbb{E}\,\|\hat{x} - x\|^2 = \sigma^2 \cdot \frac{n^2}{m} = \sigma^2 \cdot \frac{n}{r}, \qquad r := \frac{m}{n} \ \textit{(oversampling ratio)}. \tag{11.3}$$

For comparison, a basis ($m = n$, so $r = 1$) achieves expected squared error $\sigma^2 n$. The tight frame therefore gives a factor-of-r improvement:

$$\frac{\mathbb{E}_{ONB}\,\|\hat{x} - x\|^2}{\mathbb{E}_{frame}\,\|\hat{x} - x\|^2} = r = \frac{m}{n}.$$

Proof. Combine Propositions 11.2.2 and 11.2.3. For the ONB baseline: $F_{\mathrm{ONB}} = I_n$, canonical dual $= I_n$, $\|I_n\|_F^2 = n$, so expected error $= \sigma^2 n = \sigma^2 n/1$ (i.e. $r = 1$). □

Key Idea

Doubling the number of frame vectors halves the noise floor. For a tight unit-norm frame, the expected squared reconstruction error under additive coefficient noise is exactly $\sigma^2 n/r$, where $r = m/n$ is the oversampling ratio. Every doubling of r gives a 3 dB improvement in signal-to-noise ratio — the same gain as averaging r independent measurements, which is exactly what the canonical dual is doing: averaging over the m "views" of x provided by the frame.

Example 11.1: Hexagonal frame in $\mathbb{R}^2$ ($m = 6$, $r = 3$)

The regular hexagon ($m = 6$, $n = 2$) is a tight unit-norm frame with $A = 3$ and canonical dual norms $\left\|\tilde{f}_i\right\| = 1/3$. By Theorem 11.2.4, the expected squared error is $\sigma^2 \cdot 2^2/6 = 2\sigma^2/3$, compared to $\sigma^2 \cdot 2 = 2\sigma^2$ for the standard ONB. The improvement factor is exactly $r = 3$. Equivalently: to achieve the same noise floor as the ONB at σ^2, the hexagonal frame only needs noise standard deviation $\sigma\sqrt{r} = \sigma\sqrt{3}$ — a 73% looser noise tolerance.

11.3 The Dual-Norm Connection

Chapter 5 (the remark following Theorem 5.5.1) promised that "smaller dual-vector norms translate directly into better noise suppression." Propositions 11.2.2 and 11.2.3 now make this precise: the expected squared reconstruction error is $\sigma^2 \sum_i \|g_i\|^2$, and the canonical dual achieves the minimum possible value of this sum, n^2/m, among all valid duals (Theorem 5.5.1 of Chapter 5).

Corollary 11.3.1 (Canonical dual is optimal for additive noise)**.** *Among all dual frames of a given frame $\{f_i\}_{i=1}^m$, the canonical dual $\{\tilde{f}_i = S^{-1} f_i\}$ uniquely minimizes expected squared reconstruction error under independent, equal-variance additive noise.*

Proof. Immediate from Proposition 11.2.2 (error $= \sigma^2 \|G\|_F^2$) and Theorem 5.5.1 ($\left\|\tilde{F}\right\|_F^2 \le \|G\|_F^2$ for every dual G, with equality only when $G = \tilde{F}$). □

This corollary completes the loop opened in Chapter 5: the canonical dual is optimal not merely in the abstract Frobenius-norm sense, but concretely in the sense of minimizing the expected squared error a receiver will observe when reconstructing from noisy measurements.

11.4 Scalar Quantization

Additive noise models corruption by continuous random variables. A different and equally important model is *scalar quantization*: each measured coefficient $\langle x, f_i\rangle$ is rounded to the nearest multiple of a step size δ:

$$\tilde{c}_i = \mathcal{Q}_\delta(\langle x, f_i\rangle) := \delta \left\lfloor \frac{\langle x, f_i\rangle}{\delta} + \frac{1}{2} \right\rfloor .$$

The quantization error $\varepsilon_i := \tilde{c}_i - \langle x, f_i\rangle$ satisfies $|\varepsilon_i| \le \delta/2$ deterministically, but the errors are neither independent nor Gaussian.

Proposition 11.4.1 (Quantization error bound)**.** *For any dual frame $\{g_i\}$ and any scalar quantizer with step δ, the reconstruction error satisfies*

$$\|\hat{x} - x\| = \|G\varepsilon\| \le \frac{\delta}{2}\sum_{i=1}^{m} \|g_i\| \le \frac{\delta}{2}\sqrt{m}\ \|G\|_F\,, \tag{11.4}$$

where the first inequality uses $|\varepsilon_i| \le \delta/2$ and the triangle inequality, and the second uses Cauchy–Schwarz on the sum $\sum_i |\varepsilon_i|\,\|g_i\|$.

Proof. $\|G\varepsilon\| \le \sum_i |\varepsilon_i|\,\|g_i\| \le \frac{\delta}{2}\sum_i \|g_i\| \le \frac{\delta}{2}\sqrt{m}\big(\sum_i \|g_i\|^2\big)^{1/2} = \frac{\delta}{2}\sqrt{m}\,\|G\|_F$, by Cauchy–Schwarz. $\square$

11.4.1 Tight Unit-Norm Frames: The Error Bound

Proposition 11.4.2 (Quantization error bound for tight unit-norm frames)**.** *For the canonical dual of a tight unit-norm frame with m vectors in $\mathbb{F}^n$ and step size δ, the worst-case squared reconstruction error satisfies*

$$\|\hat{x} - x\|^2 = \|G\varepsilon\|^2 \le \|\varepsilon\|^2\ \left\|\tilde{F}\right\|_F^2 \le m\left(\frac{\delta}{2}\right)^2 \frac{n^2}{m} = \frac{n^2\delta^2}{4}. \tag{11.5}$$

For a basis ($m = n$, $G = I_n$, $\|\varepsilon\|^2 \le n(\delta/2)^2$), the corresponding bound is also $n(\delta/2)^2 \cdot n = n^2\delta^2/4$. Both bounds are equal: no factor-of-m/n improvement is guaranteed by the worst-case bound alone.

Proof. By the Cauchy–Schwarz inequality for vectors,

$$\|G\varepsilon\|^2 = \left\|\sum_{i=1}^{m} \varepsilon_i \tilde{f}_i\right\|^2 \le \|\varepsilon\|^2 \sum_{i=1}^{m} \left\|\tilde{f}_i\right\|^2 = \|\varepsilon\|^2\ \left\|\tilde{F}\right\|_F^2,$$

using $\langle \sum_i a_i u_i, \sum_j a_j u_j\rangle \le \big(\sum_i |a_i|^2\big)\big(\sum_i \|u_i\|^2\big)$, which follows from Cauchy–Schwarz on $\mathbb{R}^m$. Since $|\varepsilon_i| \le \delta/2$ for all i, we have $\|\varepsilon\|^2 \le m(\delta/2)^2$. Substituting $\left\|\tilde{F}\right\|_F^2 = n^2/m$ (Proposition 11.2.3) gives the bound $n^2\delta^2/4$. The ONB bound follows identically: $m = n$, $G = I_n$, $\|I_n\|_F^2 = n$, $\|\varepsilon\|^2 \le n(\delta/2)^2$, product $n^2\delta^2/4$. $\square$

Remark. The worst-case bound $n^2\delta^2/4$ does *not* improve with m. This contrast with the additive-noise result requires explanation. In the additive noise model, each ε_i is an independent random variable with $\mathbb{E}[\varepsilon_i^2] = \sigma^2$; the m error contributions $\varepsilon_i\tilde{f}_i$ partially cancel in expectation, giving the average squared error $\sigma^2\sum_i \left\|\tilde{f}_i\right\|^2 = \sigma^2 n^2/m$ — a clear $1/m$ gain. In the quantization model with a *fixed* signal x, the errors $\varepsilon_i = \mathcal{Q}_\delta(\langle x, f_i\rangle) - \langle x, f_i\rangle$ are deterministic and can — in the worst case — be systematically aligned to constructively reinforce rather than cancel, eliminating the averaging effect. In practice, for generic signals x, quantization errors do tend to behave more like independent noise and the observed error is often close to $n^2\delta^2/(4m)$ (a factor-of-m smaller); but this is an empirical observation about typical behavior, not a guaranteed bound. The fundamental difference between average and worst-case guarantees is a recurring theme in information theory and signal processing.

Key Idea

The additive-noise floor $\sigma^2 n^2/m$ has clear $1/m$ scaling: every extra frame vector reduces expected squared error by $1/m$. The quantization bound $n^2\delta^2/4$ does *not* improve with m in the worst case — but in practice, quantization errors behave more like noise, and empirical errors are often close to $n^2\delta^2/(4m)$. **Both models are controlled by the same quantity, $\|\tilde{F}\|_F^2 = n^2/m$,** but the additive-noise gain is a guaranteed bound while the quantization gain is an empirical pattern. This distinction between average and worst-case guarantees is fundamental in information theory.

11.5 Comparing Additive Noise and Quantization

Table 11.1 summarizes the two models for a tight unit-norm frame with m vectors in $\mathbb{F}^n$ and its canonical dual.

	Additive noise	**Scalar quantization**
Error measure	$\mathbb{E}\,\|\hat{x}-x\|^2$	$\|\hat{x}-x\|^2$ (worst-case bound)
Controlling quantity	$\sigma^2\,\|\tilde{F}\|_F^2$	$m(\delta/2)^2\,\|\tilde{F}\|_F^2$
For tight unit-norm frame	$\sigma^2 n^2/m$	$n^2\delta^2/4$ (w.c.); $\approx n^2\delta^2/(4m)$ (typ.)
For ONB ($m=n$)	$\sigma^2 n$	$n^2\delta^2/4$
Improvement over ONB	m/n (guaranteed)	$1\times$ (w.c. bound); $m/n\times$ (empirical)
Minimized by	Canonical dual	Canonical dual

Table 11.1: Additive noise vs. scalar quantization for tight unit-norm frames. The additive-noise gain is a proven average-case result; the quantization improvement of $\approx m/n$ is empirical (worst-case bound does not depend on m).

Remark. The additive-noise gain is m/n in the expected squared error because the m noise contributions $\varepsilon_i\tilde{f}_i$ add incoherently (zero-mean, independent), giving expected squared error $\sigma^2\sum_i\|\tilde{f}_i\|^2 = \sigma^2 n^2/m$ — a clear $1/m$ gain. In the quantization model, the m error terms are *deterministic* and can in principle add constructively in the worst case, eliminating the averaging effect. The proved worst-case bound $n^2\delta^2/4$ is the same for a frame as for a basis; only empirically (for generic signals) does the frame provide the same $\approx m/n$ improvement as in the noise model.

11.6 NumPy Implementation

Listing 11.1: Noise and quantization simulations for tight frames

```
import numpy as np

def tight_unit_norm_frame(n, m, seed=0):
    """
    Build a tight unit-norm frame of m vectors in R^n via gradient
    descent on the frame potential (Chapter 8).
    """
```

```python
    rng = np.random.default_rng(seed)
    V = rng.standard_normal((n, m))
    V = V / np.linalg.norm(V, axis=0)
    for _ in range(2000):
        S = V @ V.T
        V = V - 0.03 * (4 * S @ V)
        V = V / np.linalg.norm(V, axis=0)
    return V

def regular_polygon(m):
    """Tight unit-norm frame in R^2: regular m-gon (m >= 3)."""
    angles = 2 * np.pi * np.arange(m) / m
    return np.vstack([np.cos(angles), np.sin(angles)])

def canonical_dual(F):
    """Canonical dual synthesis matrix: S^{-1} F."""
    return np.linalg.solve(F @ F.T, F)

def scalar_quantize(c, delta):
    """Round each entry of c to the nearest multiple of delta."""
    return delta * np.round(c / delta)

# ------------------------------------------------------------------
# Section 11.2: Additive noise -- verifying the m/n gain factor
# ------------------------------------------------------------------
sigma = 0.3
n = 2

print("=== Additive noise: expected squared error vs m/n ===")
print(f"{'m':>4}  {'r=m/n':>6}  {'predicted':>12}  {'MC mean':>10}  {'gain':>8}")
for m in [2, 3, 4, 6, 8, 12]:
    if m == 2:
        F = np.eye(2)           # ONB: the m=2 "regular polygon" is degenerate
    else:
        F = regular_polygon(m)
    G = canonical_dual(F)
    predicted = sigma**2 * np.trace(G @ G.T)

    # Monte Carlo
    rng = np.random.default_rng(42)
    x_test = np.array([1.0, -0.5])
    errs = []
    for _ in range(100_000):
        eps = rng.normal(0, sigma, m)
        x_hat = G @ (F.T @ x_test + eps)
        errs.append(np.sum((x_hat - x_test)**2))

    mc_mean = np.mean(errs)
    gain = (sigma**2 * n) / mc_mean if m > 2 else 1.0
    print(f"{m:>4}  {m/n:>6.1f}  {predicted:>12.5f}  {mc_mean:>10.5f}  {gain:>8.3f}x")

```

```python
# ----------------------------------------------------------------------
# Section 11.4: Scalar quantization -- verifying the proved bound n^2*delta^2/4
# and comparing to the empirically observed (approx) n^2*delta^2/(4m) behavior
# ----------------------------------------------------------------------
delta = 0.2

print("\n=== Scalar quantization: proved bound vs. empirical ===")
print(f"{'m':>4}  {'proved n^2d^2/4':>16}  {'MC max':>10}  {'holds':>6}  {'n^2d^2/4m (typical)':>20}")
rng = np.random.default_rng(1)
n = 2
for m in [3, 4, 6, 8, 12]:
    F = regular_polygon(m)
    G = canonical_dual(F)
    bound = (n**2 * delta**2) / 4        # CORRECT proved bound (Prop. 11.4.1)
    typical = (n**2 * delta**2) / (4*m)  # empirical approximation only

    # Sample many x and quantize
    X = rng.standard_normal((n, 50_000))
    X = X / np.linalg.norm(X, axis=0)   # unit-norm test signals
    C_exact  = F.T @ X
    C_quant  = scalar_quantize(C_exact, delta)
    errors   = np.sum((G @ C_quant - X)**2, axis=0)
    mc_max   = errors.max()

    print(f"{m:>4}  {bound:>16.6f}  {mc_max:>10.6f}  {str(mc_max <= bound + 1e-9):>6}  {typical:>20.6f}")

# ----------------------------------------------------------------------
# Example 11.1: Hexagonal frame (m=6, n=2)
# ----------------------------------------------------------------------
print("\n=== Example 11.1: Hexagonal frame ===")
m, n = 6, 2
F_hex = regular_polygon(m)
G_hex = canonical_dual(F_hex)
S_hex = F_hex @ F_hex.T
A_hex = np.linalg.eigvalsh(S_hex).mean()
dual_norms = np.linalg.norm(G_hex, axis=0)
print(f"  Frame bound A = {A_hex:.4f}  (expected m/n = {m/n:.4f})")
print(f"  Dual-vector norms: {dual_norms.round(4)}  (expected n/m = {n/m:.4f})")
print(f"  ||tilde_F||_F^2 = {np.trace(G_hex @ G_hex.T):.4f}"
      f"  (expected n^2/m = {n**2/m:.4f})")
sigma_test = 0.3
print(f"  Predicted noise floor: sigma^2 * n^2/m ="
      f" {sigma_test**2 * n**2/m:.4f}")
print(f"  ONB comparison:        sigma^2 * n   ="
      f" {sigma_test**2 * n:.4f}")
print(f"  Improvement factor: {(sigma_test**2 * n) / (sigma_test**2 * n**2/m):.1f}x"
      f"  (= m/n = {m/n:.1f})")
```

11.7 Chapter Summary

- **Additive noise model.** The expected squared reconstruction error using dual frame G is $\sigma^2 \|G\|_F^2$ (Proposition 11.2.2). The canonical dual uniquely minimizes this (Corollary 11.3.1, closing the loop from Chapter 5).
- **Tight unit-norm frames.** The canonical dual has $\left\|\tilde{F}\right\|_F^2 = n^2/m$ (Proposition 11.2.3), giving expected squared error $\sigma^2 n^2/m = \sigma^2 n/r$ where $r = m/n$ (Theorem 11.2.4). This is a factor-of-r improvement over an ONB.
- **Scalar quantization.** With step δ and the canonical dual, the proved worst-case squared reconstruction error is bounded by $n^2\delta^2/4$ (Proposition 11.4.2) — the same as for a basis. The factor-of-m/n improvement is observed empirically (typical errors $\approx n^2\delta^2/(4m)$) but is not guaranteed in the worst case; see the remark in Section 11.4.1.
- **Both models, same mechanism.** The factor-of-m/n improvement comes from the same source in both models: dual vectors of norm n/m (rather than norm 1 for a basis), so each error contribution $\varepsilon_i \tilde{f}_i$ is smaller by n/m, and there are m of them. The controlling quantity is $\left\|\tilde{F}\right\|_F^2 = n^2/m$ in both cases (Table 11.1).

11.8 Lab Exercises

Lab Exercise 11.1: Reproducing the Noise Floor Table

Implement `regular_polygon`, `canonical_dual`, and the Monte Carlo loop from Section 11.6. For $n = 2$ and $m = 2, 3, 4, 6, 8, 12, 20$, compute and tabulate: (a) the predicted noise floor $\sigma^2 n^2/m$, (b) the Monte Carlo mean squared error (use $\sigma = 0.3$, 10^5 trials), and (c) the gain factor over the ONB. Verify that the gain equals m/n to at least two decimal places for all $m \geq 3$.

Lab Exercise 11.2: Non-Tight Frame Penalty

Consider the non-tight frame $\{h_1, h_2, h_3\}$ with $h_1 = e_1$, $h_2 = e_2$, $h_3 = (e_1 + e_2)/\sqrt{2}$ in $\mathbb{R}^2$ (frame bounds $A = 1$, $B = 2$; note this differs from Chapter 5's frame by normalizing h_3):

(a) Compute the canonical dual $\tilde{H}$ and its Frobenius norm $\left\|\tilde{H}\right\|_F^2$.

(b) Compare $\left\|\tilde{H}\right\|_F^2$ to the tight unit-norm prediction $n^2/m = 4/3$ for $n = 2$, $m = 3$. Which is larger? Does the non-tight frame get a better or worse noise floor than the tight triangular frame with the same m?

(c) Run 10^5 Monte Carlo noise trials (with $\sigma = 0.3$) and verify your prediction from (a).

(d) Explain why the non-tight frame performs worse: what is it about the frame geometry (not just the formula) that leads to a larger dual Frobenius norm?

Lab Exercise 11.3: Quantization Simulation

For $n = 2$ and $m \in \{3, 4, 6, 8, 12\}$, use the code in Section 11.6 to:

(a) Verify that the *proved* worst-case quantization bound $n^2\delta^2/4$ (Proposition 11.4.2) is never violated across 10^5 random unit-norm test signals (use $\delta = 0.2$). Confirm this holds for all m.

(b) Plot the empirical maximum error (over all test signals) as a function of m on the same axes as the proved bound $n^2\delta^2/4$ (constant in m) and the empirical approximation $n^2\delta^2/(4m)$ (decays as $1/m$). Which does the empirical maximum track more closely?

(c) The proved bound is a *constant* (independent of m), but the empirical maximum decays with m. Explain why in terms of the remark in Section 11.4.1: when does the worst-case alignment of quantization errors actually arise? Then try to construct, for a fixed frame ($m = 3$, $n = 2$), a signal x and step size δ such that the quantization errors ε_i all have the same sign and the reconstruction error is noticeably larger than $n^2\delta^2/(4m)$. Report the ratio of actual to "typical" error for your constructed x.

Lab Exercise 11.4: Higher Dimensions

Extend the analysis to $n = 3$ and $m = 6, 9, 12, 18$. Use `tight_unit_norm_frame` (gradient-descent construction from Section 11.6) to build tight frames for $m > 5$ where regular-polygon constructions no longer apply. For each (n, m) pair:

(a) Verify $\left\|\tilde{F}\right\|_F^2 \approx n^2/m$ for the gradient-descent tight frame.

(b) Run Monte Carlo noise trials and compare to the predicted noise floor.

(c) Plot the noise floor as a function of m for fixed $n = 3$, and compare the slope to the theoretical $\sim 1/m$ decay.

Lab Exercise 11.5 (Challenge): Alternate Duals and Noise

Theorem 5.5.1 of Chapter 5 guarantees the canonical dual minimizes $\|G\|_F^2$ and hence the noise floor. How much worse can alternate duals be?

(a) For the triangular frame, parametrize the full dual family $G(w) = \tilde{F} + wk^T$ (Chapter 6, Proposition 6.2.1) and compute $\|G(w)\|_F^2 = 4/3 + \|w\|^2$ as a function of $w \in \mathbb{R}^2$.

(b) For $\|w\| = 1$ (one unit away from the canonical dual in the parameter space), compute the predicted noise floor $\sigma^2(4/3 + 1)$ and compare to the canonical minimum $\sigma^2 \cdot 4/3$. What is the percentage penalty for using this sub-optimal dual?

(c) The Chapter 6 analysis showed there is a special point $w^* = (1, 1)/\sqrt{3}$ where the error ellipse for the $\{h_1, h_2, h_3\}$ frame becomes circular (isotropic). Compute $\|G(w^*)\|_F^2$ for this point and compare to the canonical dual's noise floor. This illustrates that minimizing total noise and minimizing noise anisotropy are two different objectives that generally require two different dual choices.

11.9 Supplementary Exercises

Theoretical Exercises

T1. Suppose noise is added directly to the *signal* x (i.e. the receiver observes $x + \varepsilon$ for $\varepsilon \in \mathbb{R}^n$, *before* any frame analysis takes place), rather than to the frame coefficients. Prove that in this model, redundancy provides *no* benefit whatsoever: the reconstruction error is $\|\varepsilon\|$ regardless of which frame or dual is used. Contrast with Theorem 11.2.4, and identify precisely where in the chain $x \to (\langle x, f_i\rangle)_i \to \hat{x}$ redundancy's benefit actually enters.

T2. Prove that the *worst-case* quantization error is controlled by the operator norm $\|G\|_{\mathrm{op}}$ (largest singular value) of the dual synthesis matrix, not the Frobenius norm $\|G\|_F$ used for *expected* additive-noise error: show $\|G\varepsilon\|_2 \le \|G\|_{\mathrm{op}} \|\varepsilon\|_2$ for any bounded quantization error vector ε.

T3. Prove that for a tight, unit-norm frame with m vectors in $\mathbb{R}^n$, the canonical dual's operator norm is exactly $\left\|\tilde{F}\right\|_{\mathrm{op}} = \sqrt{n/m}$. Compare the rate at which this quantity decays in m (like $1/\sqrt{m}$) to the rate at which the expected-error-controlling quantity $\left\|\tilde{F}\right\|_F^2 = n^2/m$ decays (like $1/m$), and explain the practical implication for worst-case versus average-case robustness.

T4. Prove the general (not necessarily tight) two-sided bound $1/\sqrt{B} \le \left\|\tilde{F}\right\|_{\mathrm{op}} \le 1/\sqrt{A}$ for the canonical dual's operator norm, in terms of the original frame's bounds A, B.

T5. For the $m = n+1$ dual family $G(w) = \tilde{F} + wk^T$ of Chapter 6, prove or disprove: the operator norm $\|G(w)\|_{\mathrm{op}}$, like the Frobenius norm, is uniquely minimized at $w = 0$.

Computational Exercises

C1. For $n = 2$, $m = 3, 6, 12, 24$, compute $\left\|\tilde{F}\right\|_{\mathrm{op}}$ for the regular m-gon and confirm it matches $\sqrt{n/m}$ (T3), plotting both $\left\|\tilde{F}\right\|_{\mathrm{op}}$ and $\left\|\tilde{F}\right\|_F^2$ against m on a log-log scale to compare decay rates.

C2. For $\{h_1, h_2, h_3\}$, compute $\left\|\tilde{F}\right\|_{\mathrm{op}}$ directly and confirm it lies within the bounds $1/\sqrt{B}$ and $1/\sqrt{A}$ from T4.

C3. For the triangular frame's dual family $G(w)$, compute $\|G(w)\|_{\mathrm{op}}$ over a grid of w values and confirm numerically whether it is minimized at $w = 0$, resolving T5.

C4. Simulate the signal-level noise model from T1 directly: add Gaussian noise to x itself, "reconstruct" by simply returning the noisy observation, and confirm the error equals $\|\varepsilon\|$ regardless of which frame's coefficients (if any) are computed downstream.

Chapter 12

Compressed Sensing

12.1 Overview: From Redundancy to Sparsity

Lab Exercise 5.5 of Chapter 5 ended with the following observation: given the frame $\{h_1, h_2, h_3\}$ in $\mathbb{R}^2$, there is not a unique coefficient vector $c \in \mathbb{R}^3$ reconstructing a given signal x. Rather, there is an entire affine family $c' = c + t\,k$ (where k spans $\ker H$), and different choices of t produce different coefficient vectors, all correctly representing x. The exercise asked you to minimize $\|c'\|_1$ over this family and observe that the minimizer is *sparse* — it has an exact zero entry. The comment at the end read: "this phenomenon — ℓ^1 minimization promoting exact sparsity — is the starting point of compressed sensing, which we will touch on again in Chapter 12."

This is that chapter. Compressed sensing is a framework for acquiring signals using *far fewer measurements than the ambient dimension* by exploiting the prior knowledge that the signal is sparse. Frames are central to the theory in two ways: as the *sensing matrices* (the $n \times m$ matrices A that measure a sparse signal $x \in \mathbb{R}^m$ via $y = Ax \in \mathbb{R}^n$, with $n \ll m$), and as the *dictionaries* in which sparse representations are sought. The frame-theoretic tools we have developed — coherence (Chapter 9), frame bounds (Chapter 3), and the $\ker F$ parametrization of redundant representations (Chapter 5) — appear throughout.

This chapter is an introduction rather than a complete treatment. Its goal is to show precisely how the frame-theoretic ideas of this course are the right language for stating and understanding compressed sensing, and to give you enough machinery to implement and test sparse recovery algorithms numerically.

12.2 Sparsity and Redundant Representations

Definition 12.2.1 (s-sparse vector)**.** A vector $c \in \mathbb{F}^m$ is *s-sparse* if at most s of its entries are nonzero: $|\operatorname{supp}(c)| \le s$, where $\operatorname{supp}(c) := \{i \,:\, c_i \ne 0\}$.

Definition 12.2.2 (ℓ^1 norm and ℓ^0 "norm")**.** The ℓ^1 *norm* of $c = (c_1, \ldots, c_m) \in \mathbb{F}^m$ is $\|c\|_1 := \sum_{i=1}^m |c_i|$. The ℓ^0 *count* $\|c\|_0 := |\operatorname{supp}(c)|$ counts the nonzero entries (it is not a norm, but the notation is standard).

12.2.1 Sparsity in the Frame Setting

Recall from Chapter 5 (Theorem 5.4.1) that when a frame $\{f_i\}_{i=1}^m$ has more vectors than the dimension ($m > n$), every signal $x \in \mathbb{F}^n$ has infinitely many coefficient representations: all vectors in the affine family $c + \ker F$, where $c = F^*x$ is the analysis operator's output. Among all these representations, we can ask: which one is *sparsest*?

Definition 12.2.3 (Sparse representation problem)**.** Given a frame $\{f_i\}_{i=1}^m$ (synthesis matrix F) and a signal $x \in \mathbb{F}^n$, the *sparse representation problem* is

$$\underset{c \in \mathbb{F}^m}{\text{minimize}} \, \|c\|_0 \quad \text{subject to} \quad Fc = x. \tag{12.1}$$

Problem (12.1) is combinatorially hard in general: finding the sparsest solution requires, in the worst case, checking all $\binom{m}{s}$ subsets of size s for $s = 1, 2, \ldots$ until one spanning x is found. The key insight of compressed sensing is that under mild conditions, replacing the ℓ^0 count with the ℓ^1 norm gives the same answer, and ℓ^1 minimization is a *convex optimization problem* solvable efficiently.

Definition 12.2.4 (Basis pursuit)**.** The *basis pursuit* problem is

$$\underset{c \in \mathbb{F}^m}{\text{minimize}} \, \|c\|_1 \quad \text{subject to} \quad Fc = x. \tag{12.2}$$

Remark. Basis pursuit is a convex program: $\|c\|_1$ is a convex function and $\{c : Fc = x\}$ is an affine (hence convex) constraint set. Convex programs are solvable in polynomial time (e.g. via interior-point methods), and basis pursuit can be recast as a linear program that any standard LP solver handles efficiently. In NumPy/SciPy, it can be solved via `scipy.optimize.linprog` or the more convenient `cvxpy` library. We show both approaches in Section 12.6.

Example 12.1: The triangular frame — ℓ^1 finds the sparsest expansion

Consider the triangular frame $f_1 = (1,0)^T$, $f_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})^T$, $f_3 = (-\frac{1}{2}, -\frac{\sqrt{3}}{2})^T$ and $x = f_1 = (1,0)^T$. The canonical coefficient vector is $c = \frac{2}{3}(1, -\frac{1}{2}, -\frac{1}{2})^T = (\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3})^T$ (with no zeros). The kernel of F is spanned by $k = (1,1,1)^T/\sqrt{3}$, so all representations are

$$c(t) = c + tk = \left(\tfrac{2}{3} + \tfrac{t}{\sqrt{3}},\ -\tfrac{1}{3} + \tfrac{t}{\sqrt{3}},\ -\tfrac{1}{3} + \tfrac{t}{\sqrt{3}}\right)^T.$$

The function $\|c(t)\|_1$ is piecewise linear and is minimized at $t^* = 1/\sqrt{3}$, giving

$$c^* = c(t^*) = (1,\ 0,\ 0)^T.$$

This is the unique 1-sparse solution (the sparsest possible), correctly identifying $x = f_1$ as a scalar multiple of the first frame vector. Its ℓ^1 norm is 1, compared to the canonical vector's $\|c\|_1 = \frac{2}{3} + \frac{1}{3} + \frac{1}{3} = \frac{4}{3}$.
The proof is a direct calculation: with $t^* = 1/\sqrt{3}$, the second and third entries both equal $-\frac{1}{3} + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} = -\frac{1}{3} + \frac{1}{3} = 0$, so $c^* = (1,0,0)^T$ exactly.

Key Idea

ℓ^1 minimization over $\ker F$ finds the sparsest representation of x in the frame. The canonical coefficients $c = F^*\tilde{x}$ (from the analysis operator) minimize $\|c\|_2$ (Chapter 5, Theorem 5.5.1), but not $\|c\|_1$. ℓ^1 minimization — basis pursuit — solves a different problem and generally finds a sparser answer. These are two orthogonal objectives, and which one you want depends on your application.

12.3 Compressed Sensing: The Measurement Model

Basis pursuit finds sparse expansions in a given frame. Compressed sensing goes further: it acquires a signal $x \in \mathbb{R}^m$ using far fewer than m measurements, and recovers it exactly by exploiting sparsity.

Definition 12.3.1 (Compressed sensing model)**.** Let $A \in \mathbb{R}^{n\times m}$ with $n \ll m$ be the *sensing matrix* (or *measurement matrix*). The receiver observes $y = Ax \in \mathbb{R}^n$ (noiseless) or $y = Ax+e \in \mathbb{R}^n$ (noisy, $\|e\|_2 \le \eta$). Given y, A, and the prior that x is s-sparse ($\|x\|_0 \le s$), recover x.

Without the sparsity prior, $y = Ax$ with $n < m$ is underdetermined and has infinitely many solutions. With the sparsity prior, the system can be uniquely solvable — but finding the sparsest solution is ℓ^0-hard. Basis pursuit gives the tractable surrogate:

$$\hat{x} = \operatorname*{argmin}_{z\in\mathbb{R}^m} \|z\|_1 \quad \text{subject to} \quad Az = y. \tag{12.3}$$

Remark. The columns of A form a frame for $\mathbb{R}^n$ (assuming A has rank n, which is necessary for any reconstruction to be possible). Here the frame is $\{a_i\}_{i=1}^m$ where a_i is the i-th column of A, and the measurement $y_j = \langle x, a_j^{\text{row}}\rangle$ is the inner product of the signal x with the j-th *row* of A. Both perspectives — columns as frame vectors for $\mathbb{R}^n$, rows as measurement functionals on $\mathbb{R}^m$ — appear in the literature; we use the row perspective throughout.

12.4 When Does Basis Pursuit Recover Sparse Signals?

The central question is: for which matrices A does basis pursuit (12.3) recover every s-sparse signal x exactly? Two complementary answers are the *coherence condition* and the *restricted isometry property*.

12.4.1 The Coherence Condition

Recall coherence from Chapter 9 (Definition 9.2.1): the coherence of a unit-column-norm matrix A is $\mu(A) := \max_{i\neq j} |\langle a_i, a_j\rangle|$. Small coherence means the columns of A are "maximally incoherent" — no two look alike.

Theorem 12.4.1 (Coherence condition for exact recovery)**.** *Let $A \in \mathbb{R}^{n\times m}$ have unit-norm columns and coherence μ. If $x \in \mathbb{R}^m$ is s-sparse with*

$$s < \frac{1}{2}\left(1 + \frac{1}{\mu}\right), \tag{12.4}$$

then basis pursuit (12.3) *with $y = Ax$ recovers x exactly.*

Proof sketch. We show that if $s < (1 + 1/\mu)/2$ then x is the *unique* s-sparse solution to $Az = y$, hence the ℓ^0 problem has a unique solution; a separate argument (which we omit) shows that in this case basis pursuit gives the same answer as the ℓ^0 problem.

Suppose $Az = y = Ax$ with $z \neq x$ and $\|z\|_0 \le s$. Then $A(z - x) = 0$ with $h := z - x \neq 0$ and $\|h\|_0 \le 2s$. Write $h = \sum_{i\in T} h_i e_i$ where $|T| \le 2s$. Then $0 = Ah = \sum_{i\in T} h_i a_i$. Taking the inner product with a_j for $j \in T$:

$$\|a_j\|^2 |h_j| = \left| \sum_{i\in T, i\neq j} h_i \langle a_i, a_j\rangle \right| \le \mu \sum_{i\in T, i\neq j} |h_i| \le \mu(2s-1)\|h\|_\infty ,$$

since $|\langle a_i, a_j \rangle| \le \mu$ and there are at most $2s-1$ terms in the sum. Since $\|a_j\|^2 = 1$ and this holds for all $j \in T$, we get $\|h\|_\infty \le \mu(2s-1)\|h\|_\infty$, which requires $\mu(2s-1) \ge 1$, i.e. $s \ge (1+1/\mu)/2$. This contradicts our assumption $s < (1+1/\mu)/2$. Hence no such $z \ne x$ exists, and the s-sparse solution is unique. □

Remark. Condition (12.4) is sufficient but not necessary: basis pursuit can and does recover signals even when $s \ge (1+1/\mu)/2$ in practice. The bound is pessimistic because it assumes the worst-case alignment of s columns. Empirically, random matrices with $n = O(s \log m)$ measurements recover s-sparse signals with high probability even when s is much larger than the coherence bound allows.

Example 12.2: Coherence bound for the triangular frame

The triangular frame has coherence $\mu = \frac{1}{2}$ (all off-diagonal inner products equal $-\frac{1}{2}$, so $|\langle f_i, f_j \rangle| = \frac{1}{2}$ for $i \ne j$; this is the Welch bound for $(n,m) = (2,3)$, Chapter 9). The coherence condition guarantees recovery of s-sparse signals for

$$s < \tfrac{1}{2}\big(1 + \tfrac{1}{1/2}\big) = \tfrac{1}{2}(1+2) = \tfrac{3}{2},$$

i.e. $s \le 1$. Since the ambient dimension is $n = 2$, this means basis pursuit recovers any 1-sparse signal — any multiple of a single frame vector — exactly. Example 12.2.1 confirmed this directly for $x = f_1$.

12.4.2 The Restricted Isometry Property

The coherence condition is simple to state and check, but it gives pessimistic bounds that preclude recovery for larger sparsity levels. The *restricted isometry property* (RIP) captures a more refined condition that allows recovery of much sparser signals.

Definition 12.4.2 (Restricted isometry property, RIP)**.** A matrix $A \in \mathbb{R}^{n\times m}$ satisfies the *RIP of order s* with constant $\delta_s \in (0,1)$ if, for every s-sparse vector $z \in \mathbb{R}^m$,

$$(1-\delta_s)\|z\|_2^2 \ \le\ \|Az\|_2^2 \ \le\ (1+\delta_s)\|z\|_2^2. \tag{12.5}$$

The smallest valid δ_s is called the *RIP constant* of order s.

Remark. Compare (12.5) to the frame inequality (Definition 2.1.1): $A\|z\|^2 \le \|F^*z\|^2 \le B\|z\|^2$ for all z. The RIP is a *restricted* frame inequality: it asks for the frame condition to hold, but only for s-sparse vectors z (not for all z). The RIP constant δ_s is the worst-case deviation from an isometry over all s-sparse vectors — it measures how far A is from being a perfect isometry on sparse signals. If $\delta_s = 0$, then A maps every s-sparse signal to a vector of the same norm: perfect preservation. Larger δ_s means more distortion.

In frame language: A satisfies RIP-s with constant δ_s if and only if *every* $n \times s$ submatrix of A (formed by taking s columns) has frame bounds $A_{\min} \ge 1-\delta_s$ and $B_{\max} \le 1+\delta_s$. This is exactly the condition that every s-column subframe of A is a well-conditioned frame for $\mathbb{R}^n$.

Theorem 12.4.3 (RIP guarantees exact recovery)**.** *If A satisfies the RIP of order $2s$ with constant $\delta_{2s} < \sqrt{2}-1 \approx 0.414$, then for every s-sparse $x \in \mathbb{R}^m$, basis pursuit with $y = Ax$ recovers x exactly.*

We state this theorem without proof, as the argument is substantially more involved than the coherence condition proof and requires tools beyond the scope of this course. The key point is that the threshold $\delta_{2s} < \sqrt{2}-1$ is sharp: there exist matrices with δ_{2s} slightly above this threshold for which basis pursuit fails on some s-sparse signal.

Proposition 12.4.4 (Coherence and RIP)**.** *If A has unit-norm columns and coherence μ, then*

$$\delta_s \;\le\; (s-1)\mu$$

for all $s \ge 1$. In particular, if $\mu < (\sqrt{2}-1)/(2s-1)$, then $\delta_{2s} \le (2s-1)\mu < \sqrt{2}-1$, so the RIP condition for exact recovery is satisfied.

Proof. Let $z \in \mathbb{R}^m$ be s-sparse with support T, $|T| = s$. Then

$$\|Az\|^2 = \sum_{i\in T}\sum_{j\in T} z_i z_j \langle a_i, a_j\rangle = \|z\|^2 + \sum_{i\ne j,\, i,j\in T} z_i z_j \langle a_i, a_j\rangle .$$

The cross terms satisfy $|z_i z_j \langle a_i, a_j\rangle| \le \mu |z_i z_j|$, so

$$\left|\|Az\|^2 - \|z\|^2\right| \le \mu \sum_{i\ne j\in T} |z_i z_j| \le \mu(s-1)\|z\|^2 ,$$

where the last step uses $\sum_{i\ne j} |z_i z_j| \le (s-1)\sum_i z_i^2 = (s-1)\|z\|^2$ (a consequence of the AM-GM or Cauchy-Schwarz inequality). Hence $\delta_s \le (s-1)\mu$. □

Remark. Proposition 12.4.4 shows that coherence controls the RIP constant, but gives a weaker bound than direct RIP analysis. Random matrices — Gaussian, Bernoulli, or random partial Fourier — achieve RIP constants $\delta_s \lesssim \sqrt{s/n}$ with high probability when $n = O(s\log(m/s))$ measurements are taken. This is far better than coherence-based bounds, which require $n = O(s^2\mu^{-2})$.

12.5 The Frame-Theoretic View of Sensing

Compressed sensing and frame theory are two perspectives on the same algebraic structure. This section makes the connections explicit.

Proposition 12.5.1 (The sensing matrix is a frame)**.** *A sensing matrix $A \in \mathbb{R}^{n\times m}$ with* $\operatorname{rank}(A) = n$ *and columns $\{a_i\}_{i=1}^m$ forms a frame for $\mathbb{R}^n$. The RIP constant δ_s is the worst-case deviation of s-column sub-frame bounds from* $(1,1)$*:*

$$\delta_s = \max_{|T|=s} \max\bigl(1 - \lambda_{\min}(A_T^T A_T),\ \lambda_{\max}(A_T^T A_T) - 1\bigr),$$

where A_T denotes the submatrix of A with columns indexed by T.

Proof. A has rank n by assumption, so its columns span $\mathbb{R}^n$ and Theorem 2.2.1 gives a frame. The RIP formula is immediate from Definition 12.4.2: $\|Az\|^2 = \|A_T c_T\|^2 = c_T^T A_T^T A_T c_T$ where $c_T = (z_i)_{i\in T}$, and the frame bounds of $\{a_i\}_{i\in T}$ are $\lambda_{\min}(A_T^T A_T)$ and $\lambda_{\max}(A_T^T A_T)$ (by Theorem 3.5.1 applied to the sub-frame). □

Key Idea

The RIP is the frame condition, restricted to s-column submatrices. A sensing matrix A satisfies RIP-s if and only if every s-column subframe $\{a_i\}_{i\in T}$ is a $(1-\delta_s, 1+\delta_s)$-frame for $\mathbb{R}^n$. Small δ_s means all s-column subframes are nearly tight: every s columns span $\mathbb{R}^n$ approximately isometrically. The closer δ_s is to zero, the better the measurement matrix preserves the geometry of sparse signals.

Proposition 12.5.2 (ETFs are optimal sensing matrices)**.** *Among all $n \times m$ matrices with unit-norm columns, ETFs (equiangular tight frames, Chapter 9) have the smallest possible worst-case coherence $\mu = \sqrt{(m-n)/(n(m-1))}$ (the Welch bound). Since Proposition 12.4.4 bounds $\delta_s \leq (s-1)\mu$, ETFs give the best coherence-based RIP bounds among all unit-norm sensing matrices of the same size. They are therefore optimal sensing matrices in the coherence-minimization sense.*

12.6 NumPy Implementation

Listing 12.1: Basis pursuit and compressed sensing in Python

```python
import numpy as np
from scipy.optimize import linprog, minimize_scalar
from scipy.linalg import null_space

# ---------------------------------------------------------------------
# Example 12.1: L1 minimization in the triangular frame
# ---------------------------------------------------------------------
f1 = np.array([1.0, 0.0])
f2 = np.array([-0.5, np.sqrt(3)/2])
f3 = np.array([-0.5, -np.sqrt(3)/2])
F = np.column_stack([f1, f2, f3])   # synthesis matrix (2 x 3)

x = f1   # signal to represent: x = (1, 0)^T

# Canonical coefficients via the frame operator
S = F @ F.T
c_can = F.T @ np.linalg.solve(S, x)
print(f"Canonical c = {c_can.round(4)}")   # (2/3, -1/3, -1/3)
print(f"  ||c||_1 = {np.sum(np.abs(c_can)):.4f}")  # 4/3

# L1 minimization over the affine family c + t*k
k = null_space(F)[:, 0]   # unit kernel vector (1,1,1)/sqrt(3)

def l1_objective(t):
    return np.sum(np.abs(c_can + t * k))

result = minimize_scalar(l1_objective, bounds=(-3, 3), method='bounded')
t_opt = result.x
c_l1 = c_can + t_opt * k
print(f"L1-optimal c = {c_l1.round(6)}")   # (1, 0, 0) to machine precision
print(f"  ||c||_1 = {np.sum(np.abs(c_l1)):.4f}")   # 1.0
print(f"  Reconstruction ok: {np.allclose(F @ c_l1, x)}")

# ---------------------------------------------------------------------
# Compressed sensing: basis pursuit via linear programming
# ---------------------------------------------------------------------
def basis_pursuit(A, y):
    """
    Solve min ||c||_1 s.t. Ac = y via linear programming.

    Reformulation: min 1^T t
                   s.t. Ac = y,  -t <= c <= t,  t >= 0
```

```
    Variables: [c (m,), t (m,)]
    """
    n, m = A.shape
    # Objective: minimize sum of t_i
    c_obj = np.zeros(2 * m); c_obj[m:] = 1.0
    # Equality: A c = y
    A_eq = np.hstack([A, np.zeros((n, m))])
    b_eq = y
    # Inequality: c_i - t_i <= 0  and  -c_i - t_i <= 0
    A_ub = np.block([[ np.eye(m), -np.eye(m)],
                      [-np.eye(m), -np.eye(m)]])
    b_ub = np.zeros(2 * m)
    # Bounds: c_i free, t_i >= 0
    bounds = [(None, None)] * m + [(0, None)] * m
    res = linprog(c_obj, A_ub=A_ub, b_ub=b_ub,
                  A_eq=A_eq, b_eq=b_eq, bounds=bounds, method='highs')
    return res.x[:m]

def coherence(A):
    """Coherence of matrix A with unit-norm columns."""
    G = A.T @ A
    off = np.abs(G - np.diag(np.diag(G)))
    return off.max()

def rip_constant(A, s):
    """
    Estimate RIP-s constant: worst-case submatrix frame-bound deviation.
    Samples up to 2000 random s-subsets for efficiency.
    """
    from itertools import combinations
    import random
    n, m = A.shape
    all_subsets = list(combinations(range(m), s))
    if len(all_subsets) > 2000:
        all_subsets = random.sample(all_subsets, 2000)
    delta = 0.0
    for T in all_subsets:
        A_T = A[:, list(T)]
        eigs = np.linalg.eigvalsh(A_T.T @ A_T)
        delta = max(delta, abs(eigs[-1] - 1), abs(1 - eigs[0]))
    return delta

# ------------------------------------------------------------------
# Demo: recover a 2-sparse signal in R^40 from 15 measurements
# ------------------------------------------------------------------
rng = np.random.default_rng(42)
n_meas, m_sig, s_spar = 15, 40, 2

# Sensing matrix: random Gaussian with normalized columns
A = rng.standard_normal((n_meas, m_sig))
A = A / np.linalg.norm(A, axis=0)

mu_A = coherence(A)
delta_2s = rip_constant(A, 2 * s_spar)
```

```python
print(f"\nSensing matrix ({n_meas}x{m_sig}): mu={mu_A:.4f}, "
      f"estimated delta_{{2s}}={delta_2s:.4f}")
print(f"Coherence bound: s < {0.5*(1 + 1/mu_A):.2f}")
print(f"RIP recovery condition: delta_{{2s}} < {np.sqrt(2)-1:.4f}, "
      f"holds: {delta_2s < np.sqrt(2)-1}")
# Note: the RIP condition is sufficient, not necessary.
# Basis pursuit can succeed even when delta_{2s} >= sqrt(2)-1;
# the condition is a worst-case guarantee over ALL 2s-sparse signals.

# True sparse signal
x_true = np.zeros(m_sig)
x_true[5] = 3.0; x_true[17] = -2.0
y_obs = A @ x_true

# Basis pursuit
x_rec = basis_pursuit(A, y_obs)
err = np.linalg.norm(x_rec - x_true)
print(f"\nBasis pursuit recovery error: {err:.2e}")
print(f"Recovered support: {np.where(np.abs(x_rec) > 1e-3)[0].tolist()}"
      f"  (true: [5, 17])")
print("(Recovery succeeds even though RIP condition is not met --"
      " the condition is sufficient, not necessary.)")
```

12.7 Chapter Summary

- **Sparsity in the frame setting.** When $m > n$, every signal x has infinitely many representations c with $Fc = x$; they form an affine family $c + \ker F$ (Theorem 5.4.1). *Basis pursuit* — ℓ^1 minimization over this family — selects the sparsest, as Example 12.2.1 illustrates explicitly: ℓ^1 finds the unique 1-sparse expansion of $x = f_1$, with ℓ^1 norm 1 vs. the canonical vector's $\frac{4}{3}$.

- **The compressed sensing model.** A sensing matrix $A \in \mathbb{R}^{n\times m}$ ($n \ll m$) acquires s-sparse signals via $y = Ax$. Basis pursuit (12.3) recovers x exactly from y under suitable conditions on A.

- **Coherence condition.** If the column coherence $\mu(A)$ satisfies $s < \frac{1}{2}(1 + 1/\mu)$ (Theorem 12.4.1), basis pursuit is guaranteed to recover any s-sparse signal. The coherence μ is exactly the quantity studied in Chapter 9 (Definition 9.2.1), and ETFs minimize it (Proposition 12.5.2).

- **Restricted isometry property.** The RIP (Definition 12.4.2) is a *restricted frame condition*: A satisfies RIP-s iff every s-column subframe is a $(1\pm\delta_s)$-frame for $\mathbb{R}^n$ (Proposition 12.5.1). If $\delta_{2s} < \sqrt{2}-1$, exact recovery holds (Theorem 12.4.3). Coherence controls RIP via $\delta_s \le (s-1)\mu$ (Proposition 12.4.4).

- **Frames are sensing matrices.** Every full-rank $n \times m$ matrix is a frame for $\mathbb{R}^n$ (Theorem 2.2.1). The entire frame-theoretic toolkit — bounds, tightness, coherence, the Welch bound — translates directly into properties of sensing matrices. ETFs are optimal sensing matrices in the coherence sense.

12.8 Lab Exercises

Lab Exercise 12.1: Reproducing Example 12.1

Implement the ℓ^1 minimization from Section 12.6 for the triangular frame. Verify numerically that:

(a) The optimal $t^* = 1/\sqrt{3}$, giving $c^* = (1, 0, 0)^T$.

(b) For $x = \alpha f_2$ (any scalar multiple of the second frame vector), ℓ^1 minimization finds the 1-sparse representation $c^* = (0, \alpha, 0)^T$.

(c) For a *generic* signal (not a multiple of any single frame vector), the ℓ^1 minimizer has exactly two nonzero entries. (Try $x = (2, -1)^T$ and report the minimizing c^*.)

Lab Exercise 12.2: Basis Pursuit via LP

Implement `basis_pursuit(A, y)` from Section 12.6 and test it on the following:

(a) The triangular frame with $x = f_1$: verify you get the same $c^* = (1, 0, 0)^T$ as Example 12.1.

(b) A random Gaussian sensing matrix ($n = 20$, $m = 100$) with a 3-sparse true signal. Does basis pursuit recover exactly?

(c) Repeat (b) for sparsity levels $s = 1, 2, 3, \ldots, 10$. For which values of s does recovery fail? Does the coherence bound (Theorem 12.4.1) correctly predict the failure threshold?

Lab Exercise 12.3: The Coherence–RIP Connection

For a random Gaussian sensing matrix with $n = 20$ columns normalized to unit norm and $m = 80$:

(a) Compute the coherence μ and the estimated RIP-2 constant δ_2 (using `rip_constant` from Section 12.6).

(b) Verify the bound $\delta_2 \leq (2 - 1)\mu = \mu$ (Proposition 12.4.4).

(c) Repeat for $s = 3, 4, 5$ to verify $\delta_s \leq (s - 1)\mu$.

(d) Is the bound tight (close to equality) or loose? What does this suggest about the relative sharpness of coherence-based vs. RIP-based guarantees?

Lab Exercise 12.4: ETFs as Sensing Matrices

Compare the triangular frame ($n = 2$, $m = 3$, coherence $\mu = 0.5$) to two other sensing matrices of the same size:

(a) A random unit-column-norm 2×3 matrix (random Gaussian, columns normalized).

(b) The 2×3 matrix $A = [[1, 0, 1], [0, 1, 0]]$ (not normalized; normalize columns first).

For each, compute μ, the coherence bound on sparsity recovery ($s < (1 + 1/\mu)/2$), and δ_2. Rank the three matrices by their sensing quality (smallest μ). Is the triangular frame (the ETF for $n = 2$, $m = 3$) the best? If so, why — connect to the Welch bound from Chapter 9.

Lab Exercise 12.5 (Challenge): Noisy Measurements

In practice, measurements are noisy: $y = Ax + e$ with $\|e\| \le \eta$. The noisy recovery program (LASSO or basis pursuit denoising) is

$$\hat{x} = \operatorname*{argmin}_z \|z\|_1 \quad \text{subject to} \quad \|Az - y\| \le \eta.$$

If A satisfies RIP-$2s$ with $\delta_{2s} < \sqrt{2} - 1$, the recovery error satisfies $\|\hat{x} - x\| \le C\eta$ for a constant C depending on δ_{2s}.

(a) Implement noisy basis pursuit using `cvxpy` (install via `pip install cvxpy`): the constraint becomes `cp.norm(A @ z - y, 2) <= eta`.

(b) For a random 20×80 sensing matrix and a 2-sparse true signal, add Gaussian noise with $\sigma = 0.01, 0.05, 0.1, 0.2$ (set $\eta = 2\sigma\sqrt{n}$ as a rough bound on $\|e\|$). Plot the recovery error $\|\hat{x} - x\|$ as a function of σ.

(c) Verify empirically that the error scales roughly linearly with σ (i.e. $\|\hat{x} - x\| \approx C\sigma$), as the theory predicts.

12.9 Supplementary Exercises

Theoretical Exercises

T1. Recall the *spark* of a matrix A (Chapter 10, Remark following Definition 10.4.1): the smallest number of linearly dependent columns. Prove that if x is s-sparse, z is s'-sparse, $Ax = Az$, and $s + s' < \text{spark}(A)$, then $x = z$. Conclude that a full-spark sensing matrix (spark $= n + 1$, the maximum possible) gives the best possible uniqueness guarantee, recovering any s-sparse signal uniquely whenever $s \le \lfloor n/2 \rfloor$.

T2. Prove that the ℓ^0 "norm" $\|\cdot\|_0$ satisfies the triangle inequality but *fails* absolute homogeneity ($\|cx\|_0 \ne |c| \, \|x\|_0$ in general), and is therefore not a genuine norm despite the name.

T3. Explain why the coherence-based recovery threshold of Theorem 12.4.1, $s < \frac{1}{2}(1+1/\mu)$, becomes vacuous (tends to infinity) as $\mu \to 0$, and why this limiting case is consistent with, but not practically meaningful for, compressed sensing (recall $\mu = 0$ forces $m \le n$, i.e. no redundancy to exploit).

T4. Prove that the RIP constant δ_s is non-decreasing in s: $\delta_s \le \delta_{s+1}$ for all s. (Hint: use the Cauchy interlacing theorem on the Gram matrix of an $(s+1)$-subset that extends a worst-case s-subset.)

T5. Using Proposition 12.4.4 ($\delta_s \le (s-1)\mu$) together with the RIP recovery condition $\delta_{2s} < \sqrt{2} - 1$, derive a coherence-sufficient condition for RIP-based recovery, and show it gives a strictly *more conservative* (smaller) guaranteed sparsity level than the direct coherence bound of Theorem 12.4.1, confirming the chapter's remark that coherence-based RIP bounds are pessimistic.

Computational Exercises

C1. Compute the spark of the triangular frame's synthesis matrix by brute-force search over all column subsets, confirm it equals 3, and verify T1's uniqueness guarantee directly: check that no other 1-sparse vector produces the same measurement as a given 1-sparse x.

C2. For a random 5×10 sensing matrix (small enough to check *all* subsets exhaustively, not a random sample), compute the exact RIP constants for $s = 1, \ldots, 6$ and confirm they are non-decreasing, verifying T4. (Caution: using the subsampled `rip_constant` from Lab Exercise 12.3 with a small `max_subsets` can spuriously *violate* monotonicity due to sampling noise — use exhaustive `itertools.combinations` for this exercise.)

C3. For $\mu = 0.05, 0.1, 0.2, 0.3$, plot both the direct coherence threshold from Theorem 12.4.1 and the coherence-via-RIP threshold from T5 as functions of μ, confirming the direct threshold is always larger (more permissive).

C4. Construct a random 10×40 sensing matrix, and for increasing sparsity levels $s = 1, \ldots, 8$, run basis pursuit on a random s-sparse signal and record whether recovery succeeds. Compare the empirical failure threshold to both the direct coherence bound (Theorem 12.4.1) and the spark-based uniqueness bound (T1, using $2s < \text{spark}(A)$, estimated via partial search since exact spark computation is expensive for $m = 40$).